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A Design Approach for Ferrite-Based Spoke-Type Motors for Electric Vehicles

Validation and Life Cycle Assessment

Danyang Cui

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Abstract

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Permanent magnet synchronous motors are widely used in electric vehicle traction because of their high efficiency and power density. Most high-performance designs use neodymium-iron-boron (NdFeB) magnets, which contain rare-earth elements. Ferrite magnets offer a rare-earth-free alternative, and their lower magnetic performance needs to be compensated for through motor design.

This thesis investigates ferrite-based spoke-type motors, focusing on technical feasibility, traction performance, and life cycle environmental performance. A review and benchmarking study identifies traction requirements and design challenges.

A segmented spoke-type motor is developed using rectangular Y40 ferrite magnets, non-magnetic structural components, and a carbon-fiber retaining sleeve. Electromagnetic, demagnetization, and mechanical finite-element analyses are combined with prototype fabrication and no-load validation. Peak torque and power densities reach 21.24 Nm/L and 18.18 kW/L, comparable to commercial reference levels. The design reduces shaft-side leakage flux to nearly zero. Under severe low-temperature conditions, demagnetization causes a simulated torque reduction of approximately 14.7%. Measured air-gap flux density and no-load back electromotive force agree with finite-element results.

The design method from the prototype study is then applied to a separate ferrite-based motor for comparison with a production-based NdFeB reference motor. A surrogate-assisted functional-equivalence framework seeks to match their torque envelopes while minimizing changes to other e-drive components, providing a basis for life-cycle comparison. The selected ferrite design requires a longer active length and higher DC bus voltage than the reference motor. Changes in material composition and driving-cycle electricity consumption are included in a cradle-to-grave life cycle assessment. The ferrite motor has a higher manufacturing-stage climate change impact because lower magnet-related impact does not offset increased use of electrical steel, copper, aluminum, ferrite, and structural materials. However, it consumes less electricity over WLTC. The life cycle comparison therefore depends on electricity carbon intensity, with a break-even value of approximately 60 g CO_2-eq/kWh under the conditions and assumptions of this study.

The results demonstrate the technical feasibility of ferrite-based traction motors. Their technical and environmental performance depends on the redesign needed to meet comparable traction requirements. The thesis combines electromagnetic design, functional equivalence, and life cycle assessment to evaluate ferrite-based alternatives to NdFeB traction motors.

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To Xiaoliang and Yourui, with love and gratitude.

List of Papers

This thesis is based on the following papers, which are referred to in the text by their Roman numerals.

I. **Cui, D.**, Max, L., Boström, C., Ekergård, B. (2022) Design of Spoke Type Traction Motor with Ferrite Material for EV Application. *2022 International Conference on Electrical Machines (ICEM)*, Valencia, Spain, pp. 315-320, doi: 10.1109/ICEM51905.2022.9910856.

II. **Cui, D.**, Max, L., Boström, C., Ekergård, B. (2026) A Comprehensive Review of Ferrite-Based Electrical Motors for Electric Vehicle Applications. *Renewable and Sustainable Energy Reviews*, vol. 243, p. 117403, Jan. 2027, doi: 10.1016/j.rser.2026.117403.

III. **Cui, D.**, Max, L., Boström, C., Ekergård, B. (2026) A Segmented Ferrite-Based Spoke-Type Permanent Magnet Motor for Electric Vehicle Traction: Design and Experimental Validation. Under review in *IEEE Access*.

IV. **Cui, D.**, Max, L., Boström, C., Ekergård, B. (2026) Functional Equivalence Design of a Ferrite-Based Spoke-Type Traction Motor Under Torque and Power Envelope Constraints. Presented at *2026 International Conference on Electrical Machines (ICEM)*, Sep. 9, 2026, Funchal, Madeira, Portugal.

V. **Cui, D.**, Max, L., Boström, C., Ekergård, B. (2026) Rare-earth substitution as a resource strategy for EV traction motors: Design compensation trade-offs and break-even electricity carbon intensity. *Resources, Conservation & Recycling Advances*, vol. 31, p. 200361, doi: 10.1016/j.rcradv.2026.200361.

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Abbreviations

AC	Alternating current
ASR	Automobile shredder residue
BEV	Battery electric vehicle
DC	Direct current
EMF	Electromotive force
EV	Electric vehicles
FEM	Finite-element method
GWP100	100-year global warming potential
HEV	Hybrid electric vehicle
LCA	Life cycle assessment
MTPA	Maximum torque per ampere
NdFeB	Neodymium-iron-boron
PM	Permanent magnet
PMSM	Permanent magnet synchronous motor
PWM	Pulse-width modulation
RMS	Root mean square
WLTC	Worldwide harmonized light vehicles test cycle

1. Introduction

1.1. Background

The electrification of road transport has increased the demand for traction motors with high efficiency, high torque density, compact dimensions, and reliable operation over a wide speed range [1], [2], [3]. Permanent magnet synchronous motors (PMSMs) are widely used in electric vehicles (EVs) because they can meet these requirements effectively [2], [4].

Most high-performance traction motors employ neodymium-iron-boron (NdFeB) magnets, sometimes with additions of dysprosium or terbium to improve resistance to demagnetization at elevated temperatures [5], [6], [7]. These magnets provide high remanence and magnetic energy density, allowing demanding torque and power targets to be achieved within a limited motor volume [8], [9].

Ferrite magnets offer a commercially mature rare-earth-free alternative. They are inexpensive, chemically stable, widely available, and characterized by high electrical resistivity [10], [11], [12], [13]. However, their remanence, coercivity, and maximum energy product are substantially lower than those of NdFeB magnets [8], [9], [11]. Their use in traction motors therefore requires compensation through topology and electromagnetic design [14], [15].

Among the investigated ferrite-based topologies, spoke-type permanent magnet motors are particularly promising because their radial magnet arrangement enables flux concentration and accommodates a relatively large magnet volume [15], [16]. Nevertheless, challenges remain regarding flux leakage, demagnetization, high-speed mechanical integrity, manufacturability, and the material compensation required to achieve traction-relevant performance [15], [17], [18], [19].

1.2. Motivation

Why should ferrite-based traction motors be investigated? Ferrite-based traction motors are investigated here as a way to reduce dependence on rare-earth magnets, alongside the assessment of their technical and environmental performance.

Rare-earth elements are essential to a wide range of strategic technologies, including EVs, wind turbines, industrial drives, electronics, aerospace systems,

and defence applications [5], [6]. Their growing use across several rapidly expanding sectors increases competition for supply and exposes manufacturers to disruptions originating outside the automotive industry [5], [6].

Rare-earth elements used in permanent magnets are therefore classified as critical or strategic raw materials in major industrial regions. In the European Union, criticality is assessed primarily in terms of economic importance and supply risk [20]. The Critical Raw Materials Act, which entered into force in 2024, aims to strengthen domestic capacity, diversify supply, increase recycling, and improve the resilience of European value chains [21], [22]. Rare-earth elements used in permanent magnets are included among the strategic raw materials considered important for the green and digital transitions [22], [23]. Similar concerns are reflected in the United States, where rare-earth elements are included in the national critical-minerals framework [24], [25].

This criticality does not necessarily imply geological scarcity. Rather, it results from the combination of high technological importance, geographically concentrated mining and processing, limited short-term substitutability, and vulnerability to geopolitical or trade-related disruptions [5], [6], [26]. The concentration of rare-earth processing and permanent magnet production also raises broader societal concerns. Environmental burdens associated with mining and refining are often localized, whereas the economic and technological benefits are distributed elsewhere in the value chain [5], [27]. Also, the supply restrictions and price volatility may affect manufacturing investment, vehicle cost, employment, and the pace of transport electrification [5], [6]. These supply risks and unevenly distributed impacts motivate the investigation of rare-earth-free traction motors.

Rare-earth-free traction motors should therefore not be viewed as a universal replacement for rare-earth-based machines. Instead, they broaden the range of technological options available to vehicle manufacturers and may improve material diversification and supply-chain resilience [10], [28]. Within this broader role, ferrite is a candidate because it is commercially established and uses comparatively abundant raw materials [10], [12], [26].

1.3. Aim and Structure

The overall aim of this thesis is to develop and evaluate a ferrite-based spoke-type traction motor design approach for rare-earth-free EV propulsion. The technical feasibility, traction performance, and life cycle environmental performance are considered. As illustrated in Figure 1.1, the research is organized around three questions corresponding to Chapters 2-4.

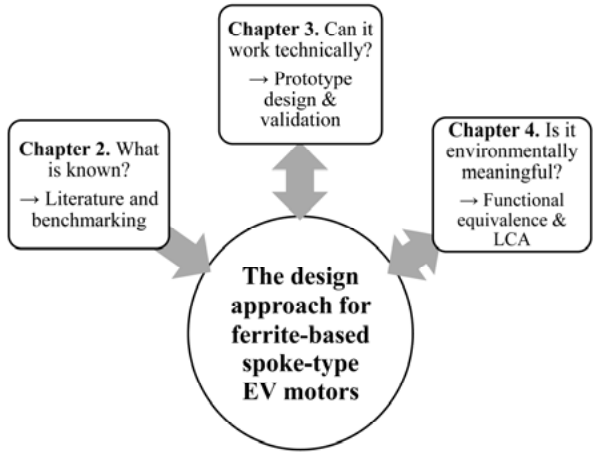


Figure 1.1. Research framework and structure of the thesis.

The first question is: What is known about ferrite-based traction motors and their potential for EV applications?

Chapter 2 addresses this question through a comprehensive literature review and systematic benchmarking study. Commercial EV powertrains are examined to establish representative requirements for torque, power, speed, and normalized motor performance. The magnetic characteristics of ferrite and experimentally validated ferrite-based motor concepts are then examined to identify suitable topologies, common design approaches, performance levels, and remaining limitations. This analysis provides the basis for selecting the spoke-type topology.

The second question is: Can the proposed ferrite-based spoke-type design approach meet traction-relevant technical requirements?

Chapter 3 investigates this question by developing and technically evaluating a ferrite-based spoke-type design approach through a prototype-oriented segmented motor. The design aims to balance traction-relevant performance with simple magnet geometry, manufacturability, demagnetization resistance, and high-speed mechanical feasibility. A segmented ferrite-based spoke-type permanent magnet motor is designed, simulated, and experimentally evaluated. The rotor uses rectangular ferrite magnets, segmented laminations, non-magnetic structural components, and a carbon-fiber retaining sleeve. Electromagnetic performance, demagnetization behavior, and mechanical integrity are assessed using finite-element method (FEM) simulations, followed by prototype-level no-load measurements of air-gap flux density and back electromotive force (EMF).

The third question is: Is the substitution of rare-earth magnets with ferrite environmentally meaningful at the motor-system level?

Chapter 4 applies the spoke-type design approach investigated in Chapter 3 to a separate ferrite-based motor for comparison with an NdFeB benchmark.

Unlike the prototype-oriented motor in Chapter 3, which was developed primarily for technical validation, the Chapter 4 motor is designed specifically for functional equivalence with an NdFeB benchmark and subsequent life cycle assessment (LCA). A surrogate-assisted functional-equivalence design framework is used to identify a ferrite-based configuration with comparable traction functionality while accounting for the geometric and electrical compensation required by ferrite substitution. The benchmark and the selected ferrite-based motor are then evaluated through a cradle-to-grave LCA covering raw-material production, manufacturing, use-phase electricity consumption, and end-of-life treatment.

1.4. Contributions

The principal contributions of this thesis are:

1. A systematic benchmark of commercial EV traction-motor performance and a focused review of ferrite magnet characteristics and ferrite-based traction-motor concepts.
2. The design, FEM analysis, and prototype-level no-load electromagnetic validation of a segmented ferrite-based spoke-type traction motor.
3. A functional-equivalence and multi-criteria design framework for comparing ferrite-based and NdFeB-based traction motors under common performance requirements.
4. A cradle-to-grave LCA linking magnet substitution with motor redesign, use-phase efficiency, electricity supply, and end-of-life treatment.

2. State of the Art

The development of a ferrite-based traction motor requires an understanding of both the performance expected from EV traction application and the design constraints introduced by ferrite magnets. Commercial EV traction systems provide practical reference points for torque, power, speed, and efficiency, and are therefore benchmarked to establish representative performance ranges for rare-earth-free traction motor development [29], [30]. The ferrite characteristics that directly influence motor design are then discussed, followed by a focused review of ferrite-based motor concepts and the design strategies used to compensate for the limited magnetic performance of ferrite.

This chapter builds primarily on the systematic review and benchmarking study reported in **Paper II** [31] and focuses on the findings most relevant to the ferrite-based spoke-type traction motor design approach developed in this thesis. Particular attention is given to spoke-type machines because their flux-concentration capability and ability to accommodate a relatively large magnet volume make them especially suitable for ferrite-based traction applications [15], [16].

2.1. Traction Motor Performance Requirements

2.1.1. Peak Torque and Power Benchmark

Figure 2.1 summarizes the peak torque and peak power specifications of 185 passenger battery electric vehicles (BEVs) collected from recent global markets. For vehicles equipped with more than one traction motor, the combined vehicle-level values are reported rather than those of an individual motor. A linear regression through the origin is included to illustrate the overall relationship between the two quantities. The complete dataset and individual sources are provided in Appendix A, based on which Figure 2.1 is reproduced.

Across the collected vehicles, peak torque increases approximately proportionally with peak power, with a fitted torque-to-power coefficient of about 2.0 Nm/kW. Using the relationship among power P, torque T, and rotational speed ω [32],

$$P[W] = T[Nm] \cdot \omega[rad/s], \tag{2.1}$$

the aggregate base speed of the collected powertrains is estimated to be approximately 4760 rpm.

Figure 2.2 presents the distribution of the same dataset and highlights the torque and power ranges in which most surveyed passenger BEVs are concentrated. Together, the two figures characterize the vehicle-level torque and power requirements relevant to ferrite-based motor development.

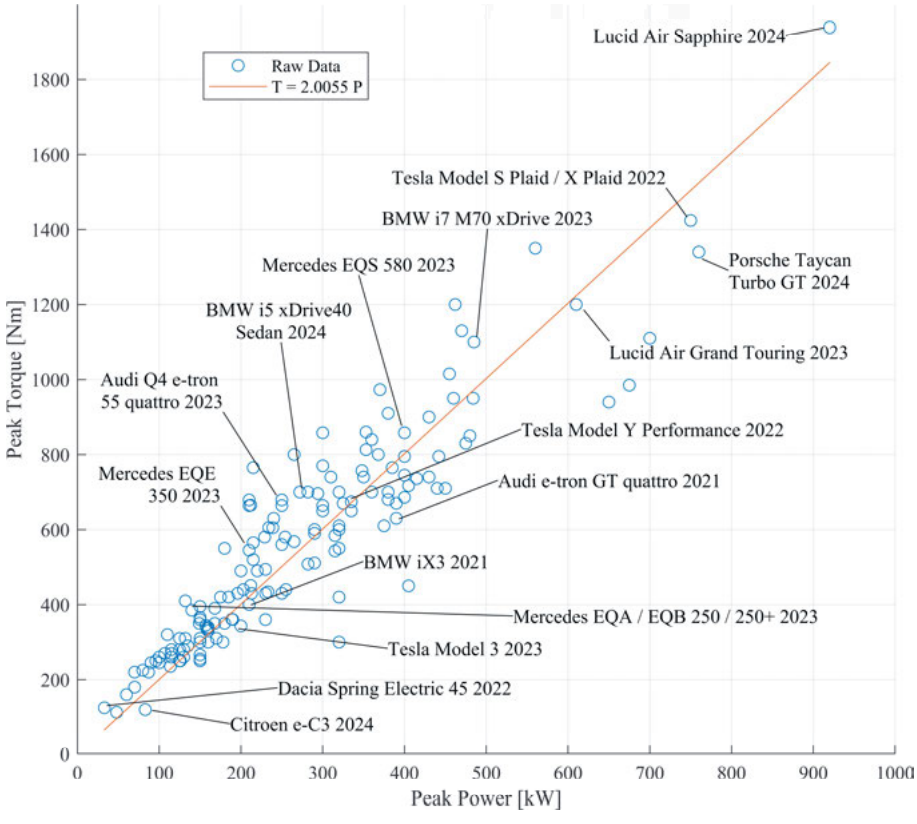


Figure 2.1. Peak torque and power of the surveyed passenger BEVs and the regression line. Adapted from **Paper II** [31] under the CC BY 4.0 license.

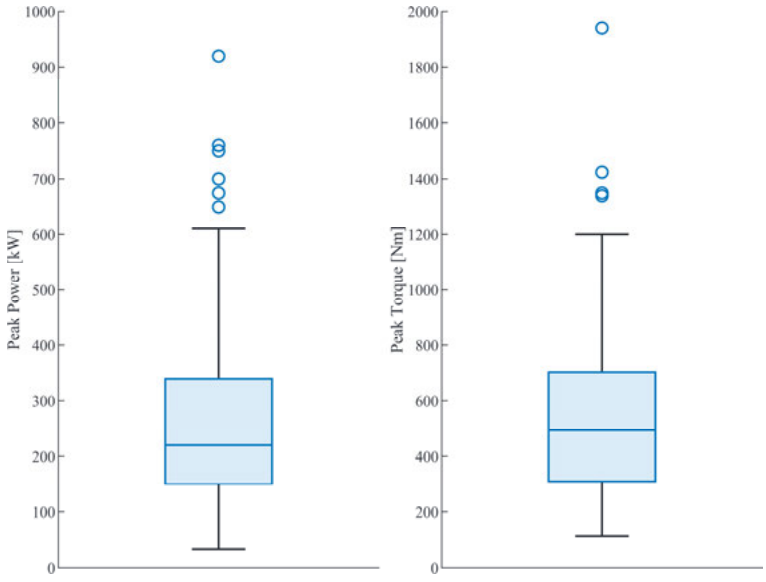


Figure 2.2. Peak power and torque distribution of the surveyed passenger BEVs. Adapted from **Paper II** [31] under the CC BY 4.0 license.

2.1.2. Normalized Performance of Traction Motors

Vehicle-level torque and power specifications alone do not indicate how effectively these requirements are achieved within the motor package. Table 2.1 therefore compares selected commercial BEV and hybrid electric vehicle (HEV) traction motors introduced between 2004 and 2017 using volumetric and mass-normalized performance indicators.

It is important to point out that the original sources use different definitions of motor volume and mass. In some cases, the reported values refer mainly to the electromagnetic active parts, whereas in others they may also include the shaft, end windings, or housing. This limits strict one-to-one comparison, but the normalized indicators remain useful for identifying broad performance levels.

For the motors in Table 2.1, the torque density ranges from 15 to 26 Nm/L, while the power density ranges from 3 to 9 kW/L. These ranges are used to evaluate the ferrite-based motor designs in the following chapters.

Table 2.1. Traction motor power, torque, and normalized performance. Adapted from **Paper II** [31] under the CC BY 4.0 license.

Year and Ref	Model	Peak Power [kW]	Peak Torque [Nm]	Max Speed [rpm]	Volume [L]	Weight [kg]	Power Density [kW/L]	Specific Power [kW/kg]	Torque Density [Nm/L]	Specific Torque [Nm/kg]
2017 [33]	Toyota Prius	53	163	17 000	9.3*	33.13*	5.7	1.6	17.53*	4.92*
2016 [33], [34]	BMW i3	125	250	11 400	13.74*	41.67*	9.1	3	18.22*	6*
2014 [33], [35]	Honda Accord	124	306	15 000	14.59*	42.76*	8.5	2.9	20.98*	7.16*
2012 [33], [36]	Nissan Leaf	80	280	10 390	19.05*	57.14*	4.2	1.4	14.7*	4.9*
2010 [33]	Toyota Prius	60	207	13 000	12.5*	37.5*	4.8	1.6	16.56*	5.52*
2008 [37]	Lexus LS600h	110	300	10 230	16.7	44.7	6.6	2.46	17.96	6.71
2007 [37]	Toyota Camry	70	270	14 000	14.8	41.7	5.9	1.7	18.25	6.47
2004 [37]	Toyota Prius	50	400	6000	15.4	45	3.3	1.1	25.97	8.89

Note: Data with asterisks (*) are estimated or calculated values derived from the original sources.

2.2. Ferrite Magnets for Traction Motor Applications

2.2.1. Magnetic Characteristics Relevant to Motor Design

Figure 2.3 compares the remanent flux density B_r and coercive field strength H_c of commonly used permanent-magnet materials [8]. For convenience, the sign of H_c is reversed in the figure. The ellipses in the figure represent the property ranges of different grades within each magnet family. These ranges arise partly from temperature dependence and partly from differences in composition and manufacturing process.

Among these materials, NdFeB provides the high remanence and coercivity required for compact high-performance traction motors, whereas ferrite exhibits substantially lower magnetic strength. A higher B_r generally supports greater air-gap flux density and torque production, while a higher H_c indicates stronger resistance to demagnetization [38]. The lower remanence and magnetic energy density of ferrite require compensation through increased magnet volume and magnetic-circuit design.

In addition, Table 2.2 compares the key magnetic properties of NdFeB and ferrite magnets. The maximum energy product of high-grade NdFeB is approximately one order of magnitude higher than that of Y40 ferrite [9], [11]. Ferrite substitution therefore cannot be treated as a one-to-one replacement by magnet volume and generally requires greater magnet volume or flux-concentrating motor topologies.

Y40 was selected for the ferrite-based motor designs in this thesis because it is shown among the highest-performance commercially available rare-earth-free ferrite grades [11], offering comparatively high remanence and maximum energy product within the ferrite magnet family.

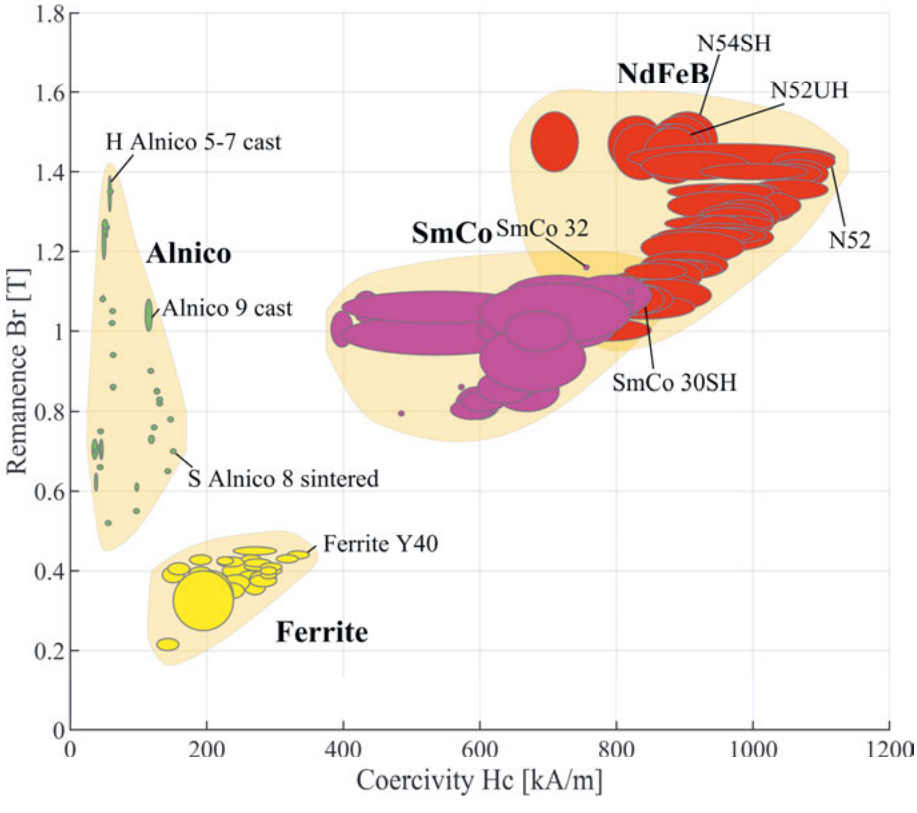


Figure 2.3. Coercivity and remanence of commonly used permanent magnets, with data reproduced courtesy of ANSYS, Inc. Adapted from **Paper II** [31] under the CC BY 4.0 license.

Table 2.2. Magnetic properties of NdFeB and ferrite permanent magnets, with data reproduced courtesy of ANSYS, Inc. unless otherwise stated. Adapted from **Paper II** [31] under the CC BY 4.0 license.

Property	Unit	NdFeB	Ferrite
Max energy product	$(BH)max\ [kJ/m^3]$	175~430	6.37~41.8
Remanence	$B_r\ [T]$	0.975~1.55	0.2~0.46
Coercivity	$H_c\ [kA/m]$	675~1120	127~350
Maximum working temperature	T_w [°C]	80~230	180~300
Curie Temperature	T_c [°C]	310~360	450~460
Temperature coefficient for B_r [39]	[%/°C]	-0.12 ~ -0.105	-0.18
Temperature coefficient for H_{ci} [39]	[%/°C]	-0.6 ~ -0.4	+0.4
Recoil permeability	μ_{rec}	1.05	1~1.1

Note: H_{ci} denotes the intrinsic coercivity.

2.2.2. Demagnetization and Manufacturing Considerations

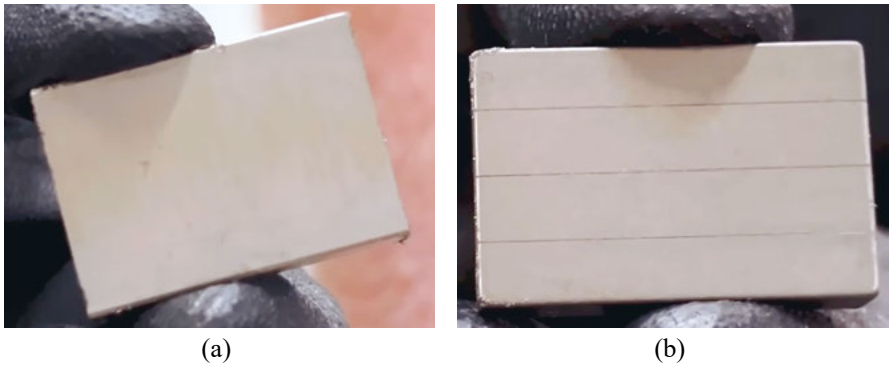
Ferrite exhibits a positive temperature coefficient of intrinsic coercivity, meaning that its resistance to demagnetization decreases as temperature falls, whereas NdFeB generally shows the opposite temperature dependence [39]. Low-temperature operation can therefore increase the risk of irreversible demagnetization in ferrite magnets, making temperature-dependent demagnetization assessment particularly important for traction applications.

Ferrite is a brittle ceramic material with limited tensile strength and is susceptible to chipping or fracture during machining and handling [40], [41]. These properties favor simple geometries and limited post-processing, motivating the use of rectangular ferrite magnets in the prototype.

Ferrite has a lower density than NdFeB. However, achieving equivalent performance may require a larger magnet volume and corresponding changes to the motor geometry, potentially resulting in a larger and heavier motor and thereby offsetting the density advantage of ferrite.

Ferrite has a much higher electrical resistivity than NdFeB and therefore experiences substantially lower magnet eddy-current loss [13]. In NdFeB-based traction motors, magnet segmentation is commonly used to reduce eddy-current heating [42]. Figure 2.4 shows segmented magnets from the Tesla Model 3 and monoblock magnets reported for the Tesla Model S and Cybertruck [43]. For ferrite magnets, the much higher resistivity reduces this constraint and can therefore support the use of larger monoblock magnets, potentially simplifying manufacturing and assembly.

In economic terms, ferrite is substantially less expensive than high-grade NdFeB [8]. The magnetic and mechanical properties discussed above inform the demagnetization assessment and the choice of rectangular magnets in Chapter 3.



(a) (b)

Figure 2.4. Neodymium magnets: (a) segmented magnets from the Tesla Model 3 and (b) monoblock magnets from the Tesla Model S and Cybertruck [43]. From **Paper II** [31] under the CC BY 4.0 license.

2.3. Ferrite-Based Traction Motor Topologies

Rare-earth-free traction motors include electrically excited synchronous machines, induction machines, reluctance machines, and PMSMs using rare-earth-free permanent magnets [2], [10], [28], [44], [45]. This thesis focuses on ferrite-based PMSMs, for which the reported traction-oriented concepts can be grouped into three principal topology families: spoke-type PMSMs, ferrite-assisted synchronous reluctance motors, and axial-flux motors.

2.3.1. Topology Comparison and Selection

Spoke-type machines accommodate a relatively large ferrite volume and concentrate the magnet flux toward the air gap, making them well suited to the low remanence of ferrite [15], [16]. Their main limitations include shaft-side leakage, demagnetization, and the mechanical retention of large radial magnets at high speed.

Ferrite-assisted synchronous reluctance motors combine permanent-magnet and reluctance torque, thereby reducing the amount of ferrite required for a given torque level [45]. Their multilayer flux-barrier rotors provide substantial saliency but also introduce structural and manufacturing complexity, particularly at high rotational speed. The reviewed designs generally exhibit slightly lower torque density than high-magnet-volume spoke-type machines, while requiring less permanent-magnet material.

Ferrite-based axial-flux motors can achieve high torque density and a compact axial package because of their large effective air-gap area and short end-winding length [46], [47]. However, axial-flux architectures generally require more demanding air-gap control, mechanical assembly, and manufacturing processes than conventional radial-flux machines [48], [49], [50]. Their short axial package is particularly advantageous where axial space is limited.

The choice of ferrite-based motor topology depends on torque density, magnet utilization, packaging, manufacturability, and mechanical requirements [28]. In this work, the spoke-type topology was selected for its strong flux-concentration capability and capacity to accommodate a large ferrite volume within a conventional radial-flux architecture. It offers greater freedom to increase magnet volume than ferrite-assisted synchronous reluctance motors. Compared with axial-flux machines, it is more closely aligned with established radial-flux manufacturing and testing methods.

2.3.2. Ferrite-Based Spoke-Type Motors

In a spoke-type rotor, the permanent magnets are arranged radially and magnetized circumferentially. This arrangement allows a relatively large

magnet volume to be accommodated within the rotor and concentrates the resulting flux toward the air gap, making the topology particularly suitable for ferrite magnets [15], [16].

Table 2.3 summarizes representative ferrite-based spoke-type motors developed for EV traction applications together with their main performance indicators. Only studies involving prototype fabrication and experimental testing were included, since prototype testing provides stronger evidence of practical feasibility than simulation alone. The reviewed studies show that increasing torque density is the primary design objective, while improving resistance to irreversible demagnetization is another important consideration.

The reported torque densities span 4-28 Nm/L, with several designs falling within or close to the 15-26 Nm/L commercial reference range established in Section 2.1.2. Differences among the reported values should be interpreted together with ferrite grade, current loading, voltage level, motor size, cooling assumptions, and rotor design.

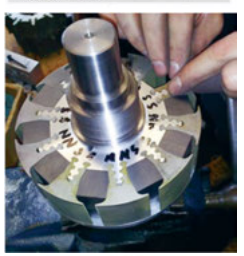
Rotor magnetomotive force is commonly enhanced through geometry optimization, including flux barriers and hybrid magnetic circuits with parallel-series flux paths. Several designs also employ auxiliary permanent magnets [19], [51], [52]. These are typically NdFeB magnets combined with a larger volume of ferrite, allowing magnetic performance to be improved while keeping rare-earth usage relatively low. In addition, reluctance torque can be increased by enlarging the difference between the d- and q-axis inductances, L_d and L_q [53], [54].

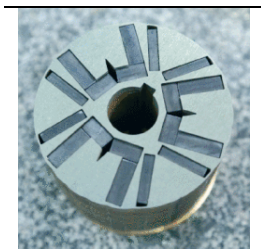
Flux leakage is reduced using several structural approaches. Some designs adopt multilayer rotor laminations with alternating core geometries to preserve mechanical strength while limiting material use and shaft-side leakage [55]. Other concepts use segmented rotors and non-magnetic structural components to interrupt unwanted leakage paths around the shaft [15], [18], [56].

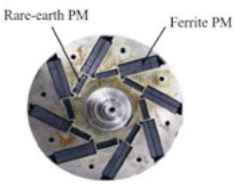
Demagnetization resistance is addressed through magnet placement, winding configuration, thermal management, and rotor geometry. Deeply buried magnets and optimized magnet layouts can reduce exposure to local demagnetizing fields [51], [52], while appropriate winding and slot-pole configurations can limit armature-field concentration [18], [57], [58]. Other approaches include improved thermal management and overhang structures that modify the local magnetic field distribution [18], [51], [52], [53], [54].

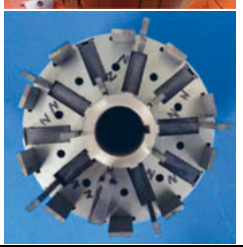
The reviewed studies therefore identify three recurring design priorities for ferrite-based spoke-type motors: increasing useful air-gap flux, suppressing leakage flux, and maintaining demagnetization resistance while preserving mechanical integrity.

Table 2.3. Selected spoke-type ferrite motors with their key performance. From **Paper II** [31] under the CC BY 4.0 license.

	Ref	Year	Ferrite grade	Volume [L]	DC bus voltage [V]	Max current [Arms]	Peak power [kW]	Peak torque [Nm]	Maximum speed [rpm]	Power density [kW/L]	Torque density [Nm/L]	Highlights and comments
	[55]	2019	NMF-9G ($B_r = 0.41\,T$ at room temperature)	0.10 L (Active part incl. shaft)		65	0.6	1.75		6.09	17.6	Classic spoke-type topology with alternately superimposed core sheets in different shapes. Rotor overhang utilized. Torque increased by 16.7%.
	Motor 1 in [15], [18], [56]	2017	FE9B ($B_r = 0.44\,T$ and $H_{ci} = 370\,kA/m$ at room temperature)	10 L (Incl. end winding, excl. housing)	400	425	117	270	15000	11.7	27	Fir-tree topology and non-magnetic steel used to mitigate flux leakage around shaft.
	Motor 2 in [15], [18], [56]	2017	FE9B ($B_r = 0.44\,T$ and $H_{ci} = 370\,kA/m$ at room temperature)	10 L (Incl. end winding, excl. housing)	400	425	93.6*	216*	15000	9.36*	21.6*	Large cavities opened to reduce weight, while is less effective in anti-leakage compared to motor 1.

	Ref	Year	Ferrite grade	Volume [L]	DC bus voltage [V]	Max current [Arms]	Peak power [kW]	Peak torque [Nm]	Maximum speed [rpm]	Power density [kW/L]	Torque density [Nm/L]	Highlights and comments
	[57]	2013	$B_r = 0.45\,T$			11		1.3964			14.691	Wing-shaped segmented magnets designed to enhance air-gap flux density.
	[58]	2016	$B_r = 0.4\,T$ (20 °C)	0.81 L (Active part incl. shaft)		7	1.5	3.4		1.85	4.2	Angle among segmented magnets optimized to improve anti-demagnetization ability.
	[53]	2012	NMF-12G $B_r = 0.45\,T$	2.55 L (Active part incl. shaft)	144	240	18.3	71.2	8000	7.18	27.92	Difference between L_d and L_q increased to improve reluctance torque.

	Ref	Year	Ferrite grade	Volume [L]	DC bus voltage [V]	Max current [Arms]	Peak power [kW]	Peak torque [Nm]	Maximum speed [rpm]	Power density [kW/L]	Torque density [Nm/L]	Highlights and comments
	[59]	2017						13.51				The DMADV design methodology utilized to significantly reduce computation time compared to traditional "one-factor-at-a-time" approach.
	[19]	2023	$B_{r,ferrite} = 0.45\,T$ $B_{r,Nd} = 1.26\,T$	0.36 L (Active part incl. shaft)		14.14		6*			16.67*	Both ferrite and Nd magnets used to balance the cost and performance.
	[51]	2022	Ferrite Y30 ($B_r = 0.4\,T$, $H_c = 188\,kA/m$ at 20°C) NdFeB35 ($B_r = 1.21\,T$, $H_c = 890\,kA/m$ at 20°C)	2.07 L (Active part incl. shaft)		28	5	39.4	1200	2.42	19.03	Nd and ferrite magnets used. Hybrid flux circuit and deeply buried magnets designed to improve the working point of the magnets, thus enhanced anti-demagnetization.

	Ref	Year	Ferrite grade	Volume [L]	DC bus voltage [V]	Max current [Arms]	Peak power [kW]	Peak torque [Nm]	Maximum speed [rpm]	Power density [kW/L]	Torque density [Nm/L]	Highlights and comments
	[54]	2015	Ferrite 12G ($B_r = 0.45\,T$)	4.08 L (Active part incl. shaft and overhang)	350		23.72	76.45	8000	5.81	18.74	Multilevel overhang structure used to deviate L_d and L_q, increase air-gap flux density, and reduce axial flux leakage.
	[52]	2022	Ferrite Y30 ($B_r = 0.4\,T$) NdFeB35 ($B_r = 1.2\,T$)	2.14 L (Active part incl. shaft)				38.5			17.99	Asymmetric hybrid PMs used to enhance reluctance torque and anti-demagnetization.

Note: Data with asterisks (*) are estimated or calculated values derived from the original sources.

2.3.3. Design Challenges Relevant to This Thesis

The reviewed spoke-type designs demonstrate that ferrite magnets can support traction-relevant performance when combined with appropriate magnetic-circuit design. Nevertheless, prototype-level evidence remains limited for traction motors that simultaneously combine efficient ferrite utilization, simple magnet geometry, suppression of shaft-side leakage flux, resistance to irreversible demagnetization, and reliable mechanical retention at high rotational speed. Balancing these electromagnetic, mechanical, and manufacturing requirements therefore remains an important design challenge.

These findings provide the technical basis for the prototype-oriented motor designed and evaluated in Chapter 3. Chapter 4 subsequently applies the design approach developed in Chapter 3 to a separate ferrite-based motor, intended for a functionally equivalent life-cycle comparison with an NdFeB benchmark.

3. Ferrite-Based Motor Design and Validation

This chapter develops and technically evaluates a ferrite-based spoke-type design approach: The aspects of efficient ferrite utilization, suppression of shaft-side leakage flux, resistance to irreversible demagnetization, mechanical integrity, and manufacturability are considered. A prototype-oriented segmented spoke-type motor is developed using rectangular ferrite magnets, segmented rotor laminations, non-magnetic structural components, and a carbon-fiber retaining sleeve. The work includes the motor design procedure, electromagnetic, demagnetization, and mechanical FEM analyses, prototype fabrication, and no-load experimental validation. The work presented in this chapter is based primarily on **Paper I** and **Paper III**.

3.1. Methodology

3.1.1. Design Requirements

The design target for the prototype combined peak power, peak torque, maximum speed, efficiency, demagnetization resistance, and manufacturability within an active volume of approximately 8 L. Meanwhile, the torque density was evaluated as one of these requirements.

The technical requirements are listed in Table 3.1 and serve as the design targets for the electromagnetic and mechanical analyses presented in the following sections.

Table 3.1. Technical requirements of the proposed motor.

Parameter	Unit	Target value
Peak power	kW	140
Peak torque	Nm	160
Active volume*	L	8
Maximum speed	rpm	30 000
Peak efficiency	%	97
Maximum current	A_{rms}	150

* The active volume is defined as the cylindrical volume calculated from the stator outer diameter and the active axial length. It therefore includes the shaft region within the active length but excludes the end windings.

The peak-power target of 140 kW, peak-torque target of 160 Nm, and maximum-speed target of 30 000 rpm define a high-speed, moderate-torque design objective. Electromagnetic feasibility is evaluated against the torque, power, speed, and efficiency requirements, while the active-volume constraint provides a consistent basis for assessing torque and power density.

3.1.2. Motor Design Procedure

Figure 3.1 summarizes the design procedure for developing the proposed motor.

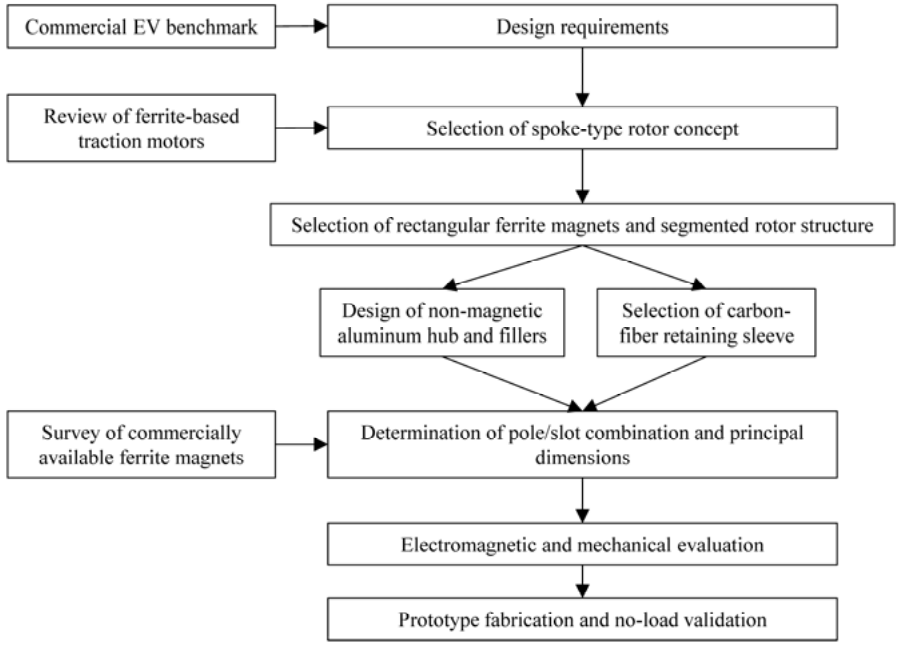


Figure 3.1. Design procedure for the proposed segmented ferrite-based spoke-type motor.

To meet the requirements in Table 3.1 while compensating for the lower magnetic performance of ferrite, a spoke-type rotor was selected for its flux-concentration capability and ability to accommodate a relatively large magnet volume. Standard rectangular ferrite magnets were chosen to simplify manufacturing and avoid customized curved or tapered geometries. A segmented rotor structure was adopted to suppress shaft-side leakage flux and facilitate the integration of non-magnetic structural components.

Non-magnetic aluminum components were then introduced at the shaft side of the rotor. The aluminum hub and filler components interrupt the low-reluctance leakage paths that would otherwise form between the inner magnet

ends and the shaft region. Because the segmented rotor does not rely on continuous ferromagnetic bridges for mechanical retention, a carbon-fiber sleeve was placed around the rotor outer surface to retain the ferrite magnets, lamination segments, and filler components during operation, particularly at high rotational speeds.

The dimensions of commercially available ferrite magnets were surveyed during the design process. Together with the required motor performance, the available magnet dimensions were used to determine the pole-slot combination and principal motor dimensions.

The motor concept was subsequently evaluated through electromagnetic, demagnetization, and mechanical FEM analyses, followed by prototype fabrication and no-load experimental validation.

3.1.3. Electromagnetic Analysis

The electromagnetic performance of the proposed motor was evaluated using a two-dimensional FEM model incorporating the temperature-dependent magnetic properties of Y40-grade ferrite magnets. The analysis included no-load magnetic behavior, loaded torque and power capability, and efficiency. Unless otherwise stated, the electromagnetic simulations were conducted at room temperature.

The no-load analysis was used to evaluate the air-gap flux-density distribution, three-phase flux linkage, and back EMF over the investigated speed range. Loaded performance was evaluated using maximum-torque-per-ampere (MTPA) control below the base speed and field-weakening control above it under the voltage constraint.

The efficiency calculation included copper loss, iron loss, and mechanical loss. A mechanical loss of 15 W per 1000 rpm was assumed to represent bearing, friction, and windage losses.

The electromagnetic model was two-dimensional and therefore did not represent end-winding fields, axial leakage flux, or other three-dimensional effects. The calculated electromagnetic quantities were consequently treated as active-region predictions.

3.1.4. Demagnetization Assessment

Irreversible demagnetization was assessed under a deliberately severe combination of low temperature and high demagnetizing current. The temperature-dependent demagnetization characteristics of the Y40 ferrite magnets were defined using manufacturer data.

Figure 3.2 shows the temperature-dependent demagnetization curves of the Y40 ferrite magnets used in the analysis. These curves were implemented in the FEM model to represent the variation in magnetic characteristics with

temperature and to determine the operating point of the magnets during the demagnetization assessment.

The demagnetization stress test was performed at 7000 rpm and -40 °C. A current of 150 Arms, corresponding to the specified maximum phase current, was applied along the negative d-axis for approximately two electrical cycles. This operating condition was selected to impose a strong demagnetizing field rather than to represent continuous normal operation.

After removal of the negative d-axis stress current, the no-load back EMF and loaded torque were recalculated under the same respective evaluation conditions used before the stress event. Irreversible demagnetization was quantified by comparing the corresponding values before and after the event.

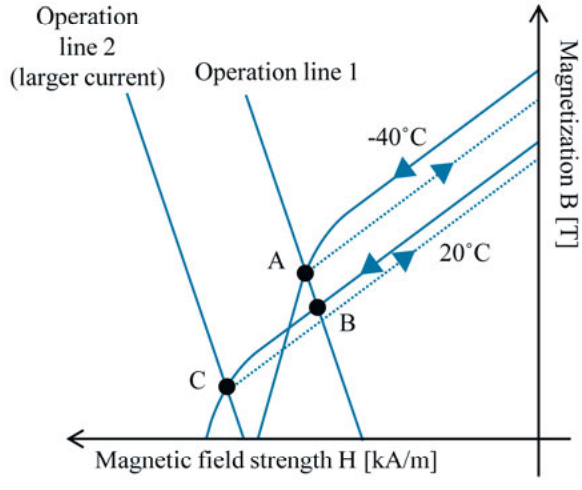


Figure 3.2. Temperature-dependent demagnetization curves and representative irreversible-demagnetization behavior of Y40 ferrite. Adapted from **Paper III**.

3.1.5. Mechanical Analysis

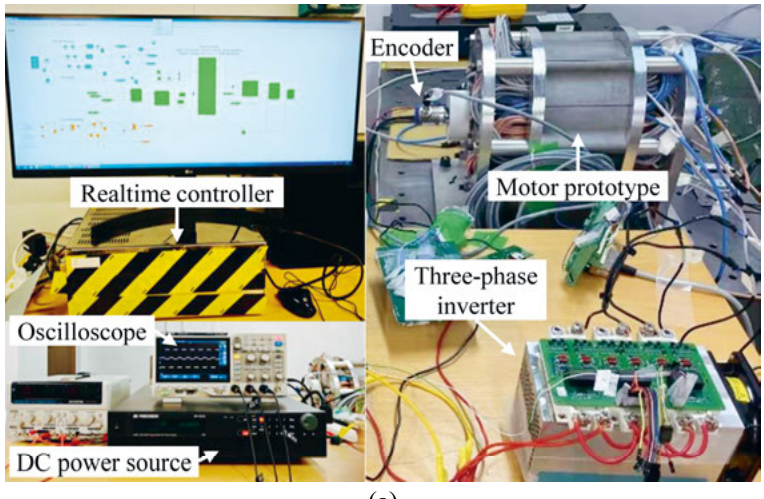
A separate mechanical FEM analysis was performed to assess the structural response of the segmented rotor and carbon-fiber sleeve under high-speed operation. Electromagnetic radial loading obtained under the maximum-current condition was combined with centrifugal loading at 30 000 rpm. Although these conditions may not occur simultaneously during normal operation, their combination was used as a conservative load case for assessing the rotor-retention structure.

The carbon-fiber retaining sleeve was represented using an equivalent isotropic material model. The analysis focused on the von Mises stress and radial deformation of the sleeve as indicators of its mechanical response under the investigated high-speed condition.

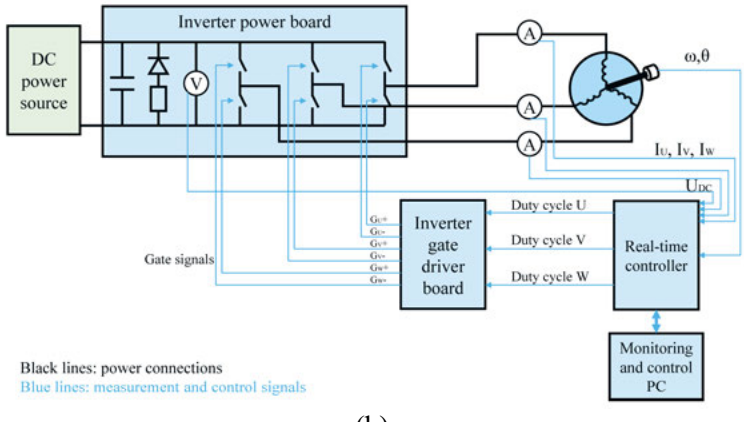
3.1.6. Prototype Fabrication and Experimental Methods

A prototype of the proposed ferrite-based spoke-type motor was manufactured for experimental validation and tested in an air-cooled laboratory environment. The prototype included the segmented rotor, stator, and complete motor assembly. Appendix B provides additional assembly views and the manufacturing drawings used to fabricate the prototype. The CAD models and drawings were created by the author.

The no-load back EMF was measured using the experimental setup shown in Figure 3.3(a), while the corresponding system configuration is illustrated in Figure 3.3(b). The induced voltage and rotational speed were recorded at 237 and 388 rpm using an oscilloscope. FEM simulations were conducted at the same speeds for direct comparison with the measured back-EMF waveforms.



(a)



(b)

Figure 3.3. (a) Experimental setup for the no-load back-EMF measurement, including the prototype motor, inverter, controller, and sensors. Adapted from **Paper III**. (b) System diagram of the motor-drive test bench.

Experimental validation was limited to no-load electromagnetic measurements. The no-load air-gap flux density was measured using a magnetic flux-density meter with a probe inserted into the physical air gap.

The experimental validation therefore focused on air-gap flux-density and no-load back-EMF measurements. Full load testing would have required additional instrumentation and the commissioning of a dynamometer test setup and was therefore beyond the experimental scope of the present work. Loaded torque and efficiency, together with irreversible demagnetization and high-speed mechanical performance, were assessed through simulations and remain to be experimentally validated.

3.2. Results

3.2.1. Final Motor Design and the Prototype

The resulting motor is a six-pole spoke-type PMSM with a segmented rotor, as shown in Figure 3.4, which also identifies the materials used in its main components. Its principal dimensions and structural parameters are summarized in Table 3.2. The manufactured rotor, stator, and complete motor assembly are shown in Figure 3.5 and Figure 3.6.

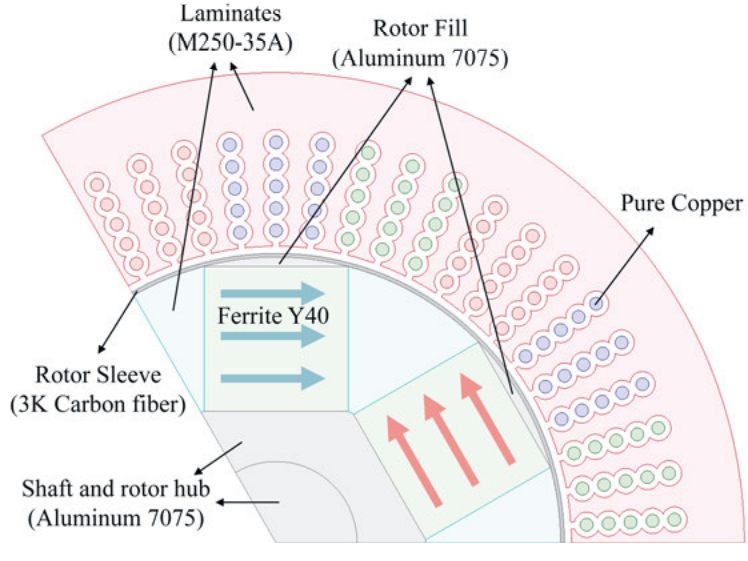


Figure 3.4. Structure and principal materials of the proposed segmented ferrite-based spoke-type motor. Adapted from **Paper III**.

Table 3.2. Principal structural parameters of the proposed motor. Adapted from **Paper III**.

Parameter	Unit	Value
Number of slots/poles	-	54/6
Stator outer diameter	mm	260
Stator inner diameter	mm	164
Rotor outer diameter incl. sleeve	mm	160
Physical air-gap thickness	mm	2
Carbon-fiber sleeve thickness	mm	1
Active axial length	mm	150
Active volume	L	7.96
Magnet cross-sectional dimensions (each)	mm × mm	40 × 40
Conductors per slot	-	5
Conductor area excl. insulation (each)	mm^2	10

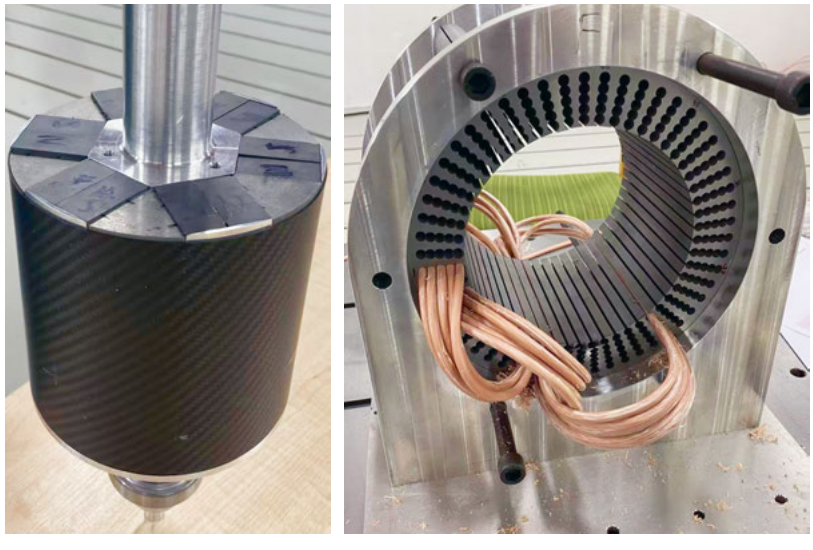


Figure 3.5. Rotor (left) and stator (right). Adapted from **Paper III**.

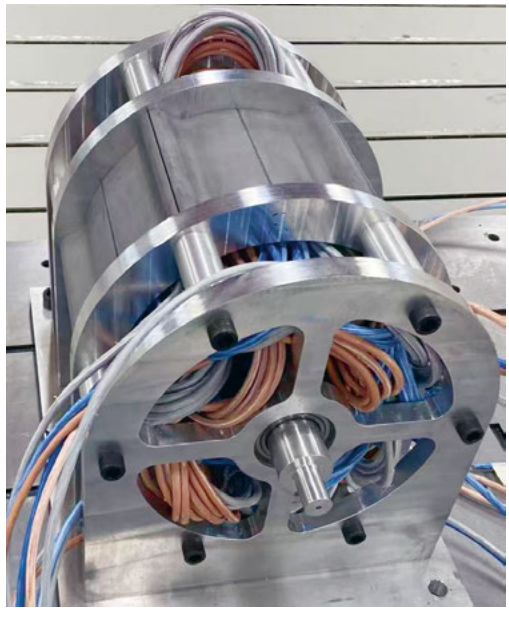


Figure 3.6. Complete motor assembly. Adapted from **Paper III**.

3.2.2. Shaft-Side Leakage Suppression

A conventional one-piece rotor was used as a comparison design, as shown in Figure 3.7. The stator geometry, winding, active axial length, principal rotor dimensions, and physical air-gap length were kept unchanged between the two motors, while the comparison rotor contained approximately 4% more ferrite. Figure 3.8 compares the no-load magnetic flux distributions of the two designs.

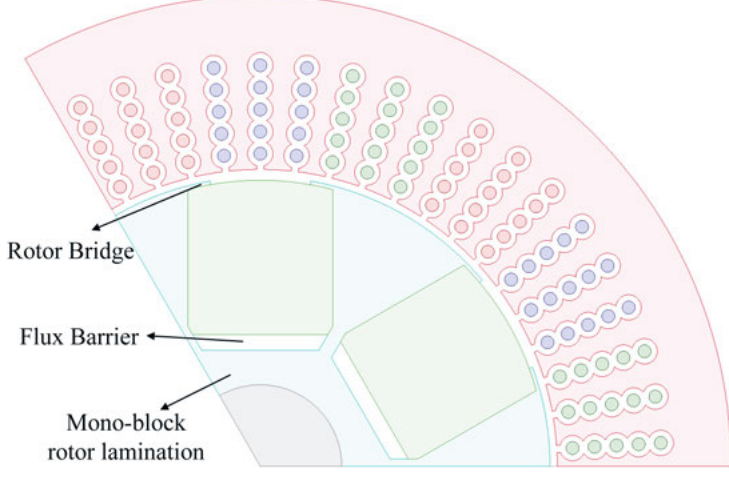
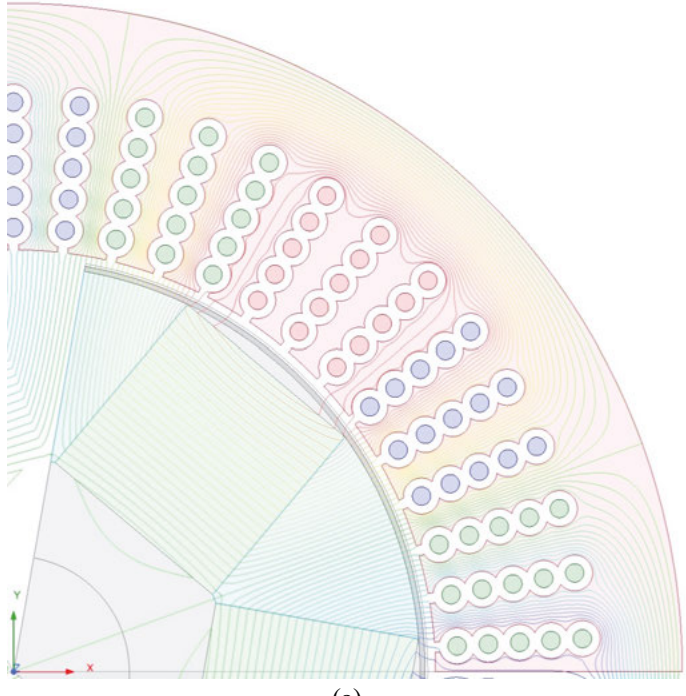
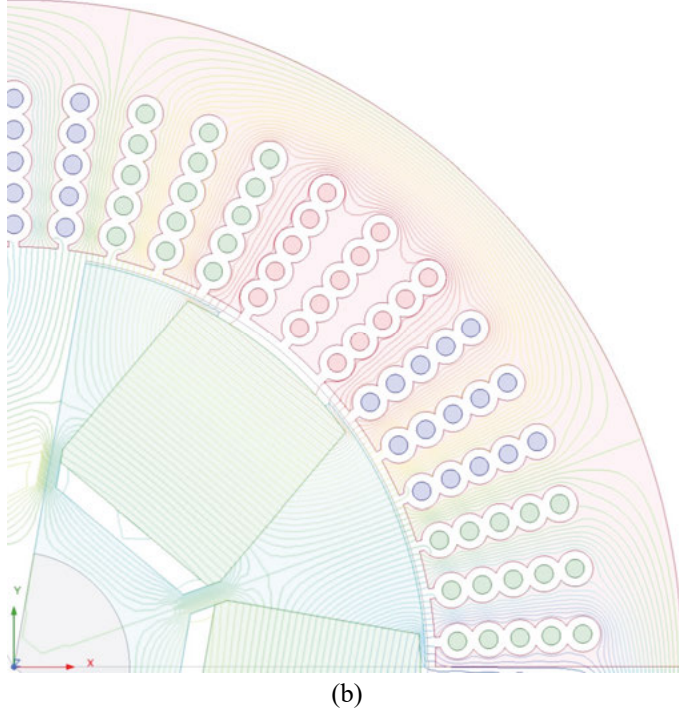


Figure 3.7. Conventional one-piece comparison rotor used to assess shaft-side leakage flux. Adapted from **Paper III**.



(a)



(b)

Figure 3.8. No-load magnetic flux distributions of (a) the proposed segmented rotor and (b) the conventional one-piece comparison rotor. Adapted from **Paper III**.

In the proposed rotor, most of the magnet flux is directed through the rotor pole and across the air gap, with only limited flux circulating around the inner magnet end. In contrast, the conventional one-piece rotor exhibits a pronounced shaft-side leakage path around the inner magnet end and the adjacent lamination. Both designs show only limited leakage near the outer magnet ends.

Despite containing more ferrite, the comparison rotor exhibits greater shaft-side leakage. The proposed segmented rotor therefore shows improved flux utilization in the two-dimensional no-load comparison.

3.2.3. Electromagnetic Performance

The no-load electromagnetic characteristics are presented in Figure 3.9, including the air-gap flux density, three-phase flux linkage, and back-EMF waveforms over the investigated speed range.

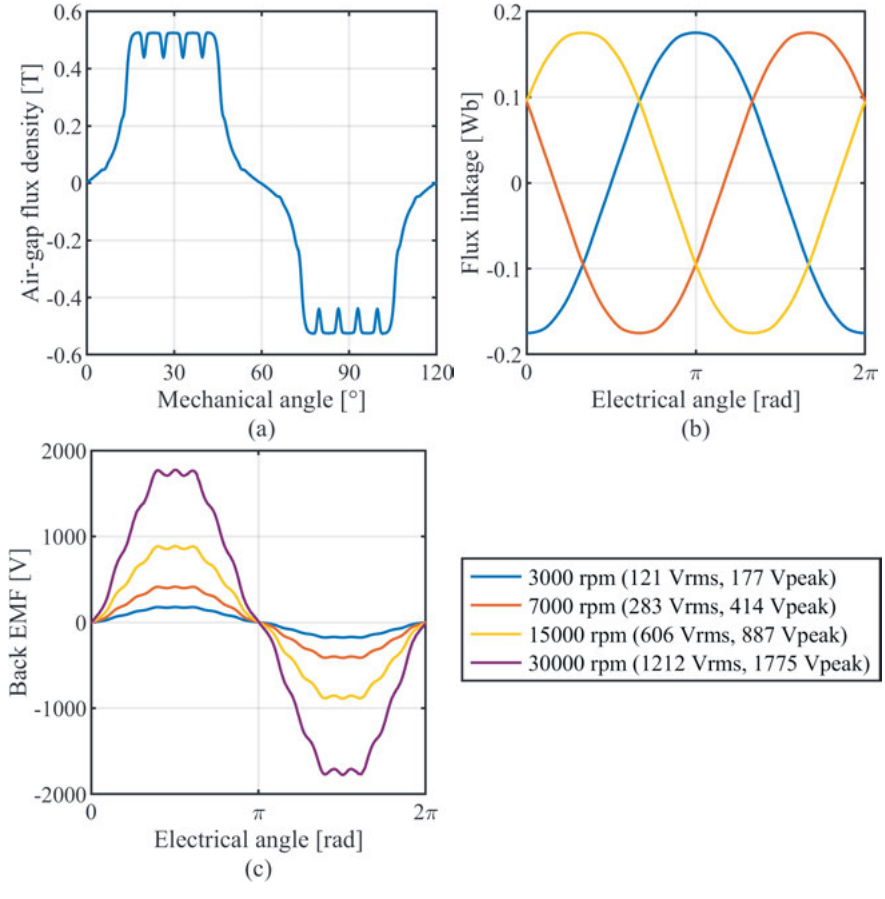


Figure 3.9. No-load simulation results: (a) air-gap flux density, (b) three-phase flux linkage, and (c) back EMF at different rotational speeds. Adapted from **Paper III**.

The simulated torque-speed and power-speed characteristics are shown in Figure 3.10, as well as the torque around base speed. Based on the active volume of 7.96 L, the simulated peak torque density and power density are 21.24 Nm/L and 18.18 kW/L, respectively. The simulated peak torque and peak power meet the specified design targets.

The corresponding efficiency map is presented in Figure 3.11. The motor achieves the specified peak-efficiency target and maintains a broad high-efficiency operating region.

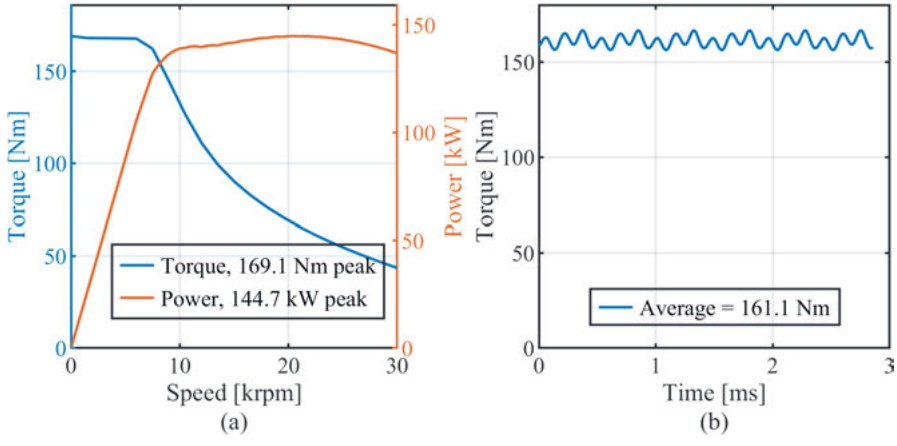


Figure 3.10. Loaded electromagnetic performance: (a) torque-speed and power-speed characteristics and (b) shaft torque at approximately the base speed and 150 Arms. Adapted from **Paper III**.

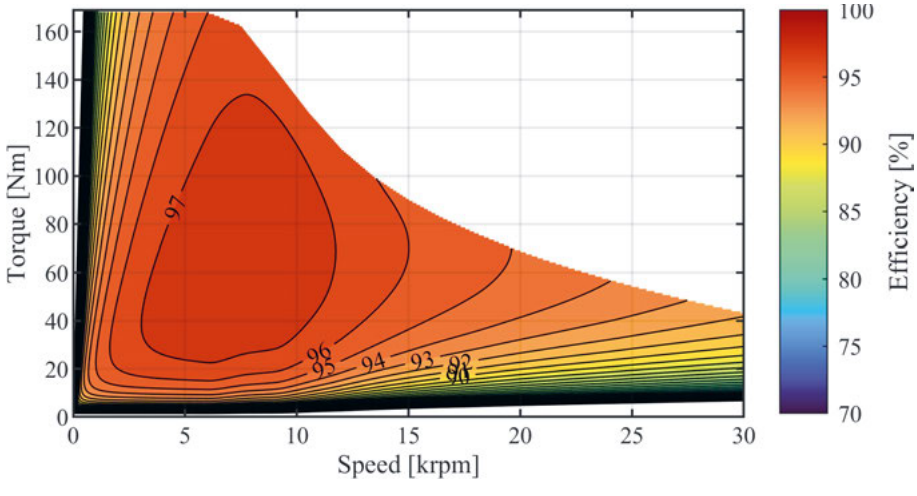


Figure 3.11. Simulated efficiency map of the proposed motor, including copper, iron, and assumed mechanical losses. Adapted from **Paper III**.

3.2.4. Irreversible Demagnetization

The effects of irreversible demagnetization on the no-load back EMF and loaded torque are shown in Figure 3.12. Both quantities decrease after the demagnetization stress event, indicating a reduction in the effective magnetic contribution of the ferrite magnets.

As shown in Figure 3.12(a), the root mean square (RMS) back EMF decreases from 317 V to 263 V, corresponding to a reduction of approximately 17.0%. Figure 3.12(b) shows that the average torque decreases from 176 Nm before the stress event to 150 Nm afterward when evaluated under the same operating condition. The corresponding reduction in average torque is approximately 14.7%.

It should be noted that the pre-demagnetization torque of 176 Nm at -40 °C is higher than the maximum torque of 169.1 Nm shown in Figure 3.10(a) at room temperature. This difference results from the negative temperature coefficient of ferrite remanence, which increases the magnet remanence and air-gap flux density as the temperature decreases. At the same time, the intrinsic coercivity of ferrite decreases at lower temperatures, increasing its susceptibility to irreversible demagnetization.

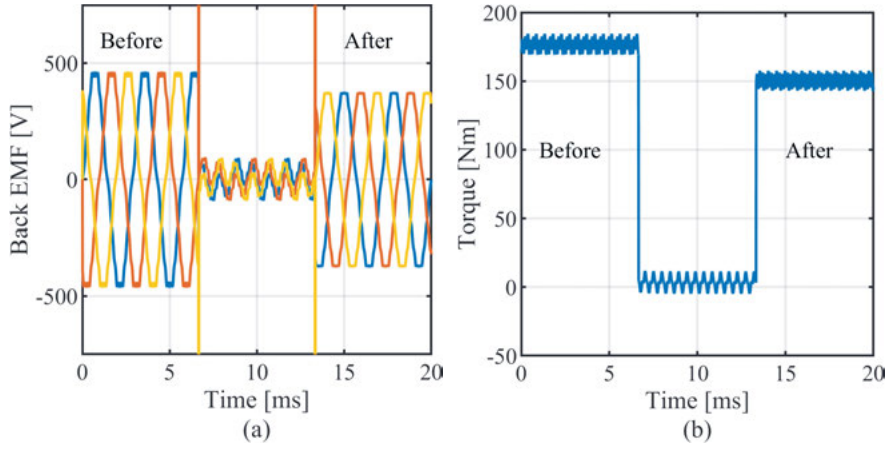
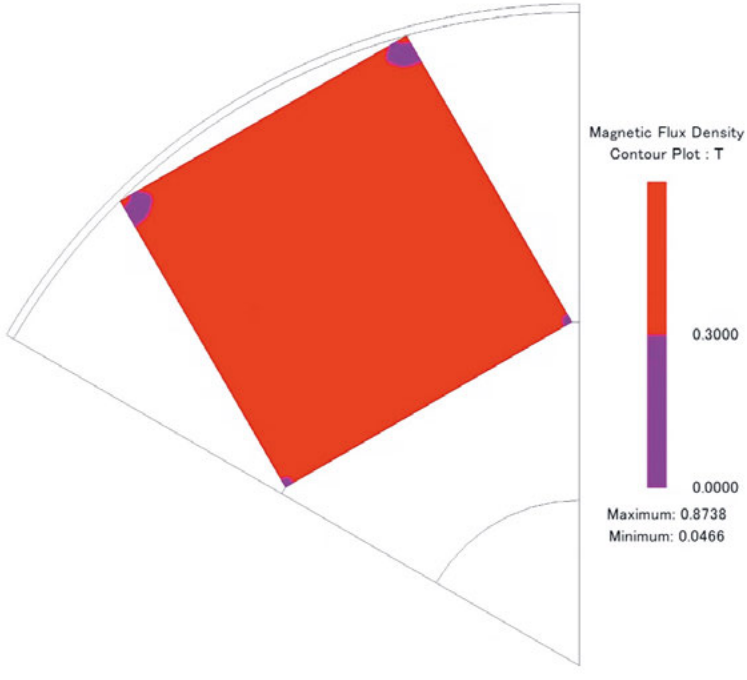
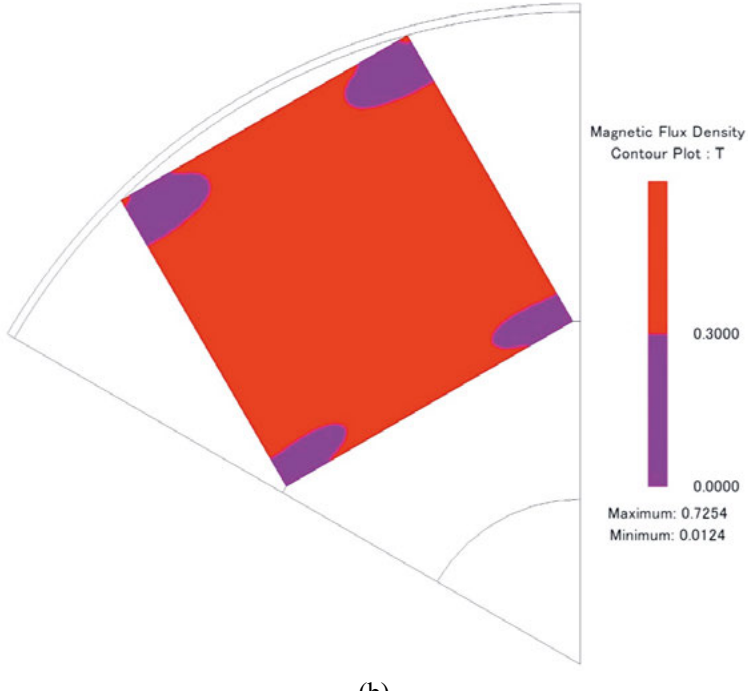


Figure 3.12. Simulated effects of irreversible demagnetization on (a) three-phase no-load back EMF and (b) loaded torque. Adapted from **Paper III**.

The corresponding flux-density distributions before and after the stress event are shown in Figure 3.13. Flux-density values below 0.3 T indicate regions affected by irreversible demagnetization, based on the knee point of the Y40 demagnetization curve at -40 °C. The post-stress distribution shows an expansion of the regions affected by irreversible demagnetization within the ferrite magnets.



(a)



(b)

Figure 3.13. No-load ferrite-magnet flux-density distributions (a) before and (b) after the simulated demagnetization stress event. Adapted from **Paper III**.

3.2.5. High-Speed Mechanical Performance

The high-speed mechanical response of the segmented rotor is shown in Figure 3.14 and Figure 3.15. Figure 3.14 presents the von Mises stress in the carbon-fiber retaining sleeve, while Figure 3.15 shows the corresponding radial deformation.

Under the equivalent isotropic model and the load case in Section 3.1.5, the maximum predicted sleeve stress is approximately 8.6 MPa and the maximum radial deformation is approximately 3.6×10^{-3} mm. These results require further assessment using a composite material model.

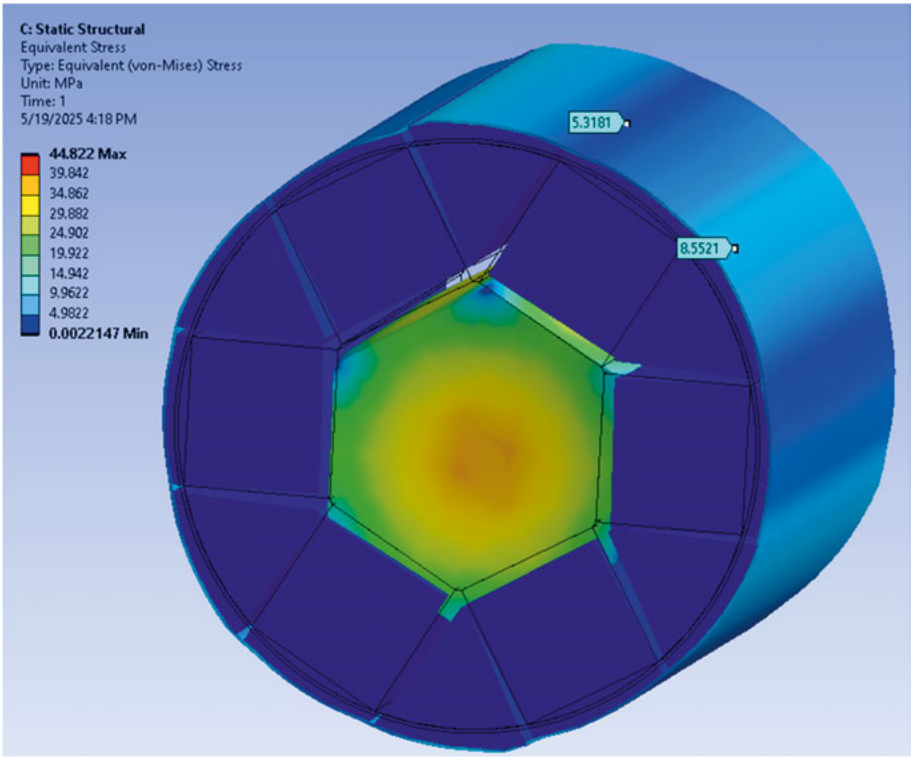


Figure 3.14. Von Mises stress in the carbon-fiber sleeve at 30 000 rpm and maximum current. Adapted from **Paper III**.

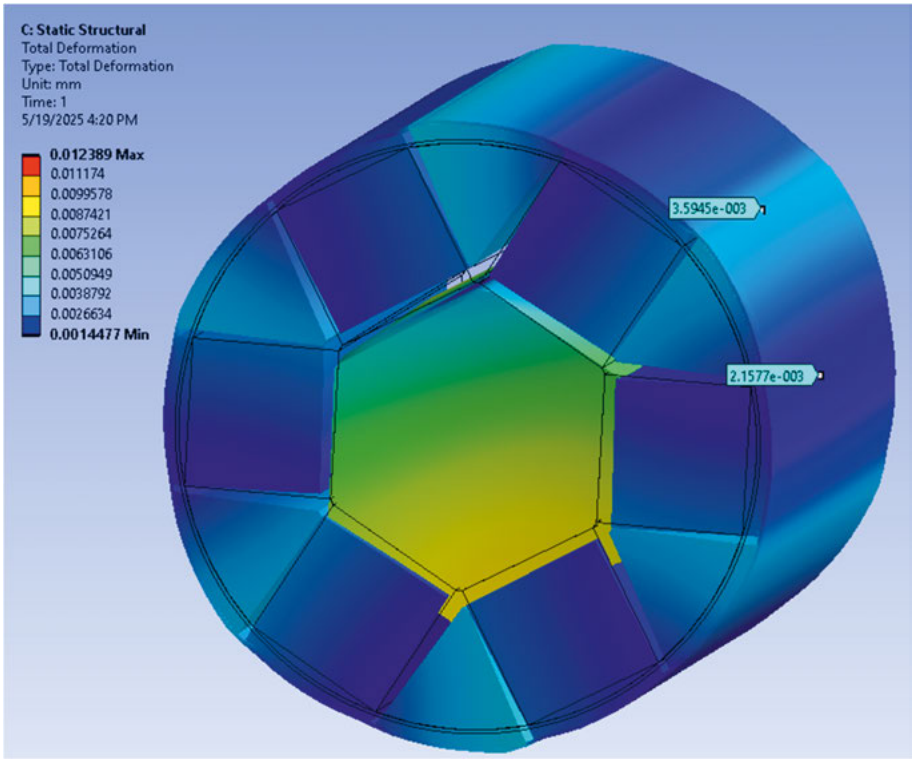


Figure 3.15. Radial deformation at 30 000 rpm and maximum current. Adapted from **Paper III**.

3.2.6. Experimental Validation

The air-gap flux-density measurement and the corresponding experimental result are shown in Figure 3.16. The measured distribution follows the overall trend predicted by the FEM model, although some localized variations are less pronounced in the measurement.

The experimental setup for the no-load back-EMF measurement is described in Section 3.1.6. The measured and simulated back-EMF waveforms at the investigated rotational speeds are compared in Figure 3.17. The measured waveforms show good agreement with the simulated results in both waveform shape and phase relationship. The measured peak back EMF is 4.4% lower than the simulated value at 237 rpm and 7.6% lower at 388 rpm.

Overall, the experimental measurements show good agreement with the corresponding FEM predictions for the no-load electromagnetic quantities evaluated.

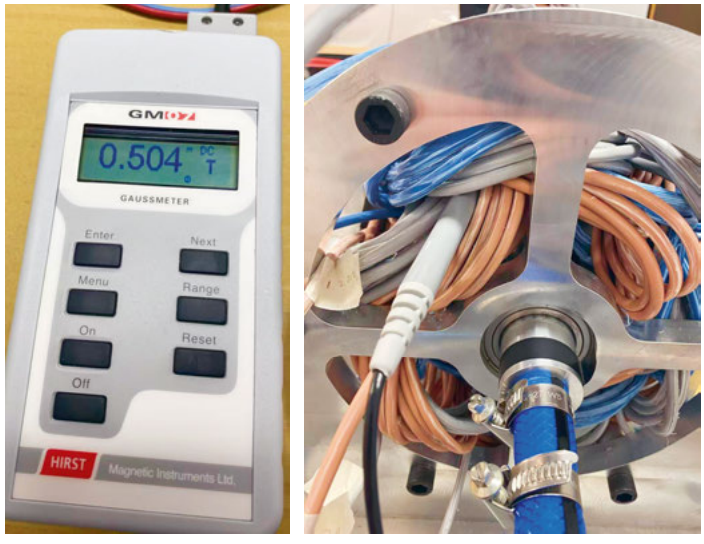


Figure 3.16. Air-gap flux-density measurement using a magnetic flux-density meter (left) with the probe inserted into the physical air gap (right). Adapted from **Paper III**.

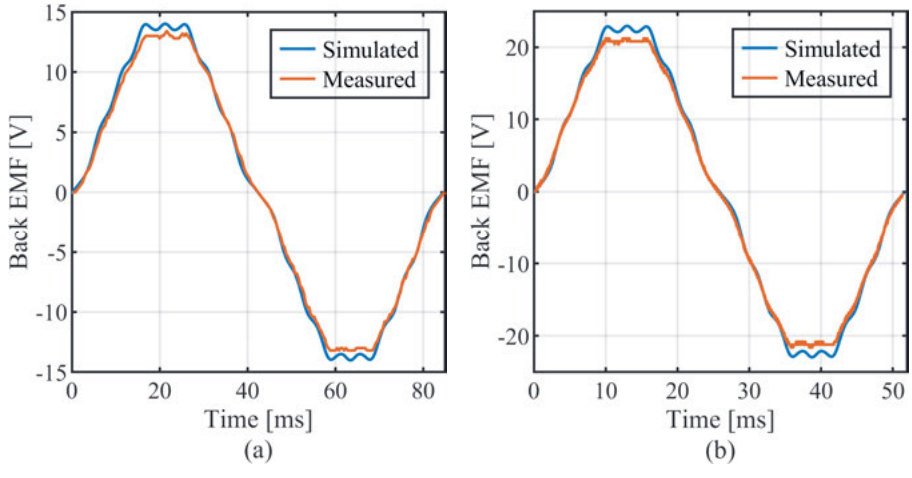


Figure 3.17. Comparison of measured and simulated no-load back EMF at (a) 237 rpm and (b) 388 rpm. Adapted from **Paper III**.

3.3. Discussion

3.3.1. Performance Relative to Traction-Motor Benchmarks

As shown by the peak torque and power results in Figure 3.10(a), the proposed motor achieves required torque and power characteristics. Based on the active volume reported in Table 3.2, the resulting peak torque density falls within the

commercial reference range established in Section 2.1.2, whereas the peak power density is higher than the corresponding range.

The efficiency map in Figure 3.11 shows a broad high-efficiency operating region. At higher rotational speeds, efficiency decreases as iron loss, mechanical loss, and the current required for field weakening become more significant. The efficiency map therefore represents predicted active-machine performance under the adopted loss assumptions. Thermal constraints and experimentally sustainable continuous operation were not established in this study.

The higher pre-demagnetization back EMF and torque at -40 °C are consistent with the increase in ferrite remanence at lower temperature. The smaller torque reduction is consistent with a remaining reluctance-torque contribution, although the torque components were not separated in this comparison.

The 17.0% reduction in back EMF can be compared with previously reported results for ferrite spoke-type machines. Kimiabeigi [18] experimentally observed a 12% reduction in the fundamental back-EMF component after partial demagnetization caused by a negative d-axis current injection of 240% of the peak rated current. Jeong and Hur [58] reported reductions in back EMF of 25.7% in simulation and 24.4% in the corresponding experiment after applying a demagnetizing current approximately three times the rated current at -40 °C. Although differences in motor design, temperature, and demagnetizing conditions prevent direct quantitative comparison, the present result lies within the range reported for other ferrite spoke-type machines. The associated torque reduction of 14.7% nevertheless indicates scope for further optimization of the magnet dimensions, rotor leakage-flux paths, and electromagnetic loading against irreversible demagnetization.

3.3.2. Design and Manufacturing Trade-Offs

The proposed rotor structure was developed primarily to satisfy electromagnetic and mechanical requirements. Nevertheless, its component geometry and assembly concept also influence manufacturability, disassembly, and material recovery. The discussion in this subsection is therefore a qualitative design-level assessment rather than a complete production-process or recycling study.

The final rotor geometry shown in Figure 3.4 uses standard rectangular ferrite magnets, repeated lamination segments, aluminum structural components, and a surrounding carbon-fiber sleeve. The use of standard rectangular ferrite magnets and geometrically simple lamination segments reduces the need for customized magnet geometries and complex rotor features. The steel segments can be manufactured by conventional stamping or laser cutting, while the aluminum hub and filler components can be

produced separately. This separation of material and function may simplify tooling and inspection during prototype and low-volume production.

The sector-shaped laminations may also offer improved sheet utilization compared with a complete one-piece rotor lamination containing a large central opening and multiple internal cavities. Individual sectors can potentially be nested more densely during sheet cutting. However, the actual production benefit would depend on the selected stamping or cutting process, tooling design, production volume, and required tolerances. A detailed manufacturing-cost analysis is therefore required before claiming a quantitative advantage.

The rotor is assembled from repeated laminated sectors, rectangular magnets, aluminum components, and a surrounding carbon-fiber sleeve. The repeated geometry reduces the number of unique component types and may support modular subassembly. At the same time, the segmented rotor structure introduces additional interfaces and requires accurate positioning of the rotor sectors and magnets to control eccentricity, imbalance, and dimensional variation.

The exploded rotor arrangement is shown in Figure 3.18. The main components are individually accessible and potentially separable, unlike in rotors where the magnets are fully enclosed within a monolithic lamination stack. The absence of a tight interference fit between the magnets and rotor core may facilitate disassembly after removal of the sleeve and bonding material.

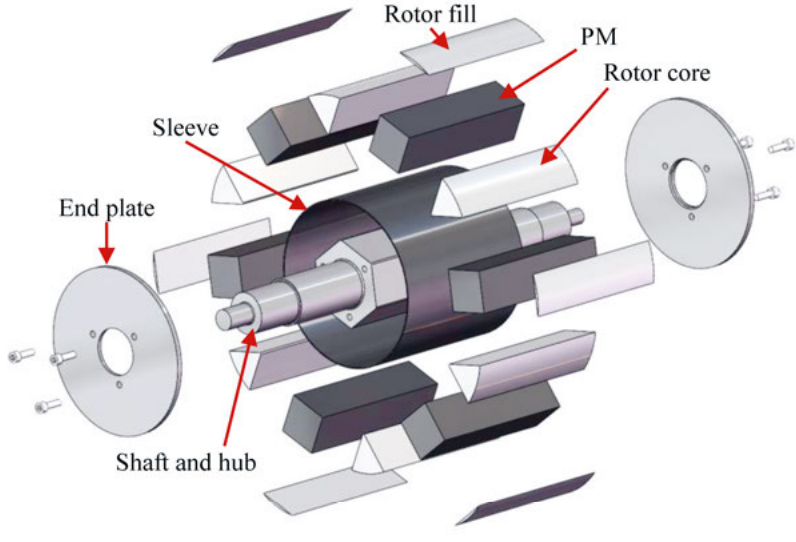


Figure 3.18. Exploded view of the proposed rotor assembly. Adapted from **Paper III**.

As reported in Table 3.2, the prototype uses a 2 mm physical air gap together with a 1 mm carbon-fiber sleeve. The physical air gap is larger than that

reported for several commercial traction motors [33], [34], [60]. Reducing the physical air gap could increase magnetic loading and potentially improve torque density, but would also require tighter manufacturing tolerances, more accurate rotor positioning, and careful control of high-speed mechanical clearance.

The non-sinusoidal air-gap flux density and back-EMF waveforms in Figure 3.9 reflect, in part, the use of rectangular ferrite magnets and the resulting rotor-pole geometry. Customized magnet or pole geometries could improve waveform quality, but would increase component complexity and reduce some of the manufacturing simplicity of the present design.

The retaining sleeve shown in Figure 3.4 and Figure 3.14 provides radial retention without introducing a ferromagnetic bridge between adjacent poles. However, because its relative permeability is close to unity, the sleeve increases the effective magnetic air-gap length. It also introduces an additional manufacturing step and requires control of sleeve dimensions, prestress, material orientation, and assembly accuracy.

The winding parameters in Table 3.2 reflect a prototype-oriented implementation using five relatively thick conductors per slot. Thick round conductors were selected to simplify manual winding, but this arrangement does not fully utilize the available slot area. A production-oriented winding design could improve slot fill, copper utilization, and thermal performance, although it would require a more specialized manufacturing process.

Adhesive removal could potentially be achieved through controlled thermal or chemical treatment [61]. The feasibility of this approach would depend on the adhesive formulation, the thermal stability of the magnets and aluminum components, and the intended recovery route. Similarly, removal of the carbon-fiber sleeve is technically possible, but the environmental and economic effectiveness of recovering or recycling the sleeve material remains uncertain [62], [63].

Recent developments in filament-wound and thermoplastic carbon-fiber-reinforced structures may improve the future detachability and recoverability of retaining sleeves [63], [64], [65]. However, the sleeve used in the present prototype should not be assumed to provide closed-loop recyclability without further process development and experimental validation.

Ferrite magnets may offer certain end-of-life advantages compared with coated NdFeB magnets. Ferrite is chemically stable, already exists in an oxidized state, and generally does not require protective metallic coatings. This may avoid coating-removal steps and reduce sensitivity to corrosion during storage and processing [61]. Nevertheless, the practical recovery value of ferrite is relatively low, and its reuse or recycling would depend on maintaining magnet integrity or establishing an economically viable material-recovery route.

Overall, the proposed rotor structure offers several potentially favorable manufacturing and end-of-life features, including standard magnet geometry,

repeated modular components, physically separable material groups, and reduced dependence on enclosed magnet cavities. These benefits are partly offset by the need for precise segmented assembly, adhesive management, and carbon-fiber sleeve removal. The design therefore offers potential advantages in manufacturing and disassembly, but it has not been fully validated for industrial production or recycling.

3.3.3. Limitations and Further Development

The experimental validation reported in Section 3.2.6 was limited to no-load air-gap flux density and back EMF. The no-load measurements provide experimental support for the principal electromagnetic behavior predicted by the FEM model, while loaded operation, thermal performance, high-speed durability, and long-term reliability remain subjects for further investigation.

The agreement between the measured and simulated results should be interpreted in light of measurement uncertainty and potential sources of experimental error. The localized differences in air-gap flux density shown in Figure 3.16 may partly result from the finite sensing area, positioning accuracy, orientation, and calibration of the Hall sensor, as well as from variations in the clearance between the sensor and rotor surface. The peak-value deviations in back EMF shown in Figure 3.17 may be influenced by rotational-speed fluctuations, voltage-probe accuracy, waveform acquisition, and manufacturing tolerances in the magnets, rotor assembly, winding, and effective air-gap geometry. In addition, the magnetic properties of the manufactured ferrite blocks may differ from the nominal manufacturer data used in the FEM model. Handling and transportation may also have caused minor mechanical damage or local magnetic degradation. The individual contributions of these factors were not measured separately. Nevertheless, the agreement in waveform shape and the peak-value deviations of less than 8% support the suitability of the FEM model for predicting the principal no-load electromagnetic characteristics of the prototype.

The numerical results are also subject to limitations arising from the two-dimensional FEM model. The model assumes a uniform electromagnetic field along the active axial length and therefore does not fully capture end effects, end-winding leakage, axial flux components, or three-dimensional manufacturing and assembly variations. Consequently, the close agreement obtained under no-load operation supports the electromagnetic validity of the proposed design, but it does not constitute full validation of its loaded, demagnetization, thermal, or high-speed mechanical performance.

The mechanical results in Figure 3.14 and Figure 3.15 provide preliminary support for the rotor-retention concept. The equivalent isotropic model predicts a maximum von Mises stress of approximately 8.6 MPa and a maximum deformation of approximately 3.6×10^{-3} mm. Carbon-fiber composites can exhibit tensile strengths above 3400 MPa and failure strains

of approximately 1.5%, depending on fiber type, winding direction, resin system, and manufacturing quality [66], [67], [68]. For the rotor radius considered here, a circumferential strain of 1.5% corresponds to a characteristic displacement scale of approximately 1.19 mm. These material values and the displacement estimate are included for reference only. A failure assessment of the sleeve requires directional material properties and composite-specific failure criteria.

However, the present mechanical model does not constitute a complete composite failure, rotor-dynamics, or fatigue assessment. The carbon-fiber sleeve was represented using an equivalent isotropic material model, whereas its actual mechanical response depends on fiber orientation and composite anisotropy. More detailed structural evaluation would require orthotropic material properties and composite-specific failure criteria, together with consideration of sleeve prestress, adhesive interfaces, thermal expansion, manufacturing defects, rotor imbalance, repeated loading, and overspeed conditions.

Overall, the prototype results provide technical support for the common ferrite-based spoke-type design approach investigated in this thesis. The prototype provides no-load experimental evidence for the segmented spoke-type design. Chapter 4 applies the same design principles to a separate motor with different functional-equivalence and LCA requirements.

4. System-Level Life Cycle Environmental Assessment

Chapter 3 examined the technical feasibility of a segmented ferrite-based spoke-type traction motor through simulations, prototype fabrication, and no-load measurements. This chapter extends the investigation from technical feasibility to life cycle environmental performance. The ferrite-based motor considered in this chapter is not the prototype developed in Chapter 3, but a separate implementation of the same spoke-type design approach developed specifically for functional-equivalence comparison with a production-relevant NdFeB benchmark.

The chapter connects motor design with cradle-to-grave LCA. A surrogate-assisted multi-criteria design framework is first used to identify a ferrite-based configuration with comparable traction functionality while accounting for the geometric and electrical compensation required by ferrite substitution [10]. The selected configuration and the NdFeB benchmark are then compared across material production, manufacturing, vehicle use, and end-of-life treatment. The analysis also examines how motor efficiency, vehicle mass, electricity supply, and material recovery affect the life cycle results. The work presented in this chapter is based primarily on **Paper IV** and **Paper V.**

4.1. Research Gap and Chapter Scope

LCA has been widely applied to electric machines, particularly industrial motors and efficiency-oriented eco-design. Representative studies include [69], [70], [71], [72], [73], [74], [75], [76], [77]. This body of work has established the importance of use-phase electricity consumption and demonstrated the environmental relevance of motor efficiency classes. However, these conclusions cannot be transferred directly to EV traction motors. Compared with many industrial machines, EV traction motors are subject to stricter requirements for torque density, package volume, overload capability, maximum speed, and field-weakening operation [78]. Their energy use is also determined by coupled motor-vehicle behavior over transient driving cycles rather than predominantly steady operating conditions [79]. Material-substitution strategies for traction motors must therefore be evaluated under traction-specific functional and operational requirements.

LCA studies focusing specifically on EV traction motors remain comparatively limited. Existing work includes cradle-to-grave assessments of individual motors [80], [81], [82], component-level studies such as an assessment of a hairpin stator [83], and comparative analyses of rare-earth-based and rare-earth-free motor concepts [84], [85], [86]. These studies provide useful case-specific inventories and environmental results, but their use in design-informed comparison is often restricted by limited geometric transparency or by inventories that are closely tied to one particular motor. In addition, some comparative studies evaluate rare-earth substitution without fully demonstrating that the substituted and benchmark motors provide equivalent traction functionality.

Nordelöf and co-workers addressed part of this limitation by developing a scalable life cycle inventory framework for automotive traction motors that links material composition and manufacturing processes to motor design parameters [87], [88]. The framework allows component inventories to be adjusted for different motor designs, providing a basis for quantitative comparison. Subsequent studies have applied or extended the framework to investigate alternative motor topologies, material choices, and design scenarios [89], [90], [91], [92], [93].

However, some important methodological limitations remain. In [89], a ferrite-assisted synchronous reluctance machine was compared with a benchmark PMSM using a common maximum-torque target, but the high-speed torque characteristics were not fully aligned. In [90], the compared rare-earth and rare-earth-free motors differed in both maximum torque and maximum speed, which would likely require different transmission matching in an actual vehicle. Under such conditions, the environmental difference cannot be attributed to the motor concepts alone because gearbox selection and drivetrain efficiency may also change. Studies [91] and [92] examined the sensitivity of environmental impacts to material masses, electricity mixes, and vehicle parameters, but they did not provide a design procedure for reducing environmental impacts while preserving traction performance. Study [93] compared rare-earth-based and rare-earth-free variants but did not include a ferrite-based motor.

Consequently, existing rare-earth-free traction motor assessments remain dominated by synchronous-reluctance-based or induction-machine concepts, while ferrite-based spoke-type PMSMs are largely absent from cradle-to-grave comparative LCA studies. More importantly, previous work has not systematically linked rare-earth substitution with a functional-equivalence design framework. They have also not carried the resulting design changes through material production, manufacturing, vehicle operation, and end-of-life treatment. In this chapter, design compensation refers to the geometric, material, structural, and electrical adaptations required to recover traction-relevant performance after replacing NdFeB magnets with ferrite. To the best of the author’s knowledge, previous studies have also not expressed the

comparison between rare-earth-based and rare-earth-free traction motors in terms of a break-even electricity carbon intensity, even though this indicator is directly relevant to the trade-off between manufacturing impacts and use-phase electricity demand. It remains unclear whether replacing rare-earth magnets with ferrite magnets can provide a net environmental benefit when the motor must remain functionally comparable to a benchmark EV traction motor. Answering this question requires the life cycle assessment to account for all changes introduced during motor design and to evaluate their effects consistently across the full life cycle.

4.2. LCA Goal and Scope Definition

4.2.1. Goal of the LCA Study

The goal of this study is to evaluate the life cycle environmental implications of rare-earth substitution in EV traction motors by comparing two functionally equivalent designs based on different permanent magnet technologies: a conventional NdFeB-based PMSM and a ferrite-based spoke-type PMSM. The assessment follows a cradle-to-grave perspective and covers raw material acquisition and processing, motor manufacturing, use-phase operation, and end-of-life treatment.

The comparative assessment is intended to support motor designers, powertrain engineers, and LCA practitioners by providing design-consistent insights into the environmental trade-offs associated with rare-earth-free traction motor solutions. The NdFeB-based benchmark is derived from the active electromagnetic components of a mass-produced Tesla Model 3 traction motor and is reconstructed within the same analytical framework as the ferrite-based motor.

The assessment builds upon the scalable life cycle inventory framework for automotive traction motors developed in previous work [87], [88]. Several targeted extensions are introduced to capture design-dependent differences between the two motor concepts. These include a geometry-based method for estimating housing mass, an analytical approach for calculating copper usage and winding resistance, and explicit foreground modelling of composite structural components not covered by the existing inventory framework, particularly the carbon-fiber rotor sleeve.

These design-specific extensions are combined with efficiency-map-based use-phase modelling under the Worldwide Harmonized Light Vehicles Test Cycle (WLTC). This allows the assessment to account for changes in motor geometry, material quantities, efficiency, and vehicle energy consumption resulting from ferrite substitution.

4.2.2. Functional Unit

The functional unit is defined as one EV traction motor providing equivalent traction service over a vehicle lifetime of 200 000 km, with comparable torque and power capability across the prescribed traction operating range.

The NdFeB-based benchmark motor and the selected ferrite-based motor are evaluated against this functional requirement. The comparison preserves low-speed torque capability and evaluates high-speed torque deviations explicitly, as defined in Section 4.3. Differences in motor efficiency are included through the electricity consumption calculated over the specified lifetime driving distance.

4.2.3. System Boundary

As illustrated in Figure 4.1, the system boundary follows a cradle-to-grave perspective and includes electricity and heat production, raw material acquisition and processing, motor manufacturing, use-phase operation, and end-of-life treatment. Waste and emissions arising from each life cycle stage are included within the boundary.

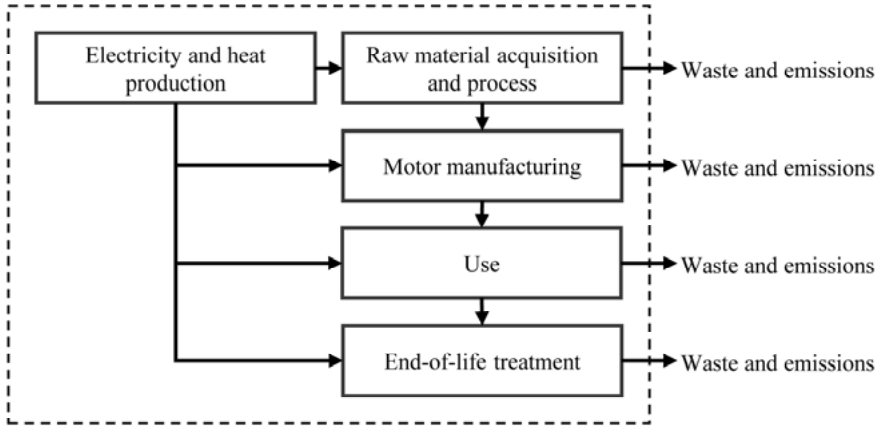


Figure 4.1. Cradle-to-grave system boundary of the traction motor comparison. From **Paper V** [94] under the CC BY 4.0 license.

Raw material acquisition and processing cover the production of all active and non-active motor materials and components. Background inventory data are obtained from Ecoinvent version 3.9.1 [95] using market-average datasets consistent with the selected cut-off system model.

The manufacturing stage includes the processing of motor materials, production of individual components, and assembly of the complete traction motor. Manufacturing inventories are based on the scalable framework developed by Nordelöf et al. [87], [88] and are extended to represent the

design-specific material compositions, component masses, and structural features of the two motors.

The use phase represents the electricity required to provide the functional unit over the defined vehicle lifetime. Differences in electricity consumption are determined from the motor-specific efficiency characteristics and their operation over the selected driving cycle.

End-of-life treatment includes dismantling, shredding, material separation, recycling, and disposal of residual waste. These processes are modelled in accordance with the Ecoinvent cut-off system model.

Vehicle-level components that are assumed to remain identical for both motor concepts, including the battery pack, inverter, and gearbox, are excluded from the system boundary. Losses associated with these shared components are treated as constant and are therefore omitted from the use-phase comparison. Infrastructure, maintenance activities, and potential second-life applications are also excluded.

The analysis is based on two specific traction motor designs. The results should therefore be interpreted as comparative indicators of the environmental consequences associated with these designs and the corresponding rare-earth substitution strategy, rather than as universally applicable results for ferrite-based or rare-earth-free traction motors.

4.3. Functional-Equivalence Motor Design and Multi-Criteria Assessment

Replacing NdFeB magnets with ferrite within an unchanged motor geometry would substantially reduce the magnetic loading and lead to lower torque capability and weaker high-speed performance. Functional equivalence therefore cannot be achieved through direct magnet substitution alone. Instead, compensatory changes in magnet geometry, active axial length, and electrical operating limits must be considered as part of the motor design.

The design framework developed in this section identifies ferrite-based spoke-type motor configurations that approach the torque-speed and power-speed characteristics of a production-relevant NdFeB benchmark. Particular emphasis is placed on preserving low-speed traction capability, while deviations in the field-weakening region are evaluated separately. Changes in active axial length and DC bus voltage are considered explicitly because they represent geometric and electrical adaptations required to compensate for the lower magnetic performance of ferrite.

Functional equivalence is formulated as a constrained torque-envelope matching problem. Below 3500 rpm, the ferrite-based motor is required to preserve the benchmark traction capability within a one-sided torque-deviation band of 0 to +1%. For designs satisfying this requirement, the

remaining trade-offs are assessed using three criteria: high-speed RMS torque deviation, normalized active axial length, and normalized DC bus voltage. These criteria are evaluated separately rather than combined into a weighted scalar objective. The overall design workflow is summarized in Figure 4.2.

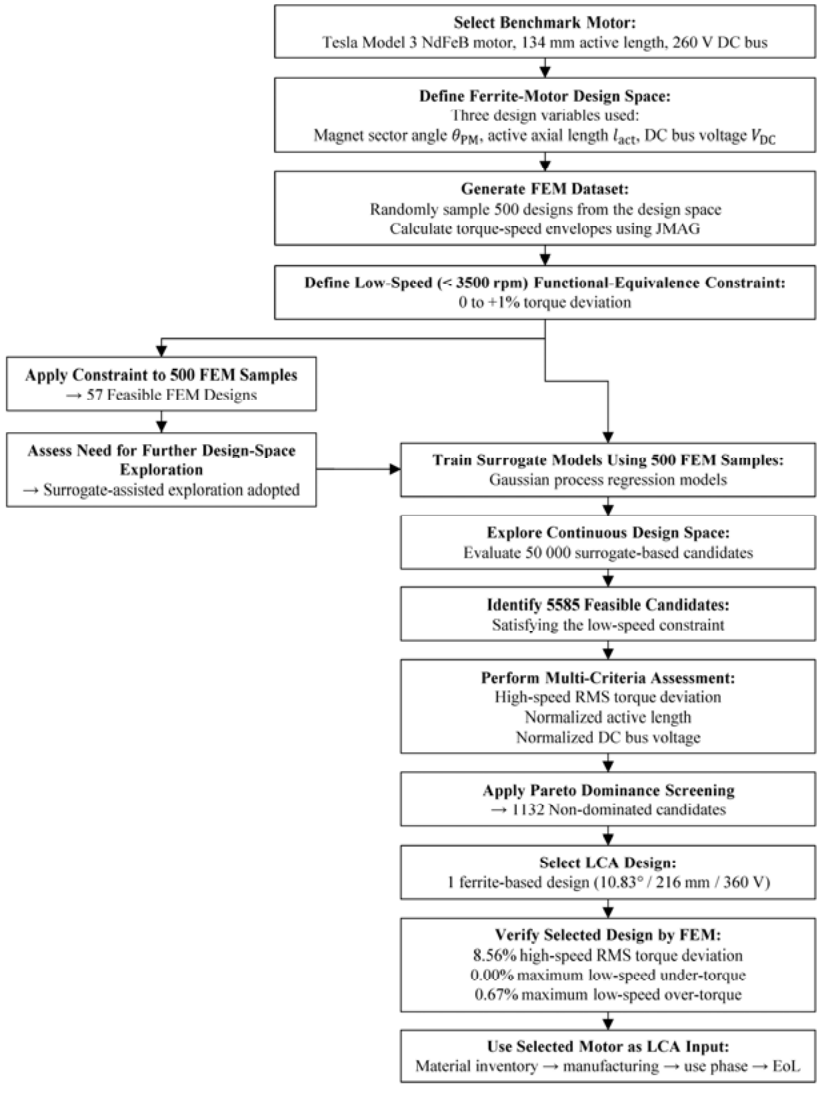


Figure 4.2. Surrogate-assisted functional-equivalence design workflow for the ferrite-based motor selected for LCA.

A FEM dataset of 500 randomly sampled ferrite-motor designs is first generated in a motor simulation software and evaluated against the low-speed functional-equivalence constraint. Since only 57 designs satisfy the prescribed low-speed torque-deviation band, surrogate models are introduced to enable

broader exploration of the continuous design space. Gaussian process regression models trained on the 500 FEM samples are then used to evaluate 50 000 candidate designs, of which 5585 satisfy the low-speed constraint. These feasible designs are further assessed using the three multi-criteria indicators, and Pareto dominance screening identifies 1132 non-dominated candidates. One ferrite-based configuration is subsequently selected for the LCA and verified by FEM analysis.

The selected design provides the geometry, material quantities, winding resistance, efficiency characteristics, and vehicle-level inputs used in the subsequent life cycle inventory and environmental assessment.

4.3.1. Methodology

4.3.1.1. Benchmark Motor and Design Space Definition

The benchmark motor is derived from the production traction motor used in the Tesla Model 3 [60], [96], [97]. Published teardown data and technical reports provide sufficient geometric and material information to reconstruct its electromagnetic active structure in the FEM environment used in this study. The model includes the stator core, rotor core, windings, and permanent magnets, with the geometry and material definitions following the reported production data as closely as possible.

The benchmark employs high-grade NdFeB magnets corresponding to the N52UH class. The magnets are divided axially into four segments. The reconstructed cross-sectional geometry is shown in Figure 4.3(a). Its maximum torque-speed envelope and the corresponding power-speed envelope, shown in Figure 4.4, define the functional-performance target for the ferrite-based alternative. The benchmark motor has an active axial length of 134 mm and a DC bus voltage of 260 V. These values are also used as reference quantities when evaluating the geometric and electrical adaptations required by the ferrite-based motor.

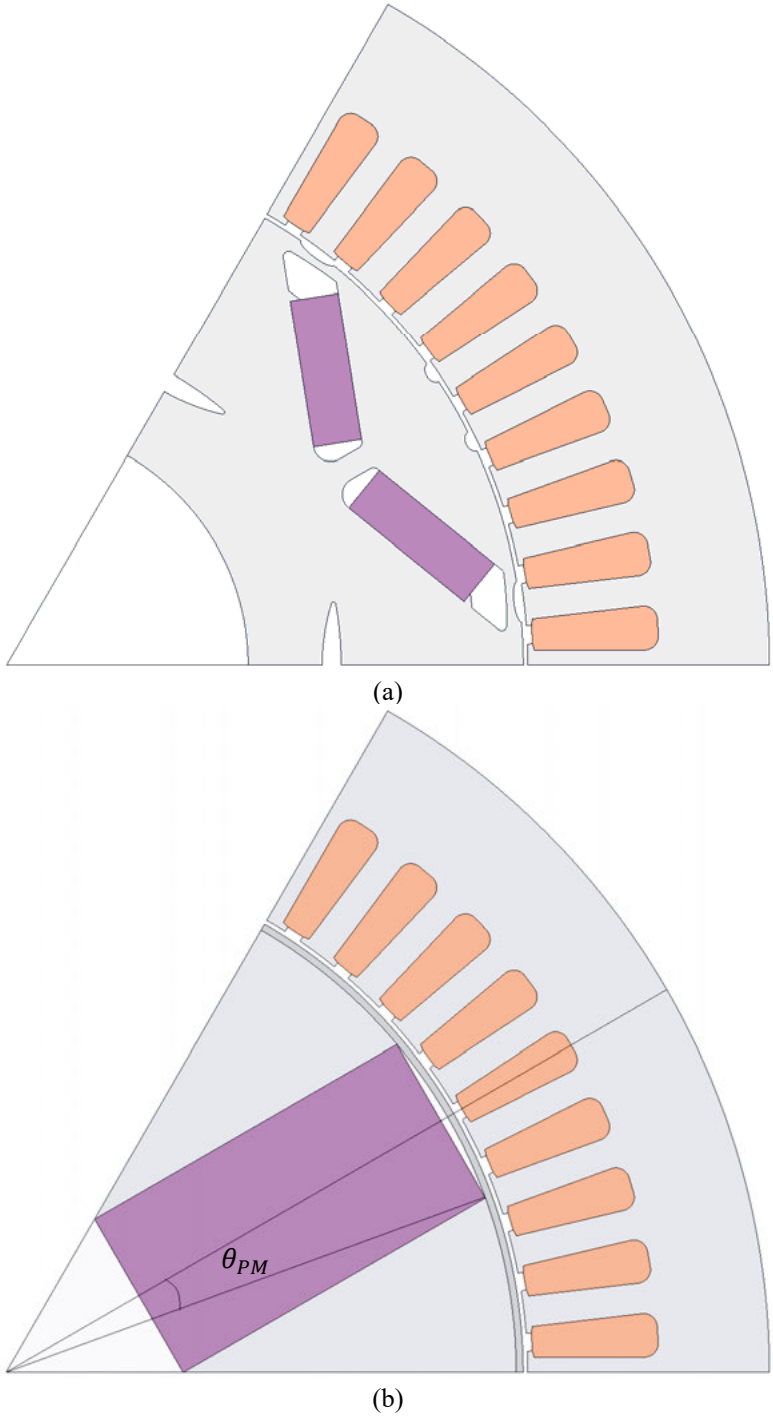


Figure 4.3. FEM models of (a) the benchmark, Tesla Model 3, and (b) the ferrite-based motor. Partially from **Paper V** [94] under the CC BY 4.0 license.

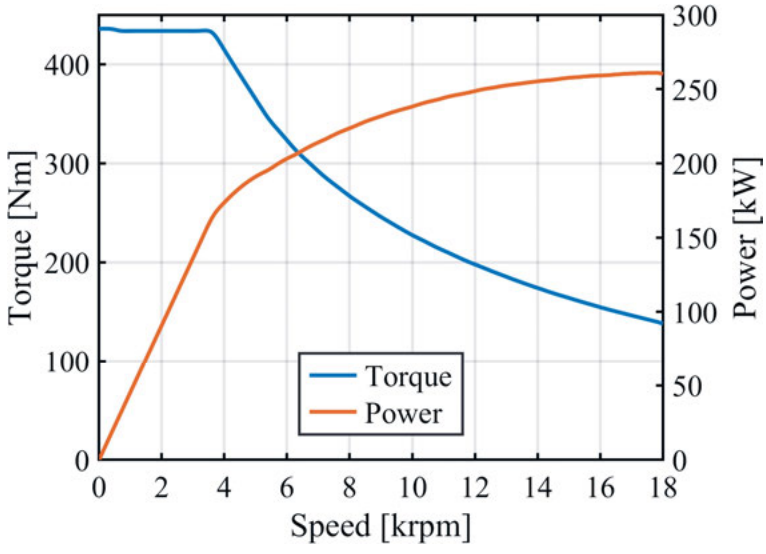


Figure 4.4. Benchmark torque-speed and power-speed envelopes. Adapted from **Paper IV**.

The alternative motor adopts a ferrite-based spoke-type rotor, as shown in Figure 4.3(b). The rotor contains Y40 ferrite magnets and a carbon-fiber retaining sleeve that supports mechanical integrity at high rotational speed. Its segmented spoke-type structure follows the common design approach established in Chapter 3, including flux concentration, suppression of shaft-side leakage, the use of ferrite magnets, and high-speed mechanical retention.

To preserve comparable vehicle integration, the ferrite motor retains the benchmark stator outer diameter, stator cross-section, slot geometry, winding layout, and physical air-gap thickness. In the ferrite design, the physical air gap is defined as the clearance between the stator inner surface and the outer surface of the carbon-fiber sleeve. The active axial length is allowed to vary because additional stack length may be required to compensate for the lower magnetic loading of ferrite. Except for the permanent magnet material and the carbon-fiber rotor sleeve, corresponding active and structural components in the two models use the same material definitions.

Both motors are evaluated using the same two-dimensional FEM procedure, including a common current limit and the same method for determining the maximum torque-speed envelope. At each predefined rotational speed, the feasible current magnitude and current angle are searched under the prescribed current and voltage constraints. This procedure captures maximum-torque operation in the low-speed region and voltage-limited field-weakening operation at higher speeds. Vehicle-level components outside the motor boundary, including the inverter and gearbox, are treated consistently as described in Section 4.2.3.

The phase resistance of each ferrite motor candidate is recalculated according to its active axial length. The calculation accounts for the conductor length within the active stack and the end-winding contribution. Updating the phase resistance for each candidate is necessary because increasing the active length affects copper usage, copper loss, efficiency, and consequently the use-phase results of the LCA.

The ferrite motor design space is defined by three variables:

- Magnet sector angle, θ_{PM}
- Active axial length, l_{act}
- DC bus voltage, V_{DC}

The magnet sector angle is varied to adjust the ferrite magnet geometry and resulting magnetic loading. The active axial length is allowed to vary to provide additional electromagnetic interaction volume when required, while the maximum DC bus voltage is included to account for electrical adaptation in the field-weakening region. These variables define the compensation measures examined in this design space, whereas the stator cross-section and current limit remain fixed.

Table 4.1 lists the investigated ranges. A total of 500 configurations are generated within this design space. Their torque-speed envelopes are calculated using the common FEM procedure and subsequently used to construct the surrogate models described in Section 4.3.1.3.

Table 4.1. Design variables and investigated ranges for the ferrite-based spoke-type motor. Adapted from **Paper IV**.

Design variables	Description	Benchmark value	Range to be investigated for the ferrite-based motor
θ_{PM}	Magnet sector angle, as shown in Figure 4.3(b)	N/A	0 - 30°
l_{act}	Axial active length	134 mm	134 - 268 mm
V_{DC}	Maximum DC bus voltage	260 V	260 - 520 V

4.3.1.2. Functional-Equivalence Formulation

The torque-speed envelope is used as the principal measure of functional equivalence because it directly represents the traction capability of the motor across the operating-speed range [98], [99]. The corresponding power-speed envelope is derived from the torque response and is used as a complementary performance indicator rather than as an independent design constraint.

Functional equivalence is treated differently in the low- and high-speed regions. At low speed, torque directly determines vehicle launch capability, acceleration, and gradeability. The ferrite-based motor is therefore required to

provide torque no lower than that of the benchmark throughout the constant-torque region. At high speed, this study evaluates RMS torque-envelope deviation as a separate design criterion rather than imposing exact pointwise matching.

Let $T_{\mathrm{ref}}(\omega)$ denote the torque-speed envelope of the benchmark motor and $T_{\mathrm{f}}(\omega, \mathbf{x})$ that of the ferrite-based motor, where the design vector is

$$\mathbf{x} = [\theta_{\mathrm{PM}}, l_{\mathrm{act}}, V_{\mathrm{DC}}]. \tag{4.1}$$

The low-speed region is defined as

$$\Omega_{\mathrm{low}} = \{\omega | \omega \leq 3500\ \mathrm{rpm}\}. \tag{4.2}$$

Within this region, the ferrite motor is constrained by a one-sided torque band:

$$T_{\mathrm{ref}}(\omega) \leq T_{\mathrm{f}}(\omega, \mathbf{x}) \leq (1+\delta) T_{\mathrm{ref}}(\omega), \quad \forall \omega \in \Omega_{\mathrm{low}}, \tag{4.3}$$

where $\delta = 0.01$ defines a 1% upper tolerance. The lower bound prevents under-torque relative to the benchmark, while the upper bound limits unnecessary over-design that could otherwise increase active length, magnet volume, and material demand without providing additional functional value.

The high-speed region is defined as

$$\Omega_{\mathrm{high}} = \{\omega | \omega > 3500\ \mathrm{rpm}\}. \tag{4.4}$$

In this region, the deviation between the two torque envelopes is quantified using an RMS relative-error metric:

$$\varepsilon_{\mathrm{high}}(\mathbf{x}) = \sqrt{\frac{1}{|\Omega_{\mathrm{high}}|} \sum_{\omega \in \Omega_{\mathrm{high}}} \left(\frac{T_{\mathrm{f}}(\omega, \mathbf{x}) - T_{\mathrm{ref}}(\omega)}{T_{\mathrm{ref}}(\omega)} \right)^2}. \tag{4.5}$$

This metric captures the overall deviation in field-weakening behavior while reducing sensitivity to isolated local differences at individual speed points.

Achieving comparable performance with ferrite magnets may require increases in active axial length and DC bus voltage. These adaptations are therefore evaluated explicitly through the normalized quantities

$$\lambda_{\mathrm{L}}(\mathbf{x}) = \frac{l_{\mathrm{act}}}{l_{\mathrm{ref}}} \tag{4.6}$$

and

$$\lambda_{\mathrm{V}}(\mathbf{x}) = \frac{V_{\mathrm{DC}}}{V_{\mathrm{ref}}} \tag{4.7}$$

where $l_{\mathrm{ref}} = 134\,\mathrm{mm}$ and $V_{\mathrm{ref}} = 260\,\mathrm{V}$ are the corresponding benchmark values. The normalized active length represents the geometric compensation

required by the ferrite-based design, while the normalized DC bus voltage represents the corresponding electrical adaptation. These quantities are evaluated separately because they influence different aspects of motor and powertrain integration and are also relevant to the subsequent LCA.

The design problem is therefore treated as a constrained multi-criteria assessment. A candidate design is first required to satisfy the low-speed traction-preservation constraint in (4.3). The feasible designs are then compared using the three criteria

$$\{\varepsilon_{\text{high}}(\mathbf{x}), \lambda_{\text{L}}(\mathbf{x}), \lambda_{\text{V}}(\mathbf{x})\}, \tag{4.8}$$

together with the design-variable bounds defined in Table 4.1.

All three criteria are minimized, favoring lower high-speed torque mismatch, shorter active length, and lower DC bus voltage. The criteria are evaluated separately without weighting, and feasible designs are compared in terms of the trade-offs among them.

A design $\mathbf{x}_a$ is considered to dominate another feasible design $\mathbf{x}_b$ if

$$\varepsilon_{\text{high}}(\mathbf{x}_a) \leq \varepsilon_{\text{high}}(\mathbf{x}_b),$$

$$\lambda_{\text{L}}(\mathbf{x}_a) \leq \lambda_{\text{L}}(\mathbf{x}_b),$$

and

$$\lambda_{\text{V}}(\mathbf{x}_a) \leq \lambda_{\text{V}}(\mathbf{x}_b), \tag{4.9}$$

with at least one of these inequalities being strict. Feasible designs that are not dominated by any other candidate form the non-dominated trade-off set. The feasible and non-dominated designs provide the reference trade-off space against which the configuration selected for the subsequent LCA is evaluated.

4.3.1.3. Surrogate Modelling and Design-Space Exploration

The initial dataset comprises 500 ferrite-based motor configurations evaluated by FEM analysis within the design space defined in Section 4.3.1.1. For each configuration, the maximum torque is calculated at the predefined speed points, subject to the previously described operating constraints.

Direct FEM evaluation of a densely sampled continuous design space would require repeated calculation of complete torque-speed envelopes and would therefore be computationally expensive. Gaussian process regression models are therefore constructed from the 500 simulated configurations. An independent surrogate model is trained at each predefined speed point, mapping the design vector in (4.1) to the corresponding maximum torque. Together, these models provide a continuous approximation of the torque-speed envelope and enable rapid evaluation of the low-speed traction-preservation constraint and the three design criteria defined in Section 4.3.1.2. The trained surrogate models are evaluated at 50 000 candidate points

generated using Latin hypercube sampling within the prescribed design-variable bounds.

4.3.1.4. Multi-Criteria Assessment and Design Selection

A ferrite-based configuration was selected from the surrogate-derived feasible trade-off region for use as the design input to the subsequent LCA. The selection considered the three criteria of high-speed torque-envelope deviation, normalized active axial length, and normalized DC bus voltage, while satisfying the prescribed low-speed traction-preservation requirement.

Pareto optimality was not imposed as a requirement for the LCA design selection. Instead, its position relative to the surrogate-derived non-dominated trade-off set was assessed to determine whether substantially better combinations of the three design criteria existed within the investigated design space. To quantify this proximity, the three criteria were individually normalized using their ranges within the feasible surrogate set, and the Euclidean distance from the selected configuration to each non-dominated candidate was calculated in the normalized criterion space.

The selected design was subsequently verified by FEM analysis before being used in the downstream LCA.

4.3.2. Results

4.3.2.1. Feasible Design Space

Application of the low-speed traction-preservation constraint yielded 57 feasible designs from the original FEM dataset and 5585 feasible designs from the surrogate-evaluated candidate set. Pareto screening identified 1132 non-dominated surrogate candidates. The resulting design-set sizes and metric ranges are summarized in Table 4.2.

Table 4.2. Summary of FEM and surrogate-assisted design-space exploration.

Design set	Number of candidates	High-speed RMS error [%]	λ_L	λ_V
Initial FEM dataset	500	-	-	-
Feasible FEM designs	57	6.40-11.80	1.6055-1.6400	1.3085-1.4552
Surrogate candidates	50 000	-	-	-
Feasible surrogate candidates	5585	5.93-13.68	1.6046-1.6417	1.3081-1.4613
Non-dominated surrogate candidates	1132	5.93-11.70	1.6046-1.6417	1.3081-1.4613

Figure 4.5 complements this analysis by providing a two-dimensional representation of the variation among the 500 FEM-derived torque-speed

envelopes. The relative torque deviations at 1000 rpm and 8000 rpm are used to represent the low-speed traction region and the high-speed field-weakening region, respectively. These two operating points are selected for visualization only, whereas feasibility is determined using the complete low-speed constraint defined in (4.3). Feasible FEM samples satisfying this constraint are highlighted in orange, while the selected LCA configuration, subsequently verified by FEM, is shown for reference.

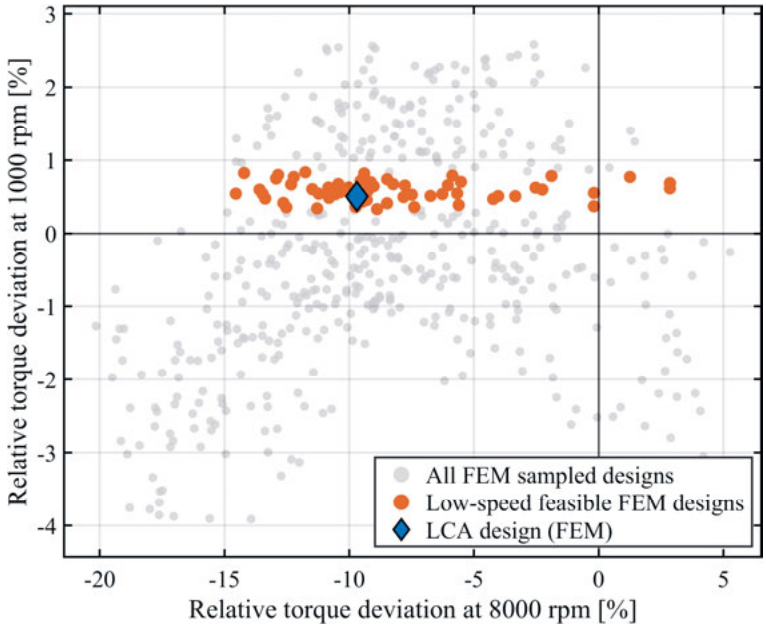


Figure 4.5. Relative torque deviations of the sampled ferrite-based designs at 1000 and 8000 rpm.

4.3.2.2. Functional Performance of the Selected Design

The selected configuration has a magnet sector angle of 10.83°, an active axial length of 216 mm, and a maximum DC bus voltage of 360 V. Its FEM-verified functional-performance metrics are summarized in Table 4.3, while the corresponding torque-speed and power-speed envelopes are compared with those of the benchmark motor in Figure 4.6.

Table 4.3. Functional-performance metrics of the selected ferrite-based motor.

Metric	Value
Low-speed maximum under-torque deviation	0.00%
Low-speed maximum over-torque deviation	0.67%
High-speed RMS relative torque deviation	8.56%
High-speed maximum absolute relative torque deviation	11.44%

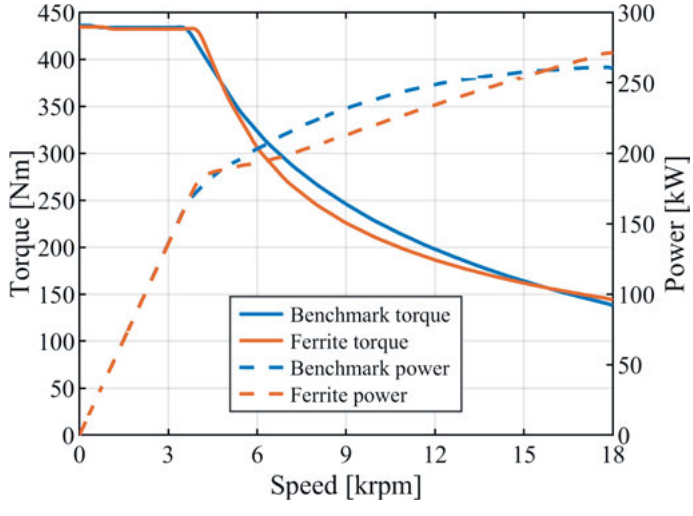


Figure 4.6. Torque-speed and power-speed envelopes of the benchmark and selected ferrite-based motors. Adapted from **Paper V** [94] under the CC BY 4.0 license.

In the FEM calculations used for design evaluation, the selected motor satisfies the prescribed 0-1% torque-deviation band at the evaluated speed points up to 3500 rpm. Above 3500 rpm, the RMS relative torque deviation is 8.56%, and the maximum absolute relative torque deviation is 11.44%.

4.3.2.3. Position of the Selected Design in the Trade-Off Space

Within the feasible surrogate set, 57 candidates dominate the selected LCA design, and the minimum normalized distance to the surrogate-derived Pareto front is 0.041. Table 4.4 compares the selected design with the nearest non-dominated candidate.

Table 4.4. Comparison between the selected LCA design and the nearest Pareto candidate.

Metric	Selected LCA design	Nearest Pareto candidate
Active axial length [mm]	216	215.97
Maximum DC bus voltage [V]	360	359.44
High-speed RMS relative torque deviation [%]	8.56	8.26

4.3.2.4. Selected Motor for LCA

The FEM-verified configuration selected above is used consistently as the ferrite-based motor in the subsequent LCA.

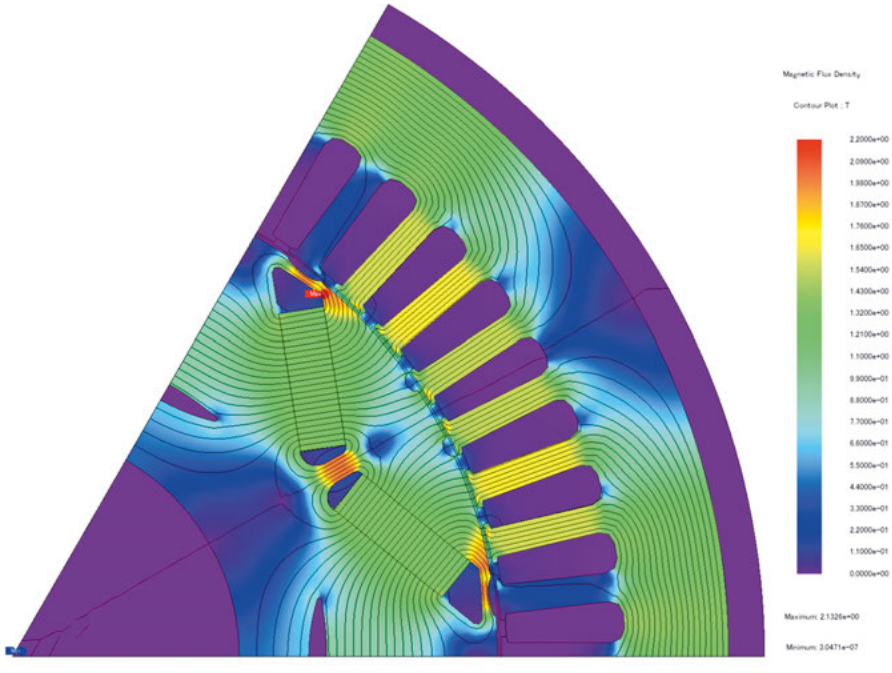
The selected motor retains the benchmark stator geometry, winding arrangement, and current loading. Functional compensation is therefore achieved primarily through the rotor magnetic design, increased active axial length, and higher allowable DC bus voltage rather than through increased current loading.

The principal differences arise from the permanent-magnet technology and the associated rotor adaptation. The benchmark employs axially segmented N52UH NdFeB magnets, whereas the selected motor uses Y40 ferrite magnets in a spoke-type rotor with a carbon-fiber retaining sleeve. Corresponding active and structural materials are otherwise kept consistent wherever the geometries permit.

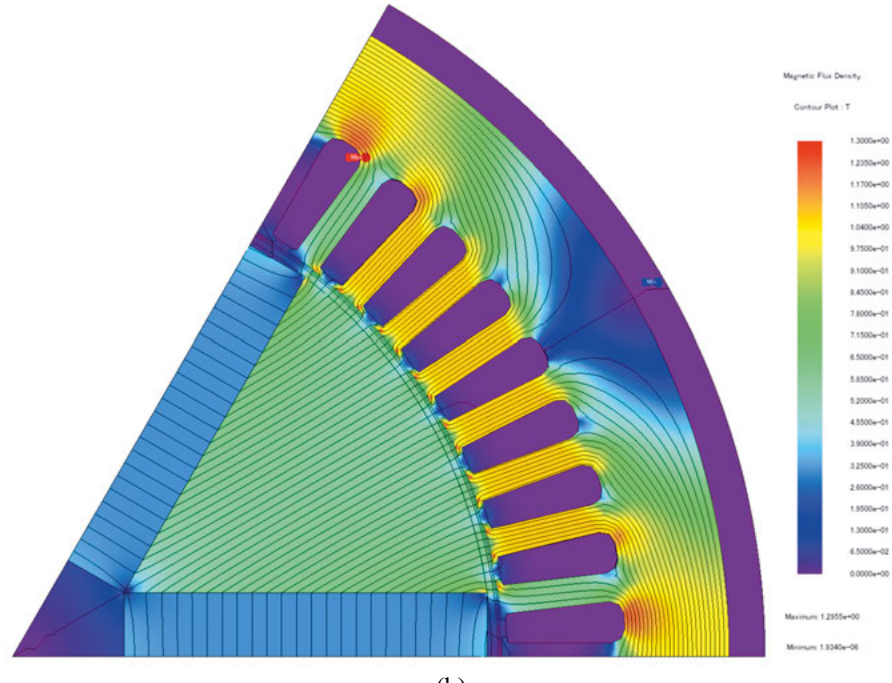
Figure 4.7 compares the no-load magnetic flux-density distributions and flux lines of the two motors. In the selected ferrite motor, the spoke-type rotor directs most of the magnet-generated flux through the rotor poles and across the air gap, while limiting shaft-side leakage. No severe local saturation is observed in the principal magnetic paths under the no-load condition.

The principal parameters used to construct the life cycle inventory are summarized in **Fel! Hittar inte referenskälla.**. Relative to the benchmark, the selected ferrite motor requires a substantially larger permanent-magnet cross-sectional area and includes the additional carbon-fiber retaining sleeve. The resulting component-level material quantities are quantified in Section 4.4 and reported in detail in Appendix C.

The torque-speed and power-speed characteristics of the two motors have already been evaluated in Section 4.3.2.2. The designs provide the level of functional comparability established by the multi-criteria framework rather than geometric or pointwise performance identity.



(a)



(b)

Figure 4.7. No-load flux density distributions and flux lines of the (a) benchmark and (b) ferrite-based traction motors. From **Paper V** [94] under the CC BY 4.0 license.

Table 4.5. Comparison of key design parameters between benchmark and ferrite-based traction motors. Adapted from **Paper V** [94] under the CC BY 4.0 license.

Parameters	Unit	Benchmark motor	Ferrite-based motor
Pole/Slot	-	6/54	Same as benchmark
Parallel paths	-	3	
Number of turns	-	5	
Peak torque	Nm	435	271
Peak power	kW	260	
Maximum speed	rpm	18 000	
Maximum current	Arms	672	Same as benchmark
Maximum current density	Arms/mm^2	26.2	
DC bus voltage	V	260	360
Stator outer diameter	mm	221	Same as benchmark
Active length	mm	134	216
Magnet cross-sectional area	mm^2	1813.74	8076.06
Total weight incl. housing	kg	58.4	78.5

4.3.3. Discussion

As shown by the trade-off-space analysis in Section 4.3.2.3, several feasible candidates dominate the selected configuration. However, its minimum normalized distance of 0.041 to the non-dominated boundary indicates that it remains relatively close to the best trade-offs identified within the investigated design space. The design is retained as a feasible, FEM-verified case for the LCA, rather than as a Pareto-optimal solution. The agreement between the surrogate prediction and the FEM result, with an RMS torque-envelope difference of 0.054% and a maximum-point difference of 0.213%, further supports the use of this configuration in the subsequent analysis.

The key design differences summarized in **Fel! Hittar inte referenskälla.** show that achieving the functional performance reported in Section 4.3.2.2 requires substantial geometric and electrical compensation relative to the NdFeB benchmark. The torque mismatch in the field-weakening region reflects the greater difficulty of reproducing the high-speed performance of the NdFeB benchmark using ferrite magnets with substantially lower magnetic energy density. Increasing active axial length and DC bus voltage can partly compensate for this limitation, but these adaptations introduce additional geometric, material, and electrical requirements. The selected ferrite-based motor has an active axial length of 216 mm, compared with 134 mm for the benchmark, corresponding to a normalized active length of

$$\lambda_{\mathrm{L}} = \frac{216}{134} = 1.61. \tag{4.10}$$

Its maximum DC bus voltage is 360 V, compared with 260 V for the benchmark, corresponding to

$$\lambda_V = \frac{360}{260} = 1.38. \tag{4.11}$$

Because the stator and rotor cross-sectional dimensions are retained, extending the active length increases the available torque-producing volume without increasing the radial package. This compensation, however, also increases the quantities of length-dependent materials, including electrical steel, copper, ferrite, and structural components. It further affects total motor mass, winding resistance, copper loss, and the resulting efficiency characteristics that are carried forward into the LCA.

The increased DC bus voltage provides additional voltage margin in the field-weakening region and thereby supports high-speed torque and power production. Relative to the 260 V benchmark, the selected 360 V limit represents a substantial electrical adaptation. At the powertrain level, such an increase may affect inverter voltage rating, semiconductor-device selection, insulation requirements, DC-link components, and compatibility with the vehicle battery architecture.

These voltage-related adaptations are not represented as additional hardware burdens in the subsequent LCA because the inverter and battery system lie outside the common system boundary defined in Section 4.2.3. The voltage increase should therefore be interpreted as an engineering integration requirement of the ferrite-based motor rather than as a directly quantified environmental burden. This modelling choice isolates the environmental consequences associated with the motor itself, but may underestimate broader powertrain modifications if the higher voltage cannot be accommodated within the existing electrical architecture.

The FEM-verified configuration is subsequently used to determine the material inventory, motor mass, efficiency map, and use-phase electricity consumption for the LCA in Sections 4.4 and 4.5.

4.4. Life Cycle Inventory Modelling

4.4.1. Material Composition

The material inventories of the benchmark and selected ferrite-based traction motors are established at the component level and derived from the electromagnetic and mechanical designs defined in Section 4.3. The foreground inventory includes electrical steel, copper, aluminum, stainless and carbon steels, permanent magnets, and auxiliary materials associated with coatings, electrical insulation, and the carbon-fiber composite sleeve used in the ferrite motor.

The material masses are calculated from the component geometries, material densities, and design-specific structural features of the two motors. Components with common geometries and materials are modelled using the same assumptions, while differences arising from the permanent magnet technology, active axial length, rotor structure, and housing dimensions are represented explicitly. The complete component-level inventories are provided in Appendix C.

Figure 4.8 compares the material composition of the two motors. The selected ferrite motor has a higher total mass than the benchmark motor, 78.5 kg compared with 58.4 kg, primarily because its longer active axial length increases the required quantities of electrical steel, copper, and aluminum. The most pronounced compositional change is the substantially larger magnet mass associated with replacing NdFeB magnets with ferrite magnets.

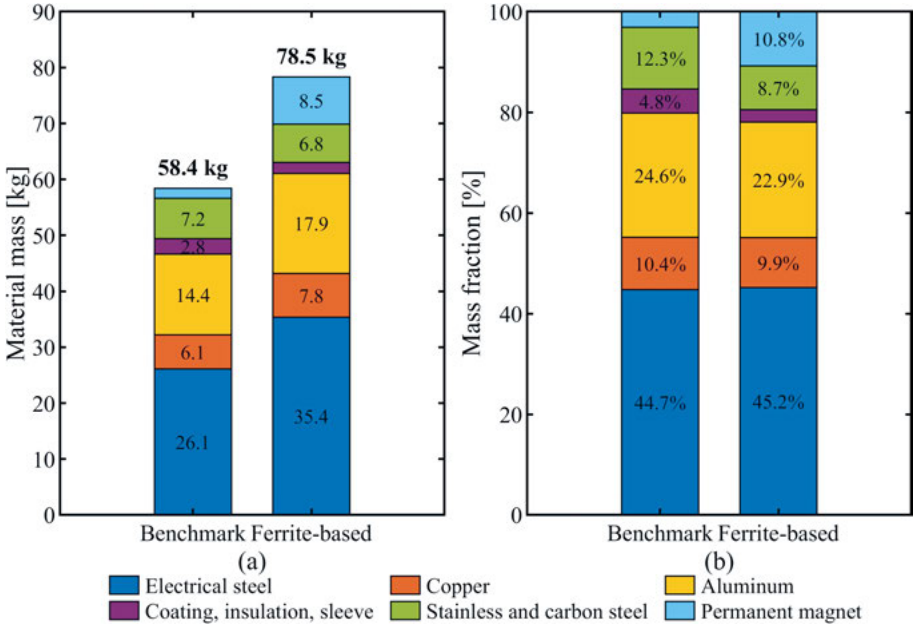


Figure 4.8. Material composition of the benchmark and ferrite-based motors: (a) absolute material mass and (b) material mass fraction. Adapted from **Paper V** [94] under the CC BY 4.0 license.

4.4.2. Manufacturing-Stage Modelling

Manufacturing-stage modelling is based on the scalable life cycle inventory framework for automotive traction motors developed by Nordelöf et al. [87], [88]. The framework represents motor production using component-level material inputs and generic manufacturing processes. It is applied consistently to both motor concepts using background data from Ecoinvent version 3.9.1

and the cut-off system model, with European production conditions assumed where applicable.

Several design-specific adaptations are introduced because the two motors differ substantially in active axial length and rotor structure. First, the housing mass is estimated using a geometry-based method rather than the power-based scaling approach in [87], [88]. Power-based scaling is less suitable here because the motors provide comparable peak power but have markedly different active lengths. The housing estimate is therefore related to the stator outer diameter and axial dimensions. The calculation procedure is provided in Appendix D.

Second, the copper mass is estimated analytically from the winding arrangement and motor geometry because copper usage is not defined explicitly in the baseline framework. The calculation includes the conductor length within the active stack and the end-winding contribution, allowing the longer active length of the ferrite motor to be reflected consistently in the material inventory.

The selected ferrite motor also contains a carbon-fiber composite retaining sleeve that is absent from the benchmark and is not represented in the baseline inventory framework. Sleeve production is therefore modelled as a separate foreground process, with its material and energy inputs assigned exclusively to the ferrite motor. Background inventories for carbon fiber and resin production are obtained from established databases, while manufacturing energy demand is scaled according to the calculated sleeve mass. The material composition, production process, and modelling assumptions for the sleeve are described in Appendix F.

All remaining motor components are modelled using the processes and assumptions of the baseline framework, adjusted according to the component masses reported in Appendix C. The same process datasets are used for corresponding components, with inputs scaled to the material quantities of each motor.

4.4.3. Use-Phase Energy Consumption Modelling

The use phase represents the electricity required by the traction motor to provide the functional unit over a vehicle lifetime of 200 000 km. Energy consumption is evaluated using a drive-cycle-based model that combines the motor efficiency maps with vehicle operating conditions and regenerative braking.

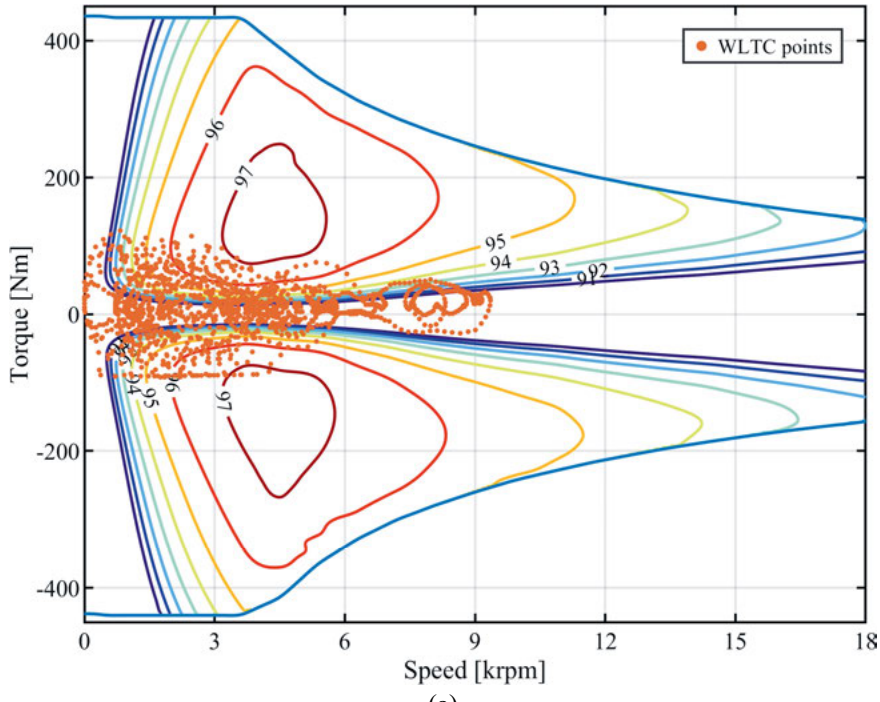
4.4.3.1. Motor Efficiency and Loss Modelling

Motor efficiency maps are derived from electromagnetic FEM simulations over the torque-speed operating range up to 18 000 rpm. This speed corresponds to the maximum operating speed of the benchmark motor and is adopted as the common upper limit for both motor concepts.

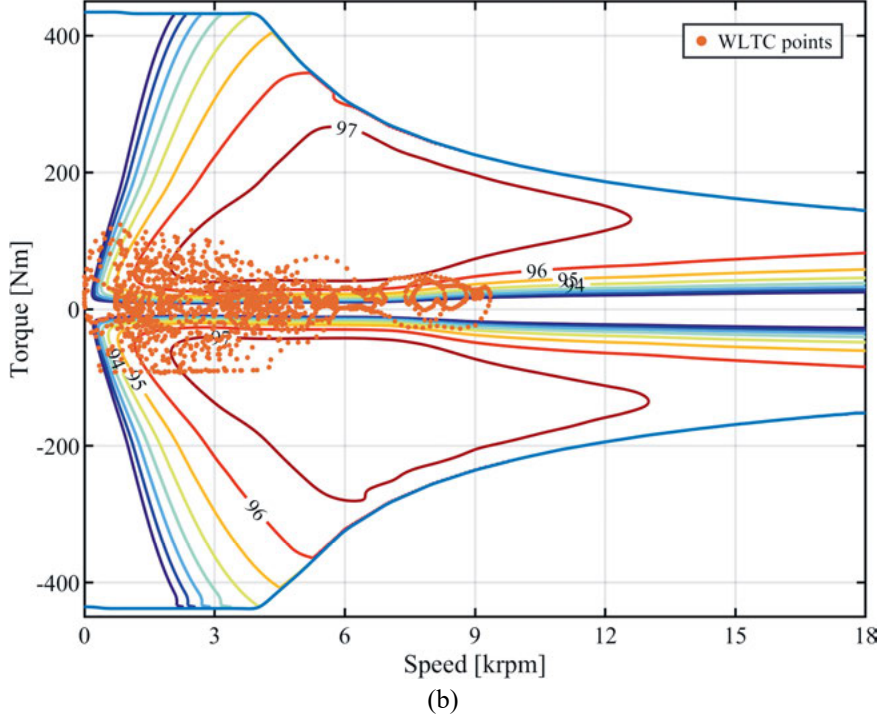
The loss calculation model includes DC copper loss, iron loss, permanent magnet eddy-current loss, AC copper loss, mechanical loss, and additional inverter-related loss associated with pulse-width-modulated operation. Iron loss is multiplied by a factor of 1.75 to account for additional losses under inverter-fed operation, following the approach reported in [100], [101].

For the selected ferrite motor, the phase resistance is recalculated using the analytically estimated conductor length. This treatment accounts for the longer active stack and maintains consistency between the motor geometry, copper inventory, winding resistance, and use-phase loss calculation. The copper-length and phase-resistance calculations are provided in Appendix E.

The resulting efficiency maps are shown in Figure 4.9. The efficiency maps are subsequently used to calculate the electrical energy associated with each operating point of the driving cycle.



(a)



(b)

Figure 4.9. Efficiency maps and WLTC operating points of (a) the benchmark and (b) the ferrite-based motor. Adapted from **Paper V** [94] under the CC BY 4.0 license.

4.4.3.2. Drive-Cycle and Operating-Point Mapping

Vehicle operation is represented using the Worldwide Harmonized Light Vehicles Test Cycle (WLTC) [102]. The vehicle-speed profile is converted into motor-speed and motor-torque demands using a backward-facing vehicle model.

The benchmark case is based on a Tesla Model 3 vehicle platform. For the ferrite-motor case, the total vehicle mass is adjusted to account for the difference in motor mass, while the remaining vehicle parameters are kept unchanged. This isolates the effects of motor mass and efficiency from other differences in vehicle design. The parameters used in the vehicle model are summarized in Table 4.6.

Table 4.6. Vehicle parameters used in the use-phase energy consumption modelling. From **Paper V** [94] under the CC BY 4.0 license.

	Benchmark	Ferrite-based	Sources and notes
Tire diameter [mm]	673	Same as benchmark	[103] The median value was used for various available tire sizes.
Gear ratio [-]	9.0363		[104]
Drag coefficient [-]	0.23		[105]
Frontal area [m^2]	2.22		[105]
Rolling resistance coefficient [-]	0.01		[105]
Curb weight [kg]	1776	1796	[106] Average of the four available car models was used.

The resulting operating points are mapped onto the corresponding motor efficiency maps, as shown in Figure 4.9. More than 64% of the operating time occurs below 4000 rpm, whereas operation above 8000 rpm accounts for less than 7% of the total cycle duration.

Both traction and generating operation are included. During positive-torque operation, the electrical energy drawn from the battery is calculated from the mechanical demand and the motoring efficiency at the corresponding operating point. During negative-torque operation, regenerative braking energy is calculated using the generating efficiency of the motor. The net electrical energy of one WLTC cycle is obtained by combining the traction-energy demand and the recovered braking energy.

The energy consumption per cycle is normalized by the WLTC driving distance and subsequently scaled to the 200 000 km vehicle lifetime defined by the functional unit. This procedure provides the lifetime electricity demand used in the use-phase impact assessment in Section 4.5.2.

4.4.4. End-of-Life Stage Modelling

At the end-of-life stage, both motors are assumed to undergo dismantling, shredding, and material separation. Ferrous and non-ferrous metals are recovered through conventional recycling processes, while the remaining fraction is treated as automobile shredder residue (ASR). The end-of-life stage is modelled using the Ecoinvent cut-off system model and is reported separately from the manufacturing stage.

Steel, aluminum, and copper are recovered according to the shredding and material-recovery efficiencies reported in [107], [108], [109]. The effective recovered mass is calculated from the original material mass, the shredding recovery rate, and the material-specific recovery efficiency. The detailed calculation procedure and the adopted recovery factors are provided in Appendix G.

Large-scale industrial recycling of NdFeB magnets is not yet widely established [110]. The NdFeB magnets in the benchmark motor are therefore assumed to enter the ASR stream. Ferrite magnets are treated in the same way because of their comparatively low economic value and the absence of established recycling routes under current market conditions [111]. Other non-metallic materials, including polymers, coatings, insulation, resin, paint, and the carbon-fiber composite sleeve, are also assigned to ASR.

The ASR mass includes all non-metallic materials originally present in the motor together with the fraction of metallic materials not recovered during shredding and separation. Material losses that do not enter the ASR stream are excluded. Consequently, the sum of the recovered material masses and ASR does not equal the original motor mass.

ASR is assumed to be incinerated. The treatment is represented using the Ecoinvent process “market for residue from shredder fraction from manual dismantling” as a proxy. The resulting masses of recovered steel, copper, aluminum, and ASR for the benchmark and selected ferrite-based motors are summarized in Table 4.7.

The selected ferrite motor generates a larger ASR mass because of its greater ferrite content, longer active structure, and carbon-fiber retaining sleeve. The environmental burdens associated with ASR treatment and the reported benefits associated with recovered metals are evaluated in Section 4.5.3.

Table 4.7. Masses of recovered materials and ASR. From **Paper V** [94] under the CC BY 4.0 license.

Recovered material	Weight per benchmark motor [kg]	Weight per ferrite-based motor [kg]	Sources in the motor
Steel	30.36	37.13	Stator and rotor cores, rotor endplates, shaft, bearings, fasteners, etc.
Copper	4.49	5.67	Stator winding
Aluminum	13.23	16.49	Housing and end bells
ASR	7.29	14.03	Magnets, resin, coating, plastic, carbon fiber, paint, etc.

Note 1: ASR is calculated as the sum of the fraction of metals not collected in the shredding and separation processes, together with all non-metallic materials originally present in the motor. Detailed calculations are provided in Appendix G.
Note 2: The sum of recovered materials and ASR (each column) does not equal the total motor mass, as material losses during recovery do not go into ASR.

4.5. LCA Results

Life cycle impact assessment is performed using the CML v4.8 2016 method [112]. The analysis focuses on the climate change impact category, expressed as the 100-year global warming potential, GWP100 [113]. The results are normalized by the lifetime driving distance and reported in g CO_2-eq/km, unless otherwise stated.

4.5.1. Manufacturing-Stage Impacts Including Upstream Processes

The manufacturing-stage assessment includes motor production and all upstream processes associated with raw material acquisition and processing. Motor manufacturing is assumed to take place in Europe, and the corresponding electricity demand is represented by the European market-average electricity mix. European market datasets are used for raw materials where available, while global market-average datasets are applied when region-specific data are unavailable.

As summarized in Table 4.8, the manufacturing-stage climate change impact is 2.01 g CO_2-eq/km for the benchmark motor and 2.34 g CO_2-eq/km for the selected ferrite-based motor. The latter is approximately 16% higher than the benchmark value.

Figure 4.10 shows the absolute contributions of the principal material and process categories. Values in parentheses give each category's percentage of the corresponding motor's manufacturing-stage impact. Electrical steel,

aluminum, and copper account for the largest shares of the manufacturing-stage impacts in both motors. The selected ferrite-based motor exhibits higher contributions from these material categories than the benchmark. By contrast, the magnet-related contribution is lower for the ferrite motor despite its substantially greater magnet mass.

Table 4.8. Manufacturing-stage climate change impacts, including upstream material production. From **Paper V** [94] under the CC BY 4.0 license.

	Normalized CO_2-eq emissions [g CO_2-eq/km]
Benchmark	2.01
Ferrite-based	2.34

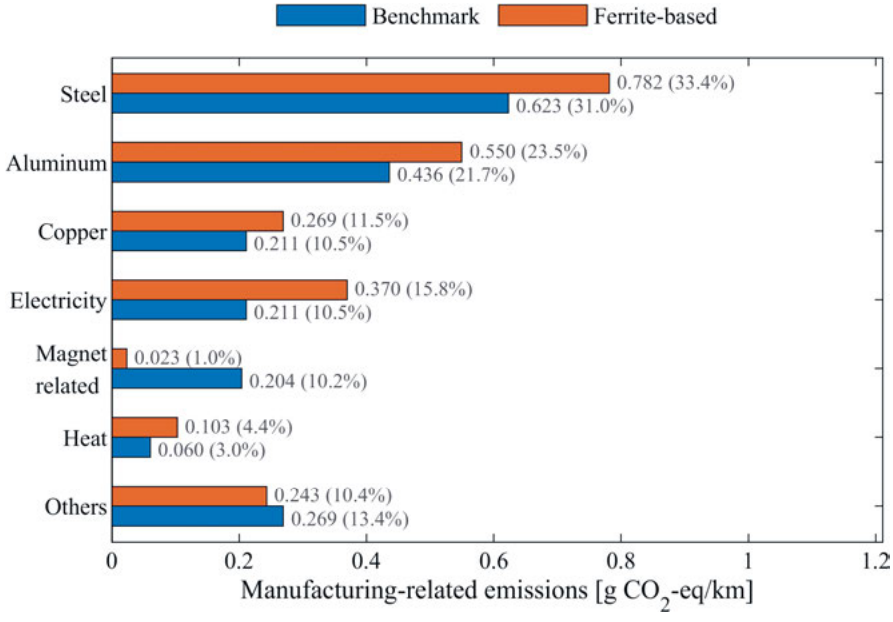


Figure 4.10. Contributions to manufacturing-stage climate change impacts. Adapted from **Paper V** [94] under the CC BY 4.0 license.

4.5.1.1. Sensitivity Analysis of Housing Mass

The aluminum housing masses are estimated using the geometry-based method described in Appendix D because complete housing geometries and disassembly data are not publicly available. To evaluate the influence of this assumption, the housing mass of each motor is varied by ±20%, while all other inventory parameters are kept unchanged.

The sensitivity results in Table 4.9 show that varying the aluminum housing mass by ±20% changes the manufacturing-stage impacts of both motors, while the selected ferrite-based motor retains a higher impact than the benchmark in all investigated scenarios.

Table 4.9. Sensitivity analysis of aluminum housing mass. From **Paper V** [94] under the CC BY 4.0 license.

		Normalized emissions [g CO_2-eq/km]	Changes
Benchmark motor	+20% Al	2.12	+5.5%
	Base case	2.01	0.0%
	-20% Al	1.90	-5.6%
Ferrite-based motor	+20% Al	2.48	+6.1%
	Base case	2.34	0.0%
	-20% Al	2.20	-5.9%

4.5.2. Use-Phase Impacts under Different Electricity Mixes

The use-phase assessment is based on the WLTC energy-consumption modelling described in Section 4.4.3. Both traction and regenerative operation are included, and the difference in vehicle mass caused by the two motor designs is retained in the calculation.

The energy flows over a single WLTC cycle are summarized in Table 4.10 and illustrated in Figure 4.11. The selected ferrite-based motor requires less net electrical energy than the benchmark and exhibits a higher cycle-averaged efficiency. The difference occurs mainly during motoring operation, while the regenerative efficiencies of the two motors remain similar.

Table 4.10. Energy consumption and efficiency results for a single WLTC cycle. From **Paper V** [94] under the CC BY 4.0 license.

	Benchmark	Ferrite-based
MOTOR PART		
Total Input Energy [kWh]	3.66	3.54
Total Output Energy [kWh]	3.33	3.35
Total Loss [kWh]	0.34	0.19
Efficiency	90.84%	94.60%
GENERATOR PART		
Total Input Energy [kWh]	1.05	1.07
Total Output Energy [kWh]	1.00	1.01
Total Loss [kWh]	0.05	0.06
Efficiency	94.87%	94.83%
MOTOR+GENERATOR OVERALL		
Total Consumed Energy [kWh]	2.66	2.53
Total Output Energy [kWh]	2.27	2.28
Total Loss [kWh]	0.39	0.25
Overall Efficiency	85.36%	90.26%

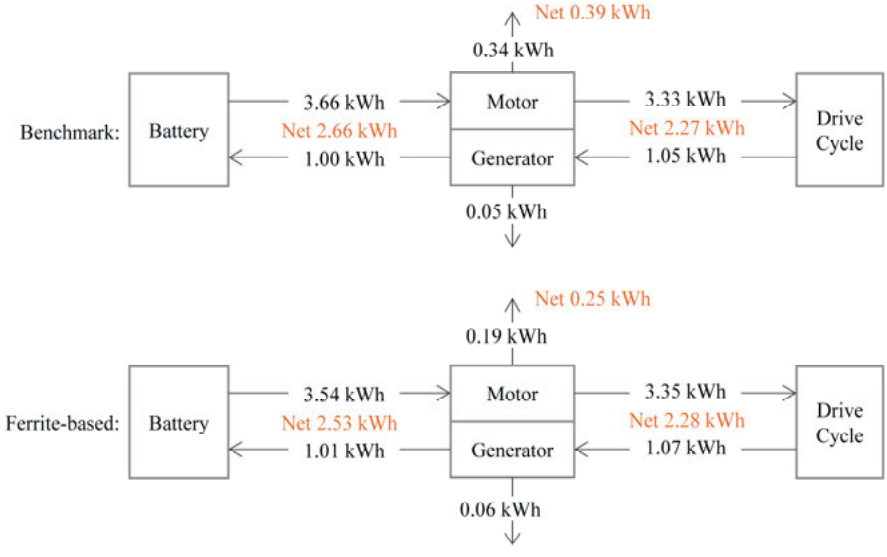


Figure 4.11. Energy flows of the two motors over a single WLTC cycle. Adapted from **Paper V** [94] under the CC BY 4.0 license.

The lifetime-normalized electricity consumption is 108.78 Wh/km for the ferrite motor and 114.33 Wh/km for the benchmark in Table 4.11. It means that when normalized by driving distance and scaled to the assumed vehicle lifetime, the selected ferrite motor retains a lower total electricity demand.

Table 4.11. Lifetime-normalized energy consumption under WLTC operation. From **Paper V** [94] under the CC BY 4.0 license.

	Benchmark	Ferrite-based
WLTC cycle length [m]	23 266	Same as benchmark
Assumed life cycle driving distance [km]	200 000	
Number of WLTC cycles	8596	
Lifetime total consumed energy [Wh/km]	114.33	108.78
Lifetime total loss [Wh/km]	16.74	10.59
Lifetime total output energy [Wh/km]	97.59	98.19
Lifetime overall efficiency	85.36%	90.26%

Figure 4.12 presents the composition of the lifetime-normalized motor losses. Iron loss is the largest loss category for both motors, particularly the eddy-current component in the electrical steel. The selected ferrite motor exhibits lower total iron loss despite containing more electrical steel. Copper and mechanical losses are of similar magnitude in the two motor concepts. Magnet eddy-current loss occurs in the NdFeB-based benchmark but is negligible in the ferrite motor.

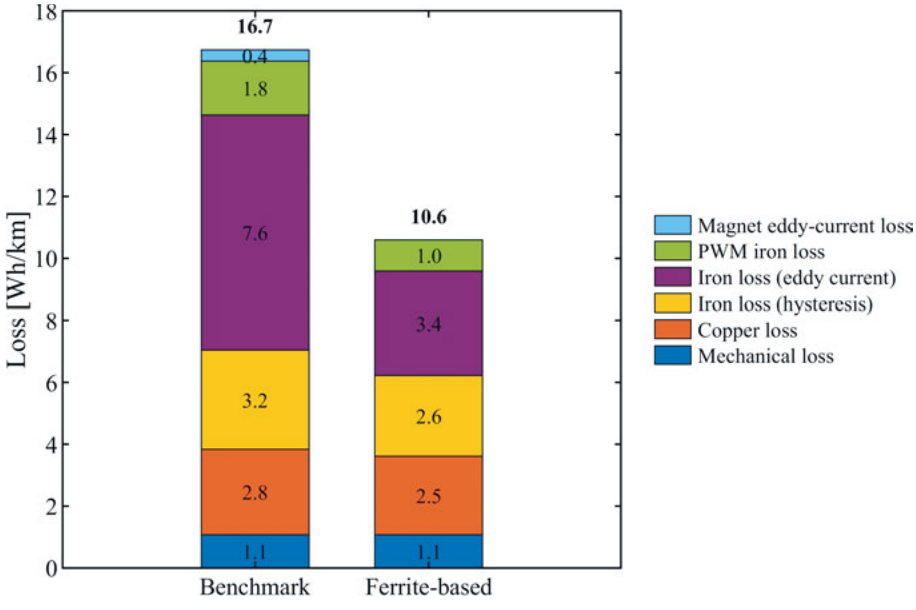


Figure 4.12. Lifetime-normalized motor losses under WLTC operation. Adapted from **Paper V** [94] under the CC BY 4.0 license.

The use-phase climate change impacts are evaluated using the electricity carbon intensities listed in Table 4.12. Four electricity mixes are considered: the global average, the United States average, the European average, and the Swedish electricity mix.

Table 4.12. Electricity carbon intensities of the regional electricity mixes considered in the use-phase assessment [95]. Adapted from **Paper V** [94] under the CC BY 4.0 license.

Region	Electricity carbon intensity [g CO_2-eq/kWh]
Global average	726.67
United States average	481.21
European average	354.80
Sweden	43.92

Figure 4.13 compares the resulting lifetime-normalized use-phase impacts. The selected ferrite motor exhibits lower use-phase climate change impacts under all four electricity mixes. The absolute impact varies substantially with electricity carbon intensity, but the relative difference between the motor concepts remains similar. The regional electricity mix therefore scales the magnitude of the use-phase impact but does not change the ranking of the two motors at this life cycle stage.

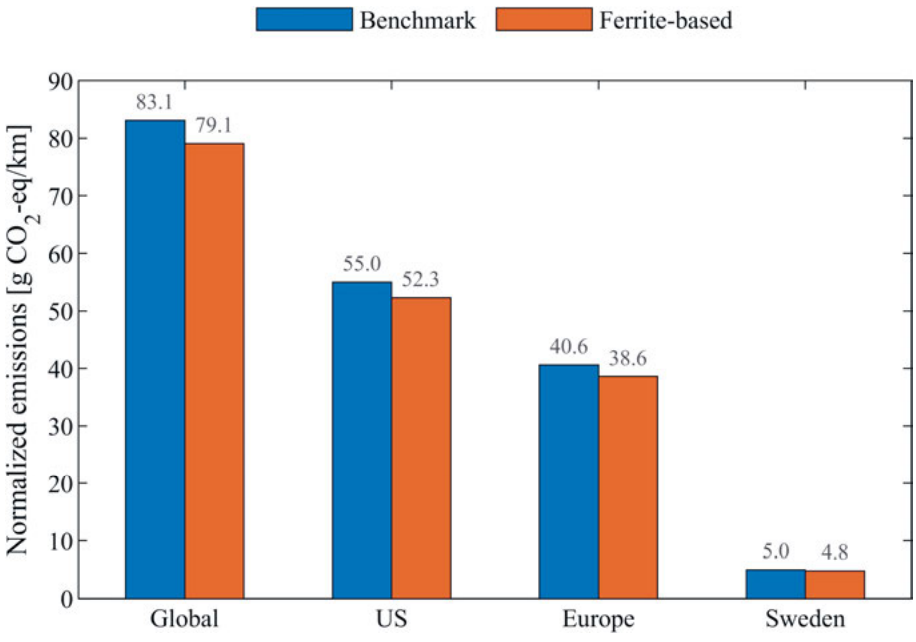


Figure 4.13. Lifetime-normalized use-phase climate change impacts under different electricity mixes. Adapted from **Paper V** [94] under the CC BY 4.0 license.

4.5.2.1. Sensitivity Analysis of Vehicle Mass

The base-case assessment represents a vehicle configuration comparable to the Tesla Model 3 platform. The base-case WLTC operating points lie within only part of each motor's torque-speed envelope. The following sensitivity analysis examines the effect of higher vehicle mass on these operating points and on electricity consumption.

To evaluate the sensitivity of the use-phase result to vehicle mass, two additional bounding scenarios are considered in which the base vehicle mass is doubled and tripled. All other vehicle parameters are kept unchanged. The motor-mass difference between the benchmark and ferrite-based cases is retained in each scenario. The corresponding vehicle masses are summarized in Table 4.13.

Table 4.13. Vehicle mass scenarios used in the sensitivity analysis. From **Paper V** [94] under the CC BY 4.0 license.

Curb weight [kg]	Benchmark	Ferrite-based
Base case	1776	1796
Doubled weight	3552	3572
Tripled weight	5328	5348

The resulting operating points are mapped onto the original efficiency maps, as shown in Figure 4.14 and Figure 4.15. Increasing the vehicle mass shifts the WLTC operating points toward higher torque levels while leaving the motor-speed distribution largely unchanged.

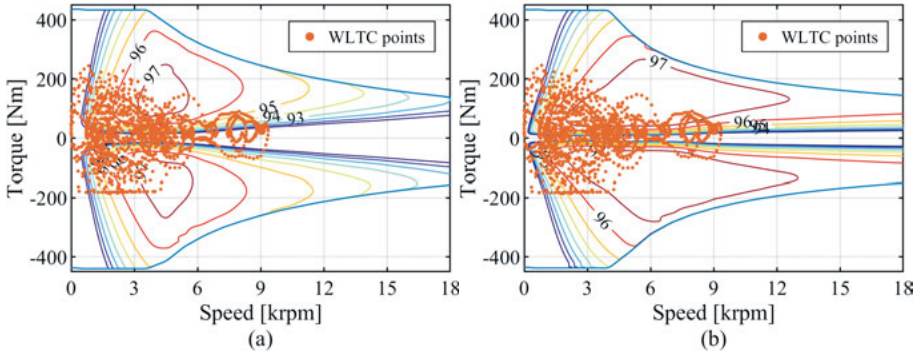


Figure 4.14. Efficiency maps and WLTC operating points for the doubled-mass scenario: (a) benchmark motor and (b) selected ferrite-based motor. Adapted from **Paper V** [94] under the CC BY 4.0 license.

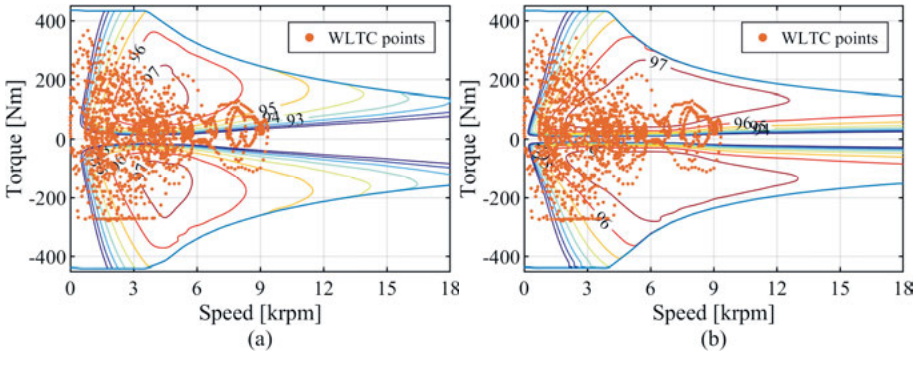


Figure 4.15. Efficiency maps and WLTC operating points for the tripled-mass scenario: (a) benchmark motor and (b) selected ferrite-based motor. Adapted from **Paper V** [94] under the CC BY 4.0 license.

Increasing vehicle mass raises the electricity consumption of both motor cases. However, the selected ferrite motor continues to exhibit lower WLTC electricity consumption than the benchmark in all investigated scenarios.

Figure 4.17 shows that the ferrite motor retains the lower use-phase climate change impact across all investigated vehicle-mass and electricity-mix scenarios.

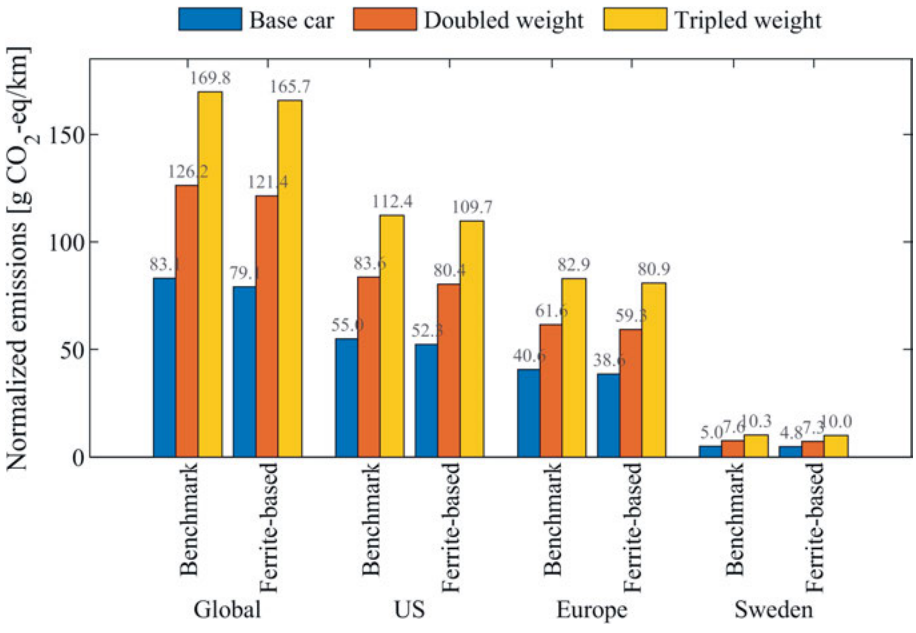


Figure 4.16. Lifetime use-phase climate change impacts under vehicle-mass and electricity-mix scenarios. Adapted from **Paper V** [94] under the CC BY 4.0 license.

4.5.3. Estimated End-of-Life Impacts

Figure 4.17 presents the estimated end-of-life climate change impacts of the benchmark and selected ferrite-based motors. Positive values represent emissions associated with dismantling, shredding, material separation, and ASR incineration. Negative values represent the estimated avoided impacts of the reference material-production processes assigned to recovered steel, copper, and aluminum.Both motor concepts exhibit a net negative end-of-life climate change impact. The selected ferrite motor shows a less favorable end-of-life result than the benchmark.

The end-of-life calculation is reported as a separate recovery scenario. It includes treatment burdens and estimated avoided impacts from recovered metals, and is not added to the manufacturing-plus-use totals reported below.

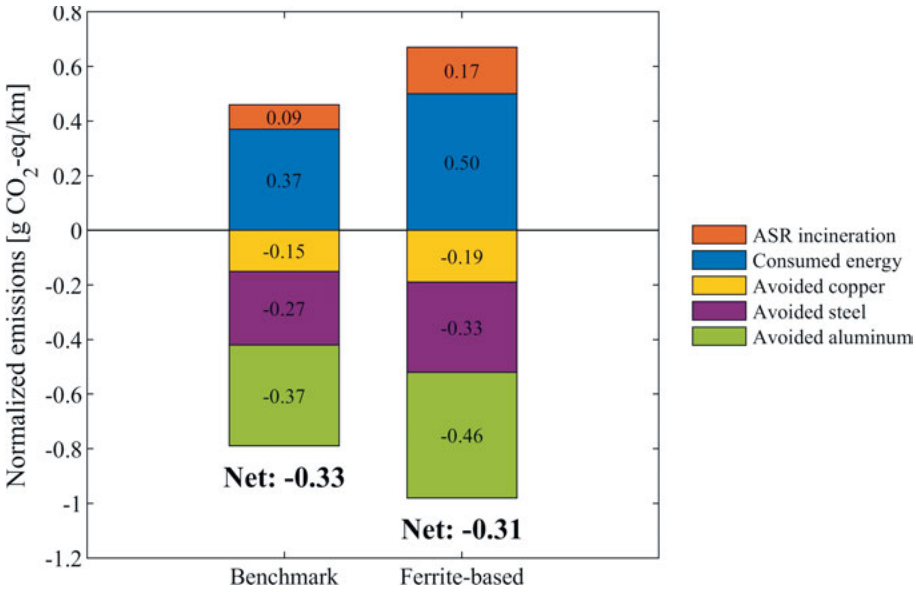


Figure 4.17. Net normalized end-of-life climate change impacts of the two motors. Adapted from **Paper V** [94] under the CC BY 4.0 license.

4.5.4. Total Life Cycle Impacts

The aggregated results combine upstream material production, motor manufacturing, and use-phase electricity consumption. The separately reported end-of-life recovery scenario is excluded from these totals.

Manufacturing is assumed to take place in Europe, while the use phase is evaluated using the global-average, United States, European, and Swedish electricity mixes. The resulting lifetime-normalized impacts are shown in Figure 4.18. Manufacturing-stage impacts include upstream material production.

Under the global-average, United States, and European electricity mixes, the selected ferrite motor exhibits a lower total life cycle impact than the benchmark. Under the Swedish electricity mix, the ranking is reversed, and the benchmark shows a slightly lower total life cycle impact than the selected ferrite motor. Across the higher-carbon electricity mixes, the use phase dominates the total result for both motors. This dependence on electricity carbon intensity is quantified further in Section 4.5.5.



Figure 4.18. Lifetime climate change impacts under different electricity mixes. Adapted from **Paper V** [94] under the CC BY 4.0 license.

4.5.5. Break-Even Electricity Carbon Intensity

To quantify the dependence of the comparative result on the electricity system, the total life cycle climate change impact is expressed as a function of electricity carbon intensity. Under the cut-off system model adopted in this study, the end-of-life results are reported separately and are therefore excluded from the break-even formulation.

For each motor concept, the total life cycle climate change impact per unit driving distance is expressed as

$$GWP_{total} = GWP_{manuf} + CI_{el} \cdot E_{use} \tag{4.12}$$

where GWP_{manuf} is the manufacturing-stage climate change impact normalized by driving distance [g CO_2-eq/km], CI_{el} is the electricity carbon intensity [g CO_2-eq/kWh], and E_{use} is the lifetime-normalized electricity consumption [kWh/km].

The break-even electricity carbon intensity CI_{el}^{*} is obtained by equating the total impacts of the benchmark and selected ferrite-based motors, which gives

$$CI_{el}^{*} = \frac{GWP_{manuf}^{Ferrite} - GWP_{manuf}^{Nd}}{E_{use}^{Nd} - E_{use}^{Ferrite}}. \tag{4.13}$$

Using the manufacturing-stage and lifetime electricity-consumption results presented in Sections 4.5.1 and 4.5.2, the break-even electricity carbon

intensity is approximately 60 g CO_2-eq/kWh. The detailed numerical substitution is provided in Appendix H.

Figure 4.19 shows the total life cycle impacts of the two motors as linear functions of electricity carbon intensity. The intersection represents the break-even condition at which the higher manufacturing-stage impact of the selected ferrite motor is exactly offset by its lower lifetime electricity consumption.

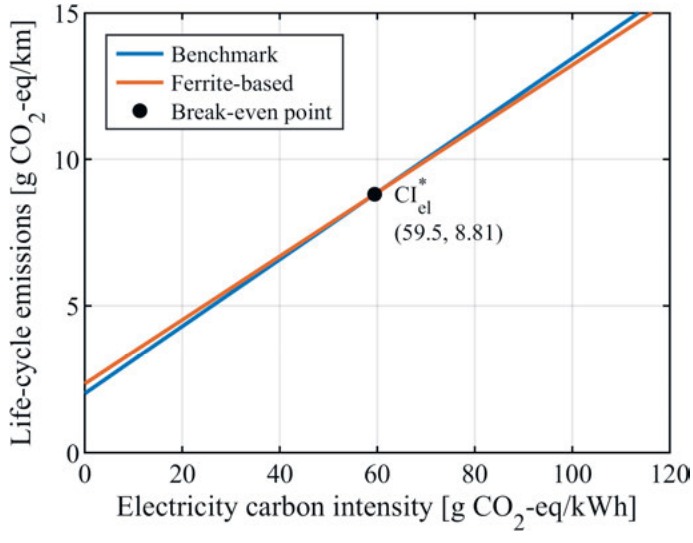


Figure 4.19. Life cycle climate change impacts as functions of electricity carbon intensity. Adapted from **Paper V** [94] under the CC BY 4.0 license.

4.6. Discussion

The LCA results show that rare-earth substitution introduces trade-offs across different life cycle stages rather than a uniform environmental benefit. These trade-offs are discussed below in relation to design compensation, electricity carbon intensity, material selection, and the assessment's limitations.

4.6.1. Life Cycle Trade-Offs

The results in Sections 4.5.1-4.5.5 reveal a shift in environmental burden from permanent-magnet production toward bulk material use and design compensation. Replacing NdFeB with ferrite reduces the magnet-related manufacturing contribution, but the longer active structure and associated increases in electrical steel, copper, aluminum, ferrite, and composite materials raise the overall manufacturing-stage impact. The environmental impact of rare-earth substitution therefore cannot be assessed from the magnet

material alone. The assessment must also account for the changes in materials and motor structure needed to achieve the required traction performance.

During vehicle operation, the selected ferrite motor exhibits lower lifetime electricity consumption over the WLTC. This result is consistent with the loss characteristics observed over the WLTC operating region, including lower total iron loss and negligible permanent-magnet eddy-current loss in the selected ferrite motor, the latter being associated with the substantially higher electrical resistivity of ferrite. During WLTC testing, the motor operates predominantly at low and medium speeds. Efficiency in these regions is therefore expected to have a greater influence on lifetime electricity consumption than efficiency near the maximum operating speed.

Whether the lower electricity consumption of the selected ferrite motor offsets its higher manufacturing-stage emissions depends on the carbon intensity of the electricity supply. Section 4.5.5 identifies the break-even carbon intensity at which these two effects balance. Above this value, the reduction in use-phase emissions exceeds the additional emissions from manufacturing. As the electricity supply becomes less carbon intensive, the emissions avoided through lower electricity consumption decrease. The implications of this dependence are discussed in Section 4.6.2.

Varying the estimated housing mass within the investigated range changes the absolute manufacturing-stage impacts but does not reverse the ranking of the two motor designs. Increasing vehicle mass raises the use-phase impacts of both motors, but the selected ferrite motor retains its lower WLTC electricity consumption in all tested scenarios. Within these ranges, the main trade-off remains between the additional materials required to achieve comparable traction performance with ferrite magnets and the lifetime electricity savings of the selected design.

4.6.2. Influence of Electricity Carbon Intensity

The relative life cycle performance of the two motor concepts depends strongly on electricity carbon intensity. Under the global-average, United States, and European electricity mixes, the lower lifetime electricity consumption of the selected ferrite motor is sufficient to offset its higher manufacturing-stage impact, whereas the benchmark becomes slightly favorable under the Swedish electricity mix. As electricity production becomes less carbon intensive, the use-phase contribution decreases and the relative importance of manufacturing increases.

The break-even electricity carbon intensity identified in Section 4.5.5 is approximately 60 g CO_2-eq/kWh. Above this value, the lower use-phase emissions of the selected ferrite motor outweigh its additional manufacturing burden. Below it, the NdFeB benchmark has a lower combined manufacturing and use-phase impact. Within the investigated range, changes in the estimated housing mass affect the magnitude of the manufacturing-stage impacts, with

the ferrite motor retaining the higher impact. Increasing vehicle mass raises the use-phase impacts of both motors, and the selected ferrite motor maintains lower WLTC electricity consumption in every scenario tested. These sensitivity results support the trade-off identified in the main analysis between the additional materials required to achieve comparable traction performance and the lifetime electricity savings of the ferrite design. The resulting environmental comparison depends on electricity carbon intensity, and the break-even value is specific to the motor designs, modelling assumptions, and operating conditions examined in this study.

4.6.3. Material Selection Implications for Motor Design

For the motor configurations investigated in this study, the selected ferrite-based design achieves traction functionality broadly comparable to that of the production-derived NdFeB benchmark. Its environmental performance, however, depends on both the design compensation required to achieve that functionality and the electricity system in which the vehicle operates.

Material selection should therefore not be based on electromagnetic performance or life cycle climate impact alone. Once the required traction functionality has been established, other factors such as material cost, availability, supply-chain concentration, price volatility, and manufacturability also become relevant. Within the investigated design space, ferrite-based solutions may be attractive where reduced dependence on rare-earth materials is prioritized and where the required design compensation can be accommodated.

The increased use of electrical steel, copper, aluminum, and structural materials in the selected ferrite motor also highlights the importance of material efficiency and circularity. Reduced active-material demand, greater use of secondary metals, improved recovery efficiencies, and higher-quality recycled material streams could lower the manufacturing-stage burden of future ferrite-based designs. The actual benefit, however, depends on the availability and environmental performance of secondary materials rather than on recycling rates alone.

Traction motor material selection should therefore account for both the motor design and its integration into the electric drivetrain. The choice of permanent-magnet material should be assessed alongside electromagnetic performance, structural requirements, packaging constraints, electrical operating conditions, efficiency across the operating range, manufacturability, and end-of-life treatment.

Design-consistent life cycle modelling could also be incorporated earlier in traction motor development. Coupling simplified environmental models with electromagnetic and mechanical design tools would allow material demand, manufacturing impacts, and use-phase consequences to be evaluated

alongside torque, power, efficiency, and packaging requirements before the motor geometry becomes fixed.

The larger active axial length required by the selected ferrite motor should also be interpreted in the context of the functional-equivalence constraints adopted in this study. To provide a consistent basis for the LCA, the ferrite-based motor was designed to reproduce the principal traction characteristics of an existing NdFeB benchmark within a largely fixed torque–speed operating range. Consequently, the lower magnetic loading of ferrite had to be compensated primarily within the motor itself, resulting in a longer active structure. In a clean-sheet, application-oriented design, however, the operating range of the motor would not necessarily be inherited from an existing powertrain. The motor maximum speed, gearbox ratio, DC bus voltage, inverter capability, and vehicle-level performance requirements could instead be designed concurrently. A higher permissible operating speed would allow the motor to deliver the same power at lower torque. This could reduce the required active-material volume or avoid the need for a longer active axial length. The dimensional increase observed in the present comparison should therefore be regarded as a consequence of the adopted functional-equivalence conditions rather than as an inherent requirement of ferrite-based traction motors. Determining whether a more compact and environmentally favorable ferrite-based solution can be achieved would require coupled optimization of the complete electric driveline.

4.6.4. Limitations of the LCA

The LCA is subject to several modelling assumptions. The housing masses are estimated from simplified geometry because complete benchmark housing data are not publicly available, although the sensitivity analysis in Section 4.5.1.1 shows that the comparative manufacturing-stage ranking is retained within the investigated ±20% variation. The vehicle-mass sensitivity analysis similarly indicates that the lower WLTC electricity consumption of the selected ferrite motor is maintained across the investigated scenarios.

The end-of-life results should be interpreted as estimates under the adopted recovery assumptions rather than as fixed future outcomes. The net negative impacts arise from avoided primary-material production associated with recovered steel, copper, and aluminum. The selected ferrite motor shows a slightly less favorable end-of-life result because the larger amount of non-recovered material, including ferrite magnets and the carbon-fiber sleeve, increases the ASR stream. Improved recovery routes for ferrite magnets, composite materials, or other non-metallic components could reduce this burden, whereas lower metal-recovery efficiencies would reduce the recycling benefits of both motors.

The break-even electricity carbon intensity is specific to the motor designs and life cycle scenarios examined in this study. Its value depends on the

manufacturing inventories, vehicle lifetime and operating conditions, and assumptions about end-of-life recovery and electricity supply.

5. Conclusions

This thesis investigated ferrite-based spoke-type traction motors as a rare-earth-free approach to EV propulsion from the perspectives of design feasibility, traction performance, and life cycle environmental performance. This work examined how motor design could compensate for the lower magnetic performance of ferrite magnets compared with NdFeB magnets. The approach included flux-concentrating rotor structures, a larger magnet volume, suppression of leakage flux, mechanical retention using a carbon-fiber sleeve, and, where necessary, geometric and electrical adaptation.

The first research question concerned what is known about ferrite-based traction motors and their potential for EV applications. The literature review and benchmarking study showed that ferrite-based motors can reach traction-relevant performance when the motor topology is designed specifically around the magnetic characteristics of ferrite. Among the reviewed concepts, spoke-type machines were identified as particularly suitable because they can accommodate a comparatively large magnet volume and concentrate the magnet flux toward the air gap. At the same time, the review identified recurring challenges related to shaft-side leakage, irreversible demagnetization, high-speed mechanical retention, and manufacturability. Benchmarking commercial traction motors provided reference values for evaluating the ferrite-based designs developed in this thesis. These findings guided the motor design work presented in the following chapters.

The second research question examined whether the proposed ferrite-based spoke-type design approach can meet traction-relevant technical requirements. A prototype-oriented segmented spoke-type motor was developed using rectangular Y40 ferrite magnets, segmented rotor laminations, non-magnetic aluminum structural components, and a carbon-fiber retaining sleeve. The segmented rotor suppressed the shaft-side leakage path observed in the conventional one-piece comparison rotor and improved utilization of the available ferrite flux. The designed motor achieved a simulated peak torque density of 21.24 Nm/L, which lies within the commercial reference range established in Chapter 2, together with a peak power density of 18.18 kW/L. The simulated motor performance also satisfied the specified peak-torque, peak-power, and peak-efficiency requirements within the investigated operating range.

The demagnetization assessment showed that ferrite remains particularly sensitive to severe low-temperature demagnetizing conditions. Under the deliberately severe stress condition applied at -40 °C, the resulting torque reduction was approximately 14.7%. The high-speed mechanical analysis also provided preliminary support for the carbon-fiber sleeve as a rotor-retention concept at 30 000 rpm. Experimental measurements of air-gap flux density and no-load back EMF showed overall agreement with the FEM predictions, with peak back-EMF deviations below 8% at the investigated speeds. Within the scope of the simulations and no-load experiments conducted here, the proposed design approach was technically feasible.

The third research question addressed whether rare-earth substitution with ferrite is environmentally meaningful at the motor-system level. A surrogate-assisted functional-equivalence framework was applied to a separate ferrite-based motor developed for comparison with a production-derived NdFeB benchmark. The selected configuration achieved a maximum low-speed over-torque deviation of 0.67% and a high-speed RMS torque deviation of 8.56%, while requiring active-length and DC-bus-voltage ratios of 1.61 and 1.38, respectively. These results show that achieving comparable traction performance requires substantial changes to the motor geometry and electrical operating conditions.

The resulting design compensation increased the demand for bulk materials, causing the manufacturing-stage climate change impact of the selected ferrite motor to be approximately 16% higher than that of the benchmark. During vehicle operation, however, the ferrite motor exhibited lower lifetime-normalized electricity consumption, at 108.78 Wh/km compared with 114.33 Wh/km for the benchmark. Whether the reduction in use-phase emissions offset the additional manufacturing emissions depended on the carbon intensity of the electricity supply. The two effects balanced at approximately 60 g CO_2-eq/kWh. Above this value, the lower electricity consumption favored the ferrite motor, whereas below it, the lower manufacturing impact favored the benchmark.

The environmental value of ferrite substitution cannot be determined by comparing magnet materials alone. Complete motor designs providing comparable traction functionality must be compared while accounting for the required geometric, structural, electrical, and operational adaptations. In the present comparison, the ferrite motor was required to reproduce the benchmark torque-speed envelope within the same maximum-speed range, which necessitated an increase in active axial length. This length increase should therefore not be interpreted as an inherent size penalty of ferrite-based motors. With a higher permissible operating speed and a suitably chosen gearbox ratio, the ferrite motor could meet the vehicle's power requirements at lower motor torque. This could avoid the need for a longer active length.

The main contribution of this thesis is to combine design principles drawn from the literature, prototype-level technical validation, design for functional

equivalence, and cradle-to-grave environmental assessment. The results show that ferrite-based spoke-type motors can provide a technically viable alternative to NdFeB traction motors. Their environmental performance depends on the design changes needed to achieve comparable traction performance and the carbon intensity of the electricity used during vehicle operation.

6. Future Work

Future work should extend the experimental validation beyond the no-load measurements conducted in this thesis. Loaded and high-speed tests should examine torque-speed characteristics, efficiency maps, field-weakening behavior, thermal performance, and low-temperature demagnetization. These measurements would help validate the electromagnetic and demagnetization predictions and quantify the effects of manufacturing tolerances and ferrite material properties.

The high-speed rotor-retention concept requires further structural assessment. The mechanical model should account for the orthotropic properties and fiber orientation of the composite sleeve, together with sleeve prestress, adhesive interfaces, and thermal expansion. Rotor imbalance, defects, fatigue, and overspeed conditions should also be examined. Experimental spin testing would help assess the mechanical integrity of the segmented rotor. Geometric development should investigate a smaller air gap, improved slot fill, and refinements to the magnets, rotor poles, and segmented assembly, with attention to manufacturability, assembly tolerances, disassembly, and material recovery.

The functional-equivalence framework developed in Chapter 4 could be extended to the optimization of the complete electric driveline. A clean-sheet design would allow the torque-speed envelope to be optimized for the vehicle's traction requirements. Maximum motor speed, gearbox ratio, DC bus voltage, inverter capability, and battery configuration could then be evaluated together. This would allow the potential for a more compact ferrite motor at higher speed to be assessed alongside the resulting demands on other driveline components. Extending the system boundary to include these adaptations would support a combined assessment of electromagnetic, mechanical, and thermal performance, manufacturing requirements, and environmental impacts.

The LCA could be extended to environmental impact categories beyond climate change. Additional comparisons should examine different driving cycles, vehicle platforms, electricity scenarios, vehicle lifetimes, manufacturing technologies, secondary-material availability, and end-of-life pathways. Recycling routes for ferrite magnets, NdFeB magnets, and carbon-fiber structures deserve particular attention.

7. Summary of Papers

Paper I

Design of Spoke Type Traction Motor with Ferrite Material for EV Application

This paper presents the initial development of a ferrite-based spoke-type traction motor for electric vehicle applications. The study investigates how flux concentration and rotor design can compensate for the lower magnetic performance of ferrite magnets, with particular attention to shaft-side leakage and magnet utilization. The results show that ferrite can support traction-relevant performance when the motor is designed specifically around its magnetic limitations, rather than treated as a direct substitute for NdFeB. The paper establishes the spoke-type design concept and identifies the main electromagnetic challenges that motivated the subsequent development of the segmented rotor structure.

Presented at *2022 International Conference on Electrical Machines (ICEM)*.

Paper II

A Comprehensive Review of Ferrite-Based Electrical Motors for Electric Vehicle Applications

This paper provides a systematic review and benchmarking study of ferrite-based traction motors. Commercial EV traction requirements, permanent-magnet characteristics, and representative ferrite-based motor topologies are compared to identify achievable performance levels and recurring design challenges. The study concludes that ferrite-based traction motors are technically feasible, but their competitiveness depends strongly on topology, magnet utilization, demagnetization resistance, and structural design. Among the reviewed concepts, spoke-type motors are identified as particularly suitable because of their flux-concentration capability and ability to accommodate a large ferrite volume. The paper defines the design requirements and provides the technical basis for the motor concepts developed in Chapter 3.

Published in *Renewable and Sustainable Energy Reviews* in 2026.

Paper III

A Segmented Ferrite-Based Spoke-Type Permanent Magnet Motor for Electric Vehicle Traction: Design and Experimental Validation

This paper develops and experimentally evaluates a segmented ferrite-based spoke-type traction motor using rectangular Y40 ferrite magnets, non-magnetic structural components, and a carbon-fiber retaining sleeve. The segmented rotor reduces shaft-side leakage while enabling mechanical retention at high rotational speed. FEM analyses of electromagnetic performance, demagnetization, and mechanical behavior are complemented by prototype-level no-load measurements. The results indicate that the proposed segmented rotor can achieve electromagnetic performance relevant to traction applications. The demagnetization and mechanical analyses support its feasibility under the conditions examined. Agreement between the no-load measurements and the corresponding FEM predictions further supports the electromagnetic design of the ferrite-based spoke-type motor.

Submitted to *IEEE Access 2026,* under review.

Paper IV

Functional Equivalence Design of a Ferrite-Based Spoke-Type Traction Motor Under Torque and Power Envelope Constraints

This paper develops a functional-equivalence design framework for comparing ferrite-based and NdFeB-based traction motors under common traction requirements. Surrogate-assisted design-space exploration is used to identify feasible ferrite-based configurations while considering high-speed torque deviation, active axial length, and DC bus voltage. For the ferrite motor examined, achieving low-speed traction performance comparable to the NdFeB benchmark requires substantial changes to the motor geometry and electrical operating conditions. Some differences remain in the high-speed field-weakening region. The paper therefore provides a functionally comparable design for the subsequent environmental assessment.

Presented at *2026 International Conference on Electrical Machines (ICEM).*

Paper V

Rare-earth substitution as a resource strategy for EV traction motors: Design compensation trade-offs and break-even electricity carbon intensity

This paper evaluates rare-earth substitution as a combined motor-design and life cycle problem. A functionally comparable ferrite-based motor is assessed against an NdFeB benchmark using a cradle-to-grave life cycle framework in which design-induced changes in material demand, motor mass, efficiency, and vehicle electricity consumption are propagated consistently. The results

show that the selected ferrite motor has a higher manufacturing-stage climate change impact because of the additional bulk materials required for performance compensation, but a lower use-phase electricity demand under the investigated WLTC conditions. Whether the ferrite motor has a lower life cycle climate change impact than the benchmark therefore depends on the carbon intensity of the electricity supply. The break-even value is approximately 60 g CO_2-eq/kWh. The paper concludes that the life cycle environmental effects of rare-earth substitution cannot be assessed from the magnet material alone. The assessment must also account for all changes to the motor design and their consequences for the wider electric drivetrain.

Published in *Resources, Conservation & Recycling Advances* in 2026.

8. Svensk Sammanfattning

Permanentmagnetiserade synkrona motorer används i stor utsträckning i drivsystem för elektriska fordon då de kan kombinera hög verkningsgrad med hög effekttäthet. De flesta högpresterande designer använder neodym-järn-bor-magneter, som ger en hög magnetisk belastning men samtidigt medför ett beroende av sällsynta jordartsmetaller. Ferritmagneter utgör ett kommersiellt moget och sällsynt-jordartsfritt alternativ, men deras lägre remanens och magnetiska energitäthet kräver betydande kompensation genom val av motortopologi och design.

I denna avhandling utvecklas och utvärderas en konstruktionsmetod för ferritbaserade elektriska motorer med avseende på teknisk genomförbarhet, teknisk prestanda och miljöprestanda över livscykeln. Inledningsvis utförs en litteraturgenomgång för att fastställa representativa krav på den elektriska motorn och identifiera de viktigaste design-utmaningarna för ferritbaserade maskiner.

Med utgångspunkt i de fastställda kraven och identifierade designutmaningarna utvecklas en elektrisk motor med rektangulära Y40-ferritmagneter. Elektromagnetiska och mekaniska finit element-analyser, kombineras tillsammans med avmagnetiseringsanalyser med fullskalig prototyptillverkning och experimentell verifiering vid tomgång. Den resulterande konstruktionen uppnår simulerade maximala moment- och effekttätheter på 21,24 Nm/L respektive 18,18 kW/L, vilket visar att prestandan är relevant för fordonsdrift och jämförbar med kommersiella referensnivåer. Den segmenterade rotorn minskar dessutom läckflödet mot axeln jämfört med en konventionell rotor i ett stycke. Under ett kraftigt avmagnetiserande belastningsfall vid låg temperatur minskar det simulerade momentet med cirka 14,7 %. Uppmätt luftgapflödestäthet och tomgångs-spänning visar god överensstämmelse med motsvarande finita element-beräkningar.

För att utvärdera ersättning av sällsynta jordartsmetaller under jämförbara förhållanden tillämpas den konstruktionsmetod som utvecklats genom prototypstudien på en separat ferritbaserad motor, utvecklad för jämförelse med en produktionsbaserad NdFeB-referensmotor. Den valda ferritbaserade konstruktionen kräver betydande geometrisk och elektrisk kompensation jämfört med referensmotorn, bland annat större aktiv längd och högre DC-spänning.

De resulterande förändringarna i motorns materialsammansättning och elanvändning under körcykeln inkluderas därefter i en cradle-to-grave-livscykelanalys. Den valda ferritmotorn har högre klimatpåverkan i tillverkningsfasen eftersom den lägre påverkan från magnetmaterialet inte kompenserar för den ökade användningen av elektroplåt, koppar, aluminium, ferrit och konstruktionsmaterial. Ferritmotorn uppvisar en lägre elanvändning under Worldwide Harmonized Light Vehicles Test Cycle. Den totala jämförelsen över livscykeln beror därför tydligt på elens klimatintensitet, med en brytpunkt på ungefär 60 g CO_2-ekv./kWh under de antaganden som används i studien.

Resultaten visar att ferritbaserade traktionsmotorer är tekniskt genomförbara. De tekniska och miljömässiga konsekvenserna bestäms i stor utsträckning av den omkonstruktion av motorn som krävs för att uppnå samma prestanda. Avhandlingen etablerar därför ett konstruktionskonsistent angreppssätt för utvärdering av ferritbaserade elektriska motorer, där elektromagnetisk konstruktion, funktionell ekvivalens och livscykelanalys behandlas inom ett gemensamt forskningsmässigt ramverk.

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Declaration on the Use of Generative AI

Generative AI tools, primarily ChatGPT 5.6 (OpenAI), were used during the preparation of this thesis to support language editing, including improvements to grammar, clarity, readability, and phrasing. The tools were also occasionally used to assist with translation and the refinement of text structure. All scientific content, analyses, results, interpretations, and conclusions were developed and critically reviewed by the author. The author takes full responsibility for the content of the thesis.

This use of generative AI is in accordance with the guidelines for the use of generative AI in doctoral education at the Faculty of Science and Technology, Uppsala University.

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Appendix

A. Commercial Passenger EV Benchmarking

This appendix provides the complete commercial passenger battery electric vehicle dataset used in Section 2.1.1 to establish representative peak torque and peak power benchmarks. The dataset reports vehicle-level peak torque and peak power for the passenger BEVs listed in Table A.1. For vehicles equipped with multiple traction motors, combined vehicle-level values are used.

The data were compiled primarily from manufacturer specifications and technical sources, supplemented where necessary by independent reports. For multi-motor vehicles, the primary motor generally refers to the rear-axle motor, while the secondary motor generally refers to the front-axle motor. Where the motor type is not reported or not specified in the source, the corresponding entry is left blank. N/A indicates that no secondary motor is applicable for a single-motor vehicle.

References in this appendix follow a dedicated numbering sequence listed at the end of Appendix A and are independent of the reference numbering used in the main thesis.

Abbreviations and notation used in this appendix:

- PMSM Permanent magnet synchronous motor
- IM Induction motor
- EESM Electrically excited synchronous motor
- N/A Not applicable

Table A.1. Commercial passenger BEV powertrain specifications used for benchmarking.

Model	Year	Peak Power [kW]	Peak Torque [Nm]	Primary Motor Type	Secondary Motor Type	Notes	Refs
Abarth 500e Hatchback	2023	113.7	235		N/A		[1]
Audi Q4 e-tron 45 (Sportback)	2023	210	545	PMSM	N/A		[2-3]
Audi Q4 e-tron 45 quattro (Sportback)	2023	210	664	PMSM			[3-4]
Audi Q4 e-tron 55 quattro (Sportback)	2023	250	664	PMSM			[3-4]
Audi Q8 e-tron 50 quattro (Sportback)	2022	250	664	IM	IM	Q8 50 and 55 quattro use the same motors; the front motor of SQ8 is same with the rear motor of 50/55, the rear motor of SQ8 is same as the front motor of 50/55 but doubled	[5]
Audi Q8 e-tron 55 quattro (Sportback)	2022	300	664	IM	IM	135 kW 309 Nm for front, 165 kW 355 Nm for rear	[5]
Audi SQ8 e-tron (Sportback)	2023	370	973	IM	IM	Three motors; 157 kW 355 Nm for front; 2x 138 kW 2x 309 Nm for rear	[5]
Audi e-tron GT quattro	2021	390	630	PMSM	PMSM	Hairpin	[6]
Audi e-tron GT RS	2021	475	830	PMSM	PMSM	Hairpin	[6]
BMW iX1 eDrive20 / iX2 eDrive20	2023 2024	150	250	EESM	N/A	Fifth-generation eDrive technology	[7]
BMW iX1 xDrive30 / iX2 xDrive30	2022 2023	230	494	EESM	EESM	Gen5, same motor in front and rear	[8]
BMW iX3	2021	210	400	EESM	N/A	Gen5	[9]
BMW iX xDrive40	2021	240	630	EESM	EESM	Gen5, 190 kW and 290 Nm for front, 200 kW and 340 Nm for rear.	[10]
BMW iX xDrive50	2021	385	765	EESM	EESM	Gen5, 190 kW and 365 Nm for front, 230 kW and 400 Nm for rear.	[10]
BMW iX M60	2022	455	1015	EESM	EESM	Gen5, 190 kW for front, 360 kW for rear.	[11]

Model	Year	Peak Power [kW]	Peak Torque [Nm]	Primary Motor Type	Secondary Motor Type	Notes	Refs
BMW i4 eDrive35	2022	210	400	EESM	N/A	Gen5	[12]
BMW i4 eDrive40	2021	250	430	EESM	N/A	Gen5	[13]
BMW i4 M50	2021	400	795	EESM	EESM	Gen 5, 190 kW and 430 Nm for front, 230 kW and 365 Nm for rear.	[13]
BMW i5 eDrive40 Sedan / Touring	2023 2024	250	430	EESM	N/A	Gen5	[14-15]
BMW i5 M60 xDrive Sedan / Touring	2023 2024	442	795	EESM	EESM	Gen 5, 192 kW and 365 Nm for front, 250 kW and 430 Nm for rear.	[14-15]
BMW i5 xDrive40 Sedan	2024	290	590	EESM	EESM	Gen5	[16]
BMW i7 eDrive50	2023	335	650	EESM	N/A	Gen5	[17]
BMW i7 xDrive60	2022	400	745	EESM	EESM	Gen5, 190 kW and 365 Nm for front, 230 kW and 380 Nm for rear.	[18]
BMW i7 M70 xDrive	2023	485	1100	EESM	EESM	Gen5, 190 kW and 365 Nm for front, 360 kW and 650 Nm for rear.	[19]
BYD DOLPHIN Active	2023	70	180	PMSM	N/A		[20]
BYD DOLPHIN Boost	2023	130	260	PMSM	N/A		[20]
BYD DOLPHIN Comfort / Design / ATTO 3	2023 2023 2022	150	310	PMSM	N/A		[20-21]
BYD SEAL Design	2023	230	360	PMSM	N/A		[22]
BYD SEAL Excellence AWD	2023	390	670	PMSM	IM		[22]
BYD HAN	2023	380	700	PMSM	PMSM		[23]
BYD TANG	2022	380	680	PMSM	PMSM		[24]
Cadillac Lyriq RWD	2024	255	440	PMSM	N/A	Hairpin	[25]
Cadillac Lyriq AWD	2024	375	610	PMSM	PMSM	Hairpin	[25]
Chevrolet Bolt EV	2023	150	360	PMSM	N/A		[26]

Model	Year	Peak Power [kW]	Peak Torque [Nm]	Primary Motor Type	Secondary Motor Type	Notes	Refs
Chevrolet Blazer EV	2024	212	451	IM	PMSM	Hairpin PMSM front, IM rear	[27]
Citroen e-C3	2024	83	120	PMSM	N/A		[28-29]
Citroen e-C4	2023	115	260	Hybrid Sync Motor (HSM)	N/A		[30-31]
CUPRA Born	2022	170	310	PMSM	N/A		[32]
Dacia Spring Electric 45	2022	33	125	PMSM	N/A		[33-34]
Dacia Spring Electric 65	2023	48	113	PMSM	N/A		[33-34]
DS 3 E-Tense	2022	115	260		N/A		[33]
Fiat 500e 24 kWh	2022	70	220		N/A		[33]
Fiat 500e 42 kWh	2020	87	220		N/A		[33]
Fiat 600e	2023	115	260		N/A		[33]
Fisker Ocean Extreme	2023	415	736				[133]
Ford Mustang Mach-E RWD	2023	196	430		N/A		[35]
Ford Mustang Mach-E AWD	2023	229	580				[35]
Ford Mustang Mach-E RWD Extended	2023	213	430		N/A		[35]
Ford Mustang Mach-E AWD Extended	2023	254	580				[35]
Ford Mustang Mach-E GT	2023	353	813				[35]

Model	Year	Peak Power [kW]	Peak Torque [Nm]	Primary Motor Type	Secondary Motor Type	Notes	Refs
Ford Mustang Mach-E GT Performance	2023	353	860				[35]
Genesis GV60 Standard 2WD	2022	168	350		N/A		[36]
Genesis GV60 Standard AWD	2022	234	605			74 kW 255 Nm front, 160 kW 350 Nm rear	[36]
Genesis GV60 Performance AWD	2022	360	700			Same for front and rear	[36]
Genesis GV70	2022	320	700			Same for front and rear	[37]
Genesis G80	2022	272	700			Same for front and rear	[38]
GWM ORA 3	2024	126	250		N/A		[39]
Honda e:Ny1	2023	150	310				[40]
Hongqi E-HS9 Comfort	2022	320	300			160 kW + 160 kW	[41]
Hongqi E-HS9 Premium / Exclusive	2022	405	450			160 kW + 245 kW	[41]
Hyundai Kona	2024	150	255	PMSM	N/A		[42]
Hyundai IONIQ 5 / 6 RWD	2022	168	350	PMSM	N/A		[43-44]
Hyundai IONIQ 5 / 6 AWD	2022	239	605	PMSM	PMSM		[43-44]
Jaguar I-Pace EV400	2023	294	696	PMSM	PMSM		[45]
Jeep Avenger Electric	2023	115	260	PMSM	N/A		[46]
Kia e-Soul	2022	150	395	PMSM	N/A		[47]
Kia Niro EV	2022	150	255	PMSM	N/A		[48]
Kia EV6 RWD	2021	168	350	PMSM	N/A		[49]
Kia EV6 AWD	2021	239	605	PMSM	PMSM		[49]

Model	Year	Peak Power [kW]	Peak Torque [Nm]	Primary Motor Type	Secondary Motor Type	Notes	Refs
Kia EV6 GT	2022	430	740				[50]
Kia EV9 RWD	2023	149	350	PMSM	N/A		[51]
Kia EV9 AWD	2023	282	700	PMSM	PMSM		[51]
Lexus UX 300e	2023	150	300	PMSM	N/A		[52]
Lexus RZ 450e	2023	233	434				[53]
Lotus Eletre / Eletre S / Emeya / Emeya S	2023 2024	450	710	PMSM	PMSM		[54-55]
Lotus Eletre R / Emeya R	2023 2024	675	985	PMSM	PMSM		[54-55]
Lucid Air Pure	2023	320	550				[56]
Lucid Air Touring	2023	462	1200				[57]
Lucid Air Grand Touring	2023	610	1200				[58]
Lucid Air Sapphire	2024	920	1940				[56]
Maserati GranTurismo Folgore	2024	560	1350	PMSM	PMSM	Three motors, 800V	[59]
Maserati Grecale Folgore	2024	368	800			400V	[60]
Mazda MX-30	2022	107	270				[61]
Mercedes EQA / EQB 250 / 250+	2023	140	385	IM	N/A		[62-63]
Mercedes EQA / EQB 300 4MATIC	2023	168	390	IM	IM		[62-63]
Mercedes EQA / EQB 350 4MATIC	2023	215	520	IM	IM		[62-63]
Mercedes EQE 300	2023	180	550	PMSM	N/A		[64-65]

Model	Year	Peak Power [kW]	Peak Torque [Nm]	Primary Motor Type	Secondary Motor Type	Notes	Refs
Mercedes EQE 350	2023	215	565	PMSM	N/A		[64-65]
Mercedes EQE AMG 53 4MATIC+ / SUV	2023	460	950	PMSM	PMSM		[64-65]
Mercedes EQE 350 4MATIC SUV	2023	215	765	PMSM	PMSM		[65-66]
Mercedes EQE 500 4MATIC SUV	2023	300	858	PMSM	PMSM		[65-66]
Mercedes EQS 450+	2023	265	568	PMSM	N/A		[65, 67]
Mercedes EQS AMG 53 4MATIC+	2023	484	950	PMSM	PMSM		[65, 67]
Mercedes EQS 450 4MATIC SUV	2023	265	800	PMSM	PMSM		[65, 68]
Mercedes EQS 580 4MATIC SUV	2023	400	858	PMSM	PMSM		[65, 68]
Mercedes EQV 300	2024	150	366	IM	N/A		[69]
MG MG4 Standard	2022	125	250	PMSM	N/A		[70]
MG MG4 Luxury	2022	150	250	PMSM	N/A		[70]
MG MG4 Trophy Extended Range	2023	180	350	PMSM	N/A		[70]
MG MG4 XPOWER	2023	320	600	PMSM	PMSM		[70]
MG MG5 Comfort	2022	130	280	PMSM	N/A		[71]
MG MG5 Luxury Long Range	2022	115	280	PMSM	N/A		[71]
MG ZS EV	2021	115	280	PMSM	N/A		[72]
MG Marvel R Luxury	2021	132	410	PMSM	N/A		[73]

Model	Year	Peak Power [kW]	Peak Torque [Nm]	Primary Motor Type	Secondary Motor Type	Notes	Refs
MG Marvel R Performance	2021	212	665	PMSM	PMSM		[73]
Mini Cooper E	2023	135	290		N/A		[74]
Mini Cooper SE	2023	160	330		N/A		[75]
Mini Countryman E	2024	150	250	EESM	N/A		[76]
Mini Countryman SE ALL4	2024	230	494	EESM	EESM		[76]
NIO ET5 / EL6	2023	360	700	PMSM	IM	150 kW 280 Nm IM at front and 210 kW 420 Nm PMSM at rear.	[77-78]
NIO ET7 / EL7	2022	480	850	IM	PMSM	180 kW 350 Nm PMSM at front and 300 kW 500 Nm IM at rear.	[79-80]
Nissan Leaf	2022	110	320	PMSM	N/A		[81]
Nissan Leaf e+	2022	160	340	PMSM	N/A		[81]
Nissan Ariya	2022	160	300	EESM	N/A		[82]
Nissan Ariya +	2022	178	300	EESM	N/A		[82]
Nissan Ariya e-4ORCE	2022	250	560	EESM	EESM		[82]
Nissan Ariya e-4ORCE+	2023	290	600	EESM	EESM		[82]
Opel Corsa-e / Mokka-e	2023	115	260		N/A		[83-84]
Peugeot e-208 50 kWh / e-2008 50 kWh	2023	100	260		N/A		[85-86]
Peugeot e-208 51 kWh / e-2008 54 kWh	2023	115	260		N/A		[85-86]
Peugeot e-308	2023	115	270		N/A		[87]
Peugeot e-3008	2023	157	345	PMSM	N/A		[88]

Model	Year	Peak Power [kW]	Peak Torque [Nm]	Primary Motor Type	Secondary Motor Type	Notes	Refs
Polestar 2 Standard Range Single Motor	2023	200	490	PMSM	N/A		[89]
Polestar 2 Long Range Single Motor	2023	220	490	PMSM	N/A		[89]
Polestar 2 Long Range Dual Motor	2023	310	740	PMSM	IM	110 kW 250 Nm IM at front and 200 kW 490 Nm PMSM at rear.	[89]
Polestar 2 Long Range Performance	2023	350	740	PMSM	IM	135 kW 250 Nm IM at front and 215 kW 490 Nm PMSM at rear.	[89]
Polestar 3 Long Range Dual Motor	2022	360	840	PMSM	PMSM		[90]
Polestar 3 Long Range Performance	2022	380	910	PMSM	PMSM		[90]
Polestar 4 Single Motor	2024	200	343	PMSM	N/A		[91]
Polestar 4 Dual Motor	2024	400	686	PMSM	PMSM		[91]
Rolls-Royce Spectre	2023	430	900	EESM	EESM	190 kW 365 Nm front, 360 kW 710 Nm rear	[92]
Porsche Taycan / Sport Turismo	2024	320	420	PMSM	N/A	800 V for all	[93-94]
Porsche Taycan 4S / Turismo	2024	440	710	PMSM	PMSM		[95-96]
Porsche Taycan Turbo / Turismo	2024	650	940	PMSM	PMSM		[97-98]
Porsche Taycan Turbo S / Turismo	2024	700	1110	PMSM	PMSM		[99-100]
Porsche Taycan Turbo GT	2024	760	1340	PMSM	PMSM		[101]
Porsche Taycan 4 Cross Turismo	2024	320	610	PMSM	PMSM		[102]

Model	Year	Peak Power [kW]	Peak Torque [Nm]	Primary Motor Type	Secondary Motor Type	Notes	Refs
Porsche Macan 4 Electric	2024	300	650	PMSM	PMSM		[103]
Porsche Macan Turbo Electric	2024	470	1130	PMSM	PMSM		[104]
Renault Twingo	2020	60	160	EESM	N/A		[105-106]
Renault Zoe R110	2018	80	225	EESM	N/A		[106-107]
Renault Zoe R135	2019	100	245	EESM	N/A		[106-107]
Renault Kangoo	2022	90	245	EESM	N/A		[106, 108]
Renault Megane EV40	2022	96	250	EESM	N/A		[106, 109]
Renault Megane EV60	2022	160	300	EESM	N/A		[106, 109]
Renault Scenic EV60	2023	125	280	EESM	N/A		[110]
Renault Scenic EV87	2023	160	300	EESM	N/A		[110]
Skoda Enyaq 60	2023	132	310	PMSM	N/A		[111-112]
Skoda Enyaq 85	2023	210	545	PMSM	N/A		[111-112]
Skoda Enyaq RS	2023	250	679	PMSM	IM	545 Nm rear, 134 Nm front	[111-112]
Smart #1 Pro / Premium	2023	200	343		N/A		[113]
Smart #1 Brabus	2023	315	584				[113]
Smart #3	2024	200	343		N/A		[114]

Model	Year	Peak Power [kW]	Peak Torque [Nm]	Primary Motor Type	Secondary Motor Type	Notes	Refs
Smart #3 Brabus	2024	315	543				[114]
Subaru Solterra	2022	160	336			Same for front and rear	[115]
Tesla Model 3	2023	190	360	PMSM	N/A		[116]
Tesla Model 3 Long Range Dual Motor	2023	290	511	PMSM	IM	87 kW 163 Nm front IM, 203 kW 348 Nm rear PMSM	[116]
Tesla Model Y	2023	189	361	PMSM	N/A		[117]
Tesla Model Y Long Range Dual Motor	2022	282	508	PMSM	IM	IM front and PMSM rear	[118]
Tesla Model Y Performance	2022	335	674	PMSM	IM	IM front and PMSM rear	[119]
Tesla Model S Plaid / X Plaid	2022	750	1424	PMSM	PMSM	PMSM x3	[120-121]
Toyota bZ4X FWD	2022	150	266	PMSM	N/A		[122]
Toyota bZ4X AWD	2022	158	336			Same for front and rear	[123]
Volkswagen ID.3 Pro / Pro S	2023	150	310	PMSM	N/A		[124]
Volkswagen ID.4 Pure	2023	125	310	PMSM	N/A		[33]
Volkswagen ID.4 Pro / ID.5 Pro	2023	210	545	PMSM	N/A	APP550 motor	[33, 125]
Volkswagen ID.4 Pro 4MOTION	2023	210	679	PMSM		APP550	[33, 125]
Volkswagen ID.4 GTX / ID.5 GTX	2023	250	679	PMSM		APP550	[33, 125]
Volkswagen ID.7 Pro	2023	210	545	PMSM	N/A	APP550	[33, 125]
Volkswagen ID. Buzz Pro	2022	150	310	PMSM			[126]

Model	Year	Peak Power [kW]	Peak Torque [Nm]	Primary Motor Type	Secondary Motor Type	Notes	Refs
Volvo EX30 Single Motor	2023	200	343	PMSM	N/A		[127]
Volvo EX30 Twin Motor	2023	315	543	PMSM	PMSM	115 kW 200 Nm for front, 200 kW 343 Nm for rear	[127]
Volvo EC40 / EX40 Single Motor	2024	175	420	PMSM	N/A		[128]
Volvo EC40 / EX40 Single Motor ER	2024	185	420	PMSM	N/A		[128]
Volvo EC40 / EX40 Twin Motor	2024	325	670	PMSM	PMSM		[128]
Volvo EX90 Twin Motor	2022	300	770	PMSM	PMSM		[129-130]
Volvo EX90 Twin Performance	2022	380	910	PMSM	PMSM		[129-130]
XPENG P7 RWD	2023	203	440	PMSM	N/A		[131]
XPENG P7 AWD	2023	348	757	PMSM	IM	145 kW IM at front, 203 kW PMSM at rear	[131]
XPENG G9 RWD Standard Range	2023	230	430	PMSM	N/A		[132]
XPENG G9 AWD Long Range	2023	405	717	PMSM	IM	175 kW 287 Nm IM for front, 230 kW 430 Nm PMSM for rear	[132]
Zeekr 001 RWD	2023	200	343		N/A		[33]
Zeekr 001 AWD	2023	400	686				[33]
Zeekr X RWD	2023	200	343		N/A		[33]
Zeekr X AWD	2023	315	543				[33]

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B. Prototype Assembly and Manufacturing Drawings

This appendix presents additional assembly views and the principal manufacturing drawings used to fabricate the prototype motor.

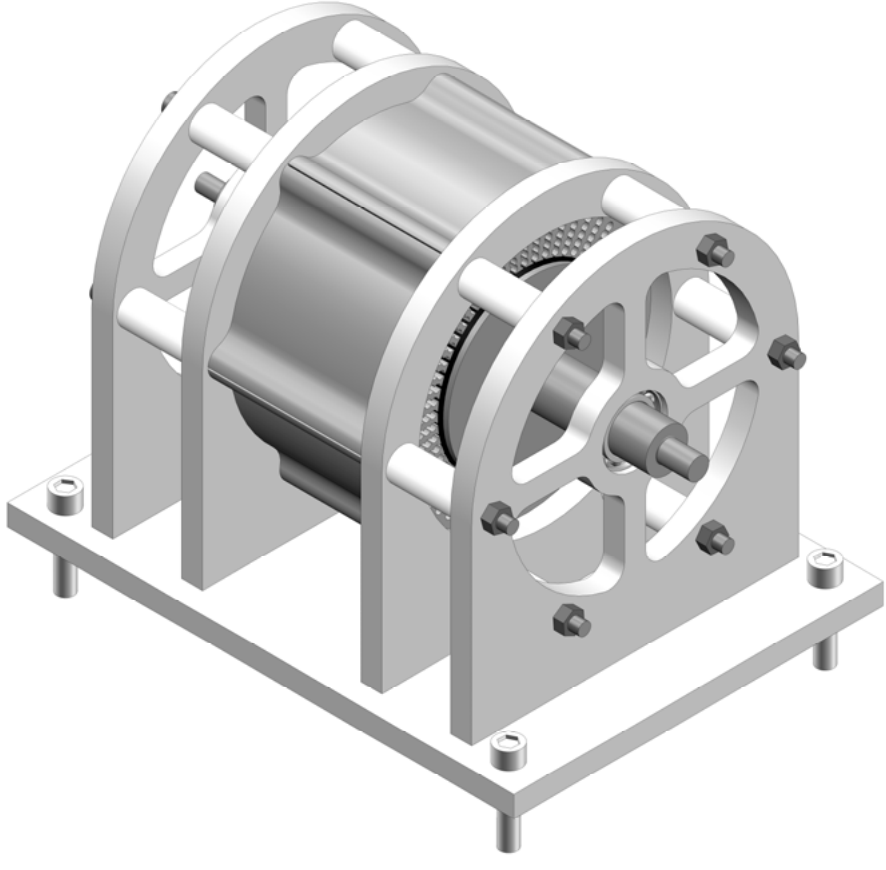


Figure B.1. Prototype motor assembly.

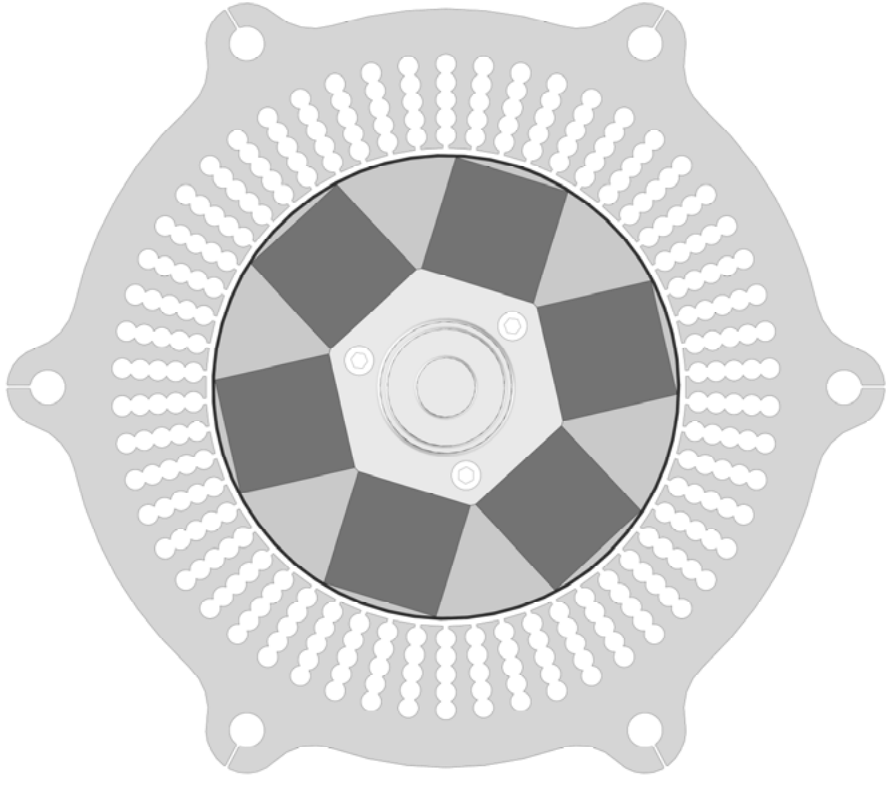


Figure B.2. Prototype motor cross-sectional view without supporting frames.

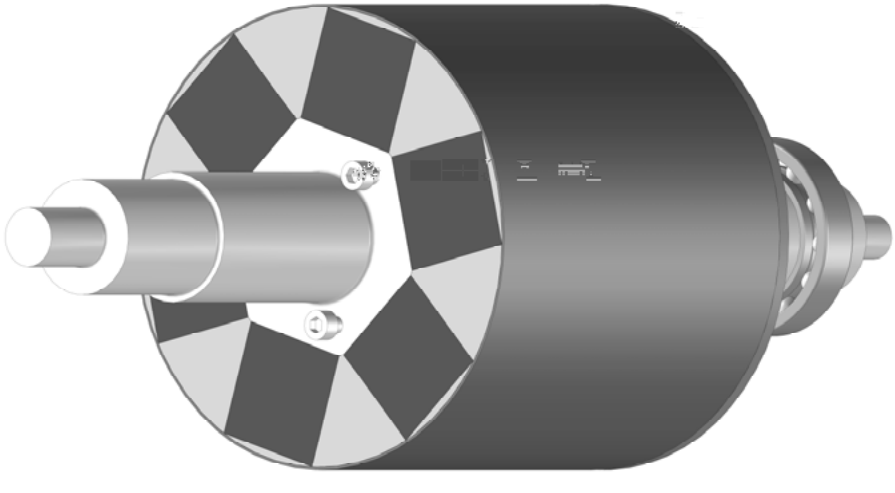


Figure B.3. Assembled segmented rotor with carbon-fiber retaining sleeve.

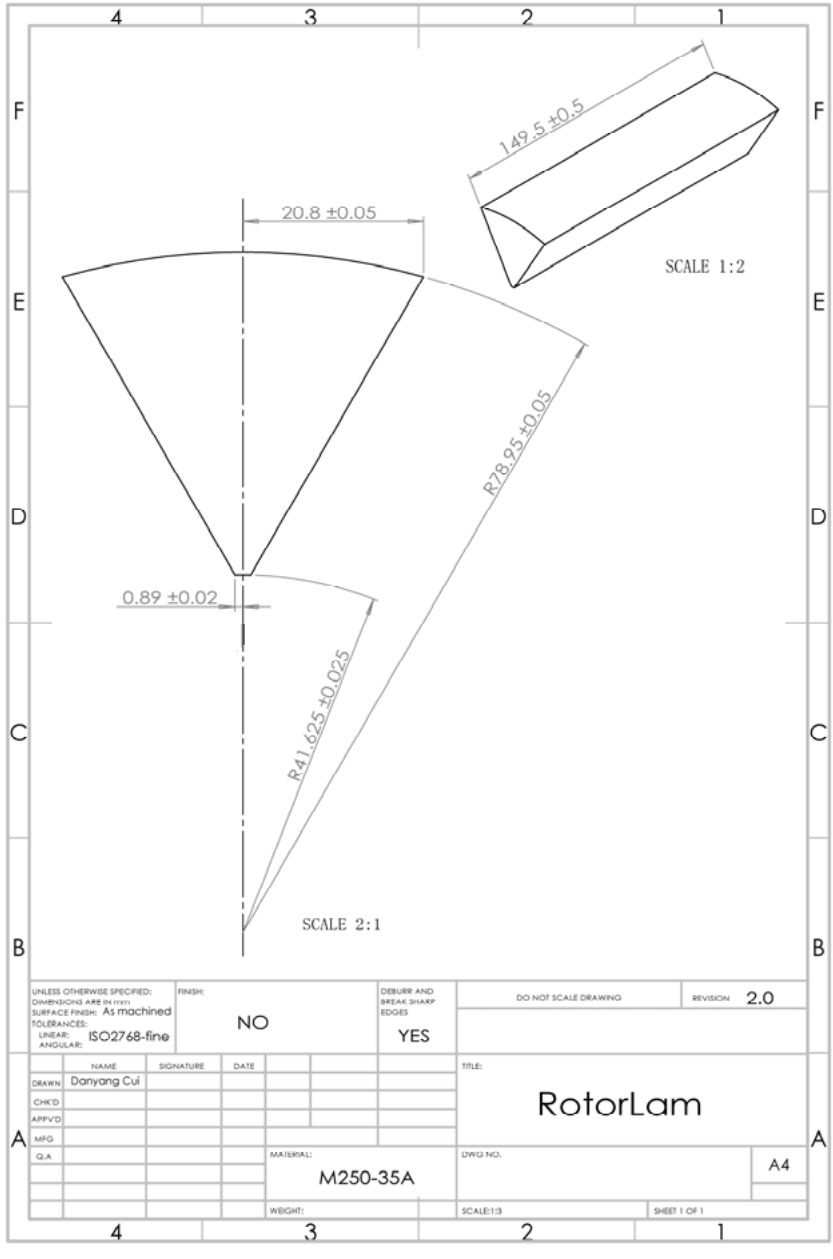


Figure B.4. Manufacturing drawing of the rotor lamination.

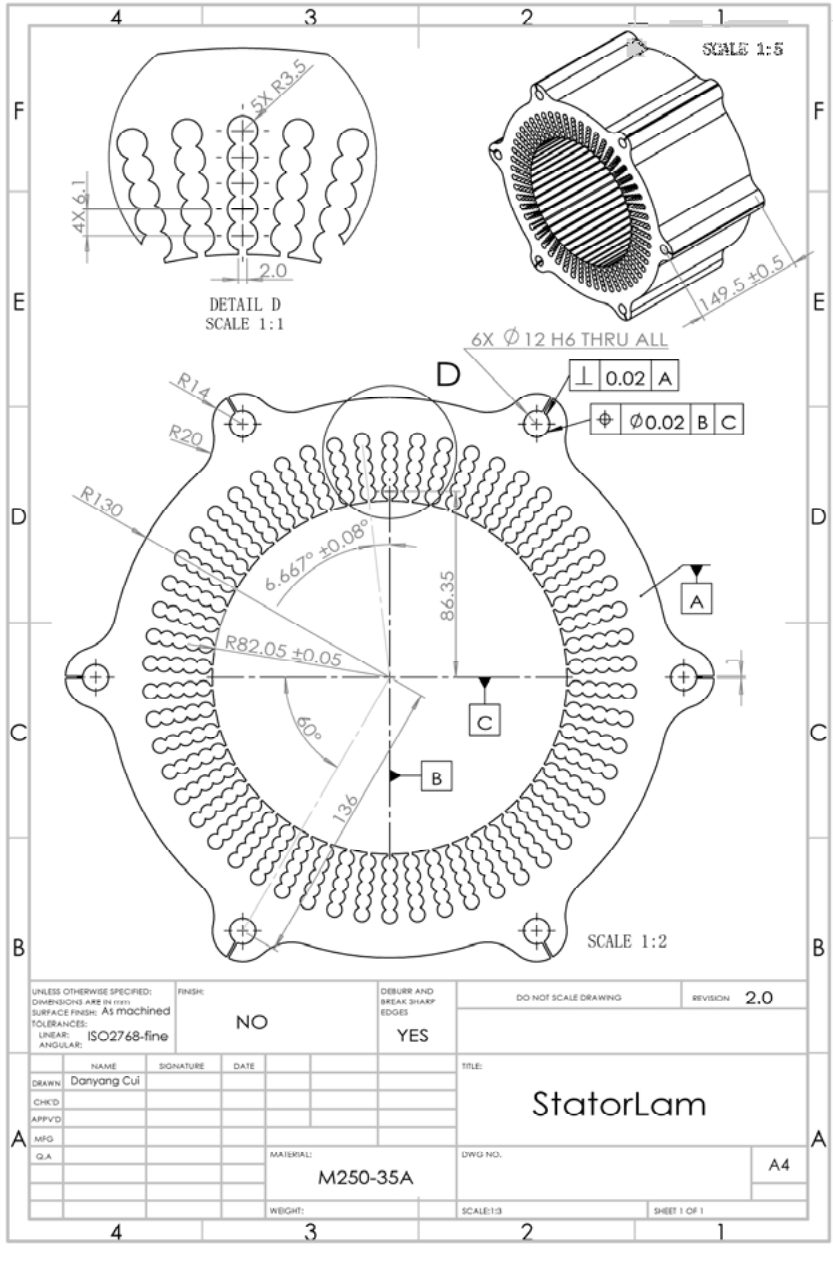


Figure B.5. Manufacturing drawing of the stator lamination.

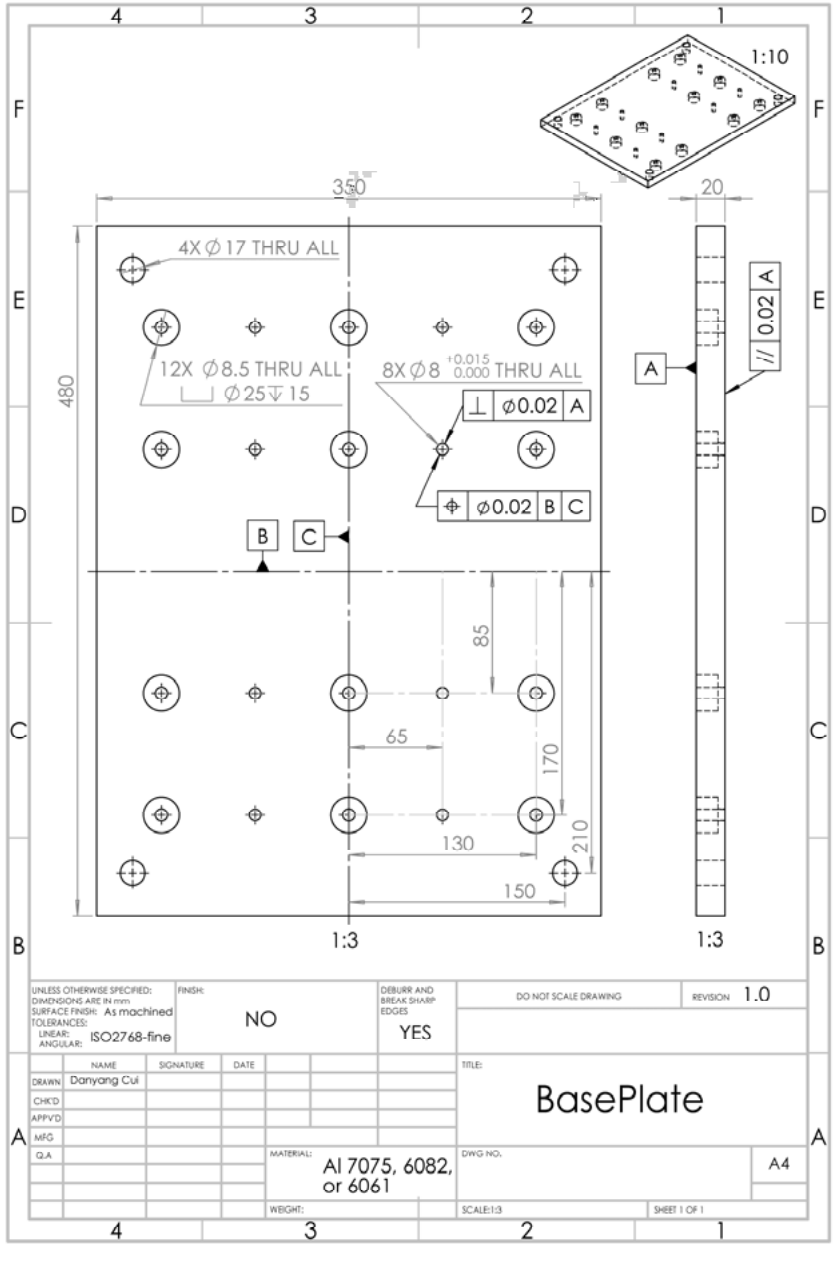


Figure B.6. Manufacturing drawing of the base plate.

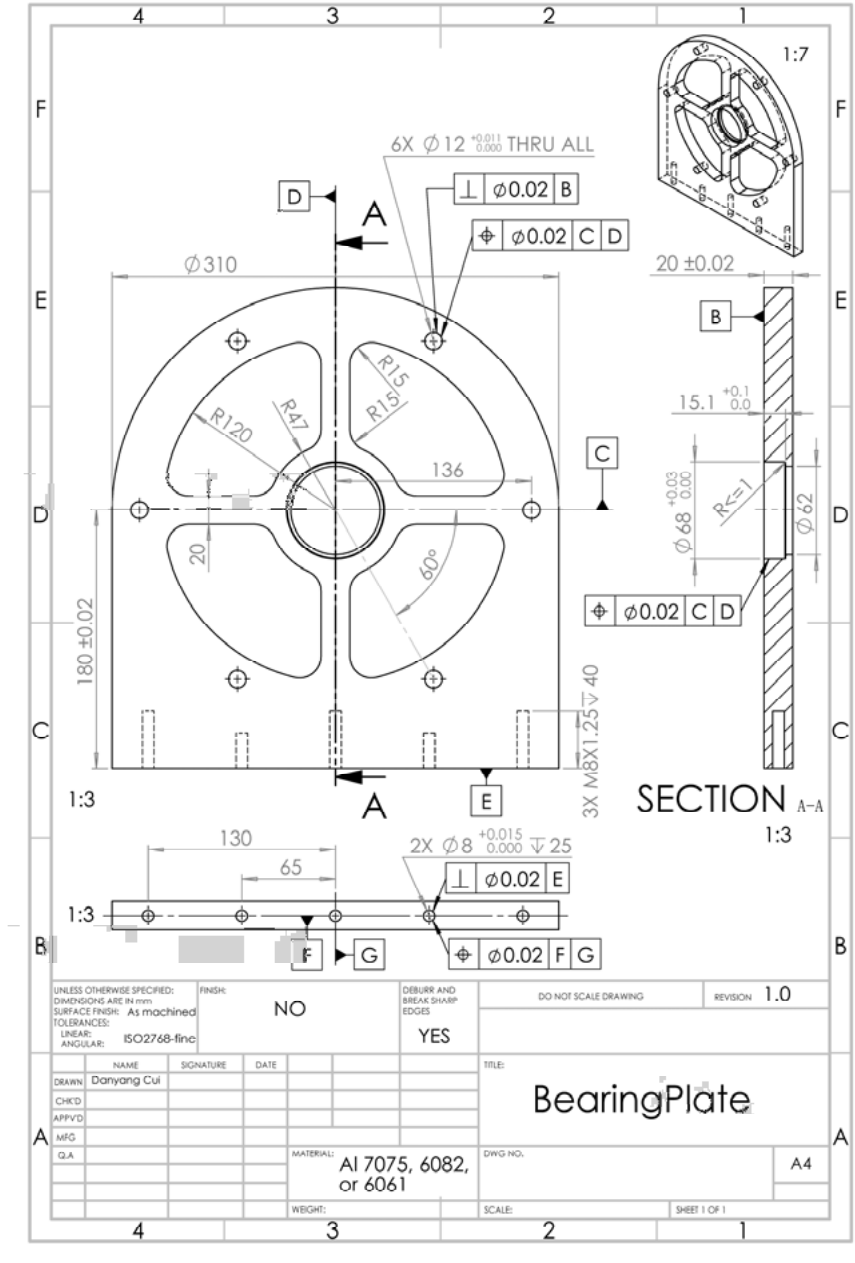


Figure B.7. Manufacturing drawing of the bearing plate.

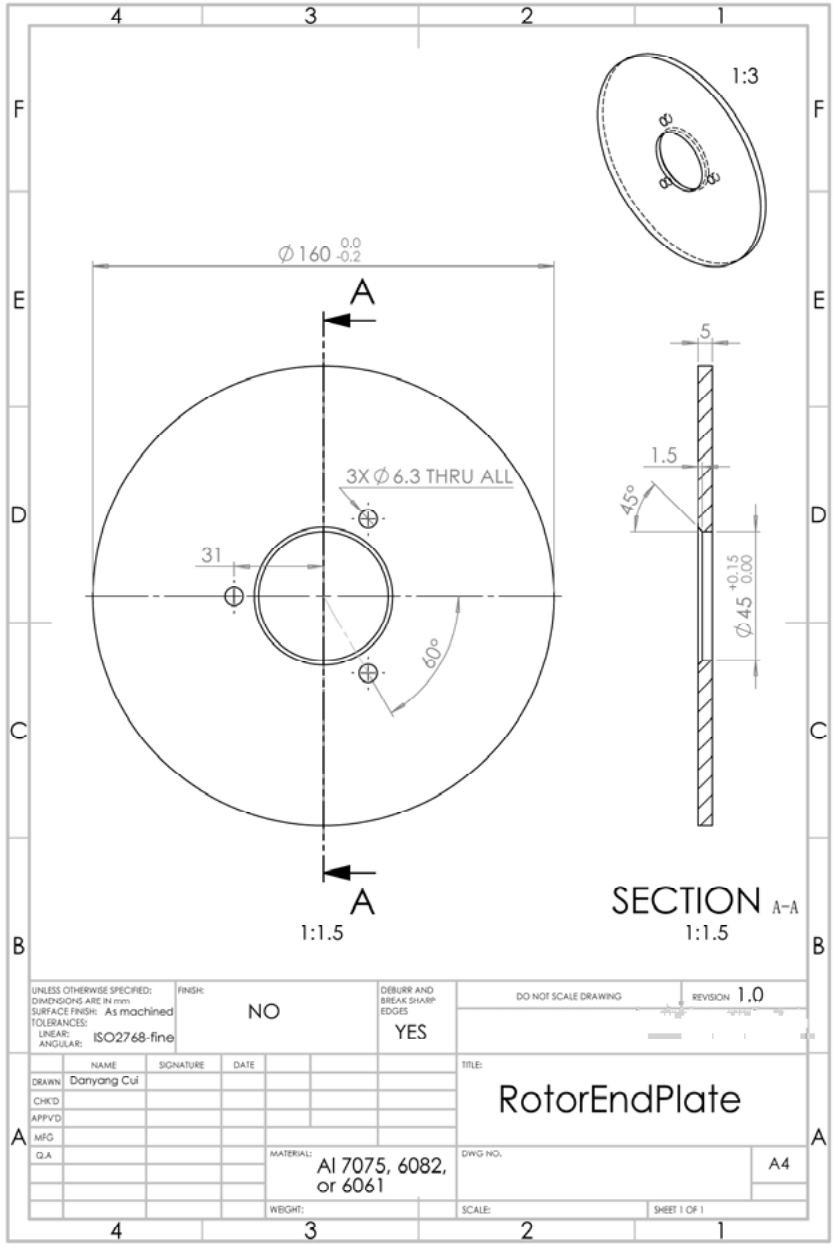


Figure B.8. Manufacturing drawing of the rotor end plate.

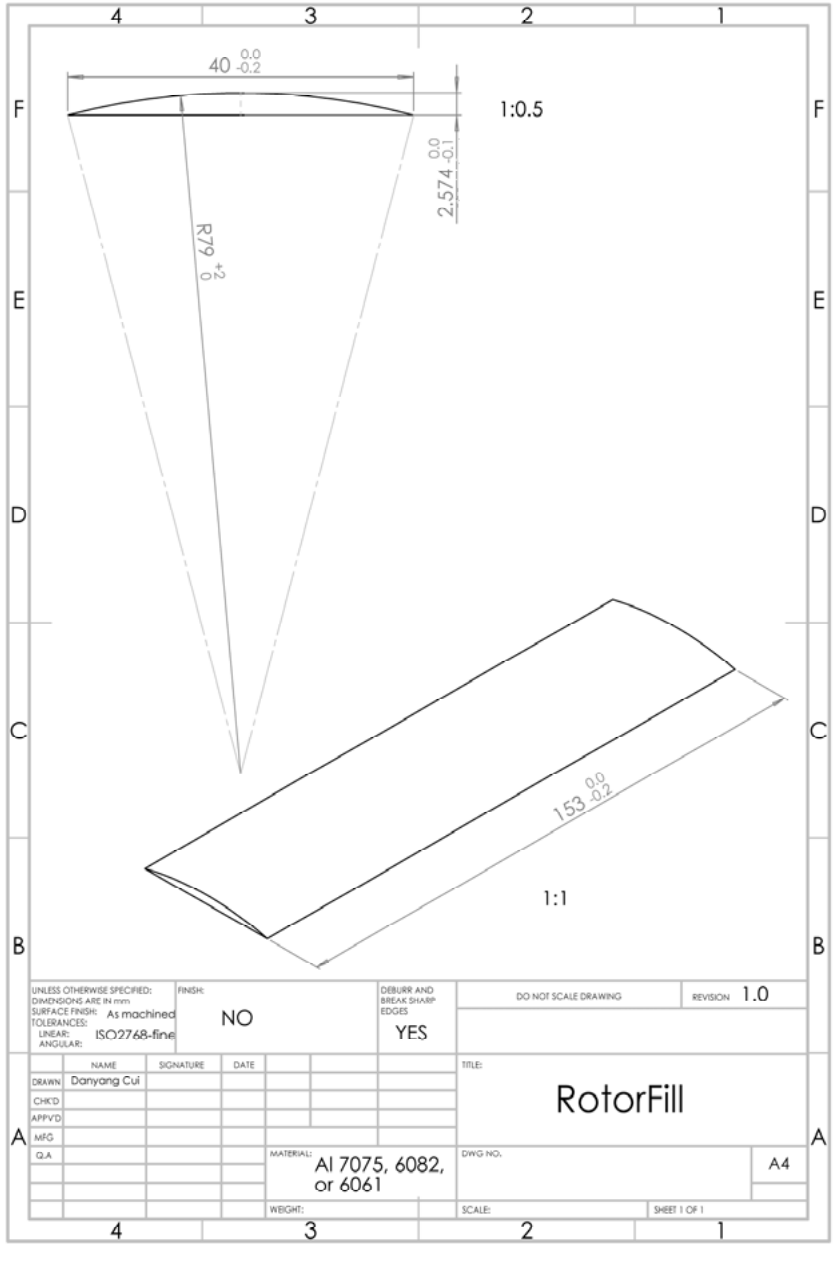


Figure B.9. Manufacturing drawing of the rotor filler components.

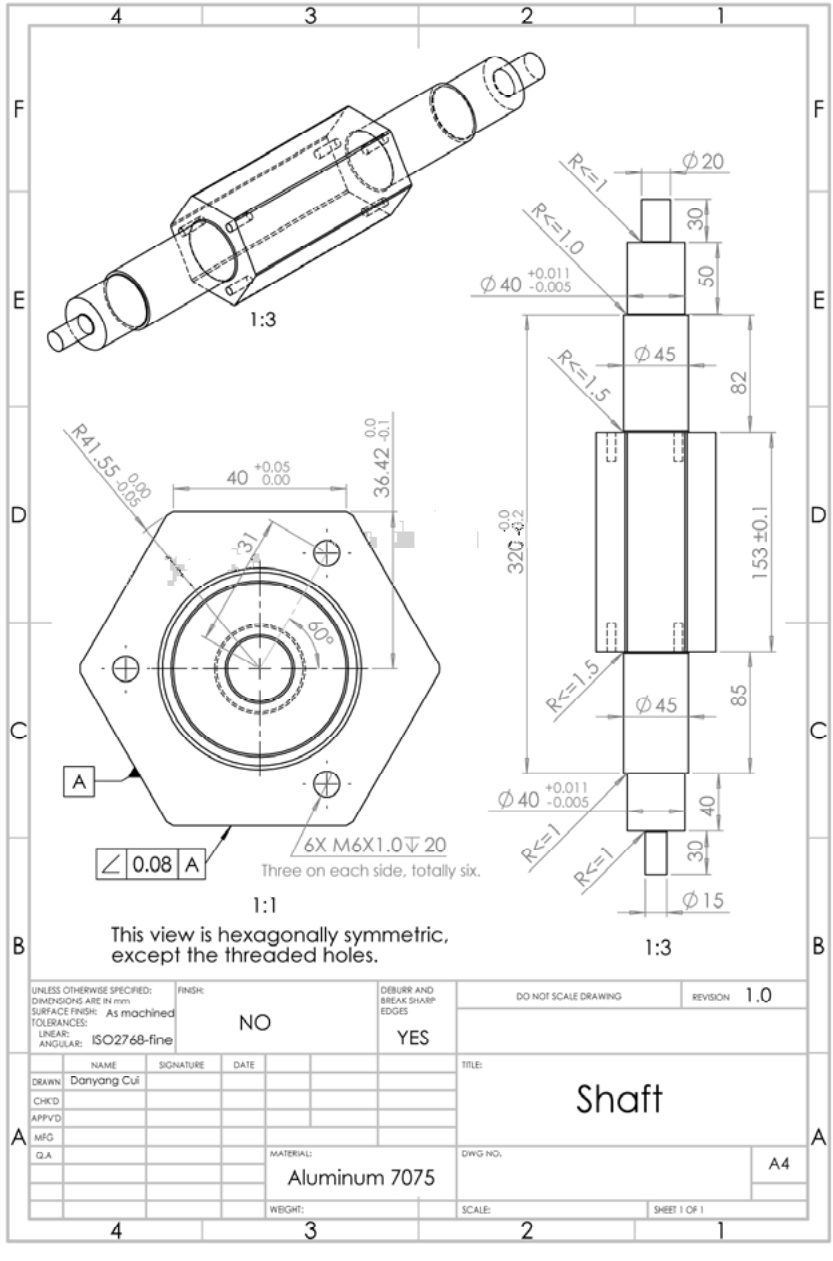


Figure B.10. Manufacturing drawing of the shaft.

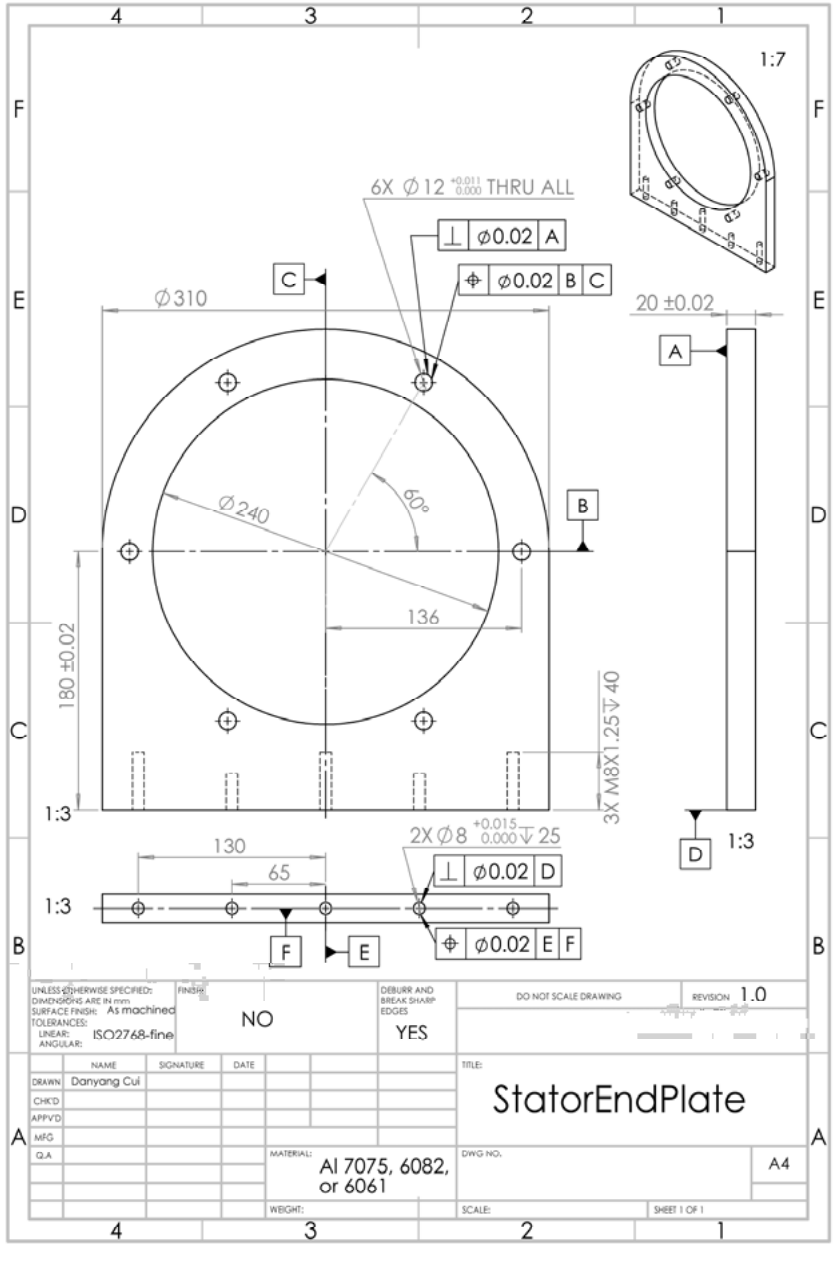


Figure B.11. Manufacturing drawing of the stator end plate.

C. Component-Level Material Inventory of the Two Motors

This appendix presents the material inventories for the benchmark NdFeB-based motor and the selected ferrite-based motor. Both inventories use the same component breakdown and provide the foreground life cycle inventory data for the manufacturing-stage modelling described in Section 4.4.2.

The inventory structure follows the scalable life cycle inventory framework for automotive traction machines developed by Nordelöf et al. [87], [88]. The framework is used as a baseline and extended where necessary to represent design-specific features of the two motors, including the longer active axial length of the ferrite-based motor and its carbon-fiber composite rotor sleeve.

Material masses are reported on a per-motor basis and are derived from the detailed motor geometries together with geometry-based estimations where required. Identical component definitions are applied to both motor concepts to maintain comparability.

The inventory also includes polymers, coatings, and insulation materials for completeness and transparency. These materials account for small fractions of the total motor mass and make only minor contributions to the total climate change impact.

Table C.1. Component-level material inventory of the benchmark and selected ferrite-based motors. From **Paper V** [94] under the CC BY 4.0 license.

Type	Part	Material	Benchmark motor	Ferrite-based motor	Unit
Active parts	Core laminations	Electrical steel	26.09	35.34	kg
		Phenolic resin (coating)	77.08	101.34	g
	Magnet wires	Copper	6.01	7.70	kg
		Enamel (coating)	272.40	349.07	g
	Permanent magnets	Nd(Dy)FeB / Ferrite	1.81	8.46	kg
		Nickel (coating only for NdFeB)	18.55	0.00	g
Stator insulation	Slot liner	Polyethylene terephthalate	138.26	216.99	g
	Phase-to-phase insulation tape	Mica	26	26	g
		Glass fiber (cloth)	7	7	g
		Silicone (bond)	4	4	g
	Impregnation	Epoxy resin	521.35	715.73	g
	Phase conductor insulation	Silicone (crosslinked)	28	28	g
	Lacing cord	Nylon	6	6	g
Rotor fastening	Rotor endplates	Stainless steel	596.33	596.33	g
	Magnet-fixation resin	Methacrylate ester resin	1367.48	0.00	g
	Sleeve	Prepreg carbon fiber	0	0.16	kg
Housing	Housing body	Aluminum	14.39	18.13	kg
		Paint/varnish	180	207	g
Shaft and bearings	Shaft and rotor fill	Low-alloy medium carbon steel	6.08	5.67	kg
	Bearings	Low-alloy medium carbon steel	280	280	g
Other	Resolver	Electrical steel	42	42	g
		Copper	25	25	g
		Enamel (coating)	1	1	g
		Polybutylene terephthalate	8	8	g
	Terminal block frame	Polybutylene terephthalate	130	130	g
	Fasteners and plates	Low-alloy medium carbon steel	240	240	g
		Zinc (galvanization)	10	10	g
	Connector lugs	Copper	51	51	g
TOTAL			58.41	78.50	kg

D. Estimation of Housing Mass

This appendix describes the method used to estimate the aluminum housing mass of the two traction motors analyzed in this study. Because detailed housing geometries and disassembly data for modern traction motors are rarely publicly available, a transparent and design-consistent estimation method is required for the comparative LCA.

In the scalable life cycle inventory framework proposed by Nordelöf et al. [87], [88], cast aluminum motor housing mass is estimated using a power-based scaling relationship. This approach is suitable for broad parametric studies across different power classes, but it is less appropriate when motors with comparable power ratings have substantially different geometries.

In the present study, the benchmark and ferrite-based motors provide comparable traction-level peak power, 260 kW and 271 kW, respectively, while their axial active lengths differ substantially. The benchmark motor has an active length of 134 mm, whereas the selected ferrite-based motor has an active length of 216 mm. A power-based scaling approach would therefore predict similar housing masses for the two motors and would not adequately capture the effect of the increased axial length of the ferrite-based design.

To address this limitation, a geometry-based housing mass model is adopted. The housing is approximated as a hollow cylindrical structure with a uniform effective wall thickness. Under this simplified representation, the housing mass is expressed as the product of the housing surface area, effective wall thickness, and aluminum density.

The housing mass can therefore be written as

$$\mathrm{HousingMass} = \left(2\pi\left(\frac{\mathrm{SOD}}{2}\right)^2 + \pi \cdot \mathrm{SOD} \cdot \mathrm{length}\right) \cdot \mathrm{thickness} \cdot \mathrm{density} \tag{D.1}$$

where SOD denotes the stator outer diameter, length denotes the axial active length, thickness denotes the effective housing wall thickness, and density denotes the density of aluminum.

For practical fitting, this expression can be reformulated as

$$\mathrm{HousingMass} = A \cdot \mathrm{SOD}^2 + B \cdot \mathrm{SOD} \cdot \mathrm{length} + C \tag{D.2}$$

where A, B, and C are fitting coefficients that implicitly capture the combined effects of housing thickness, material density, and geometric simplifications.

The proposed formulation (D.2) does not explicitly include non-active axial length such as end-winding regions. This simplification is based on the observed relationship between stator outer diameter and end-winding volume. A larger stator diameter generally allows larger slot and conductor cross-sectional areas, which can increase the volume of the end windings. The inclusion of the SOD^2 term therefore implicitly accounts for variations associated with non-active length.

Reference data for fitting the coefficients are obtained from published traction motor disassembly studies. Table 2.7 on page 46 in the literature [114] provides stator outer diameter, axial active length, and casing mass for four traction motors. These auxiliary components were assigned a conservative total mass of 1 kg based on publicly available industrial datasheets and component descriptions [115], [116]. This assumed mass was then subtracted from the reported casing masses. The reported casing masses include auxiliary components such as the resolver and oil pump, which are not included in the housing definition adopted here.

After this adjustment, four data points are initially available for fitting. One motor, corresponding to the 2004 Prius, exhibits a casing mass that deviates significantly from the remaining data and results in non-physical negative coefficients when included in the regression. This data point is therefore excluded from the fitting procedure. Using the remaining three data points, the coefficients A, B, and C are uniquely determined, resulting in positive values for all coefficients and ensuring that housing mass increases monotonically with stator outer diameter and axial active length. These three points determine the following fitted relation:

$$\begin{aligned}\mathrm{HousingMass} = {} & 9.8859 \cdot 10^{-5} \cdot \mathrm{SOD}^2 + 2.0494 \cdot 10^{-4} \cdot \mathrm{SOD} \\ & \cdot \mathrm{length} + 3.4958\end{aligned} \tag{D.3}$$

where SOD and length have the unit mm and HousingMass has the unit kg.

The fitted model (D.3) is applied to the benchmark and ferrite-based motors using a stator outer diameter of 221 mm and active axial lengths of 134 mm and 216 mm, respectively. The estimated housing masses are 14.39 kg and 18.13 kg, corresponding to approximately 25% and 23% of the respective total motor masses. These fractions lie within the range reported in previous traction-motor studies, including approximately 15.5% calculated from [82] and 32.1% reported for the Nd-PMSM in [89], providing a plausibility check for the adopted geometry-based estimation.

E. Copper Length and Phase Resistance Calculations

Both motors use the same three-phase distributed single-layer winding configuration. A simplified representation of one turn is shown in Figure E.1. Only one axial side is illustrated, while the opposite side follows the same geometry. The slot count shown in the schematic is illustrative and does not correspond to the actual 54-slot stator.

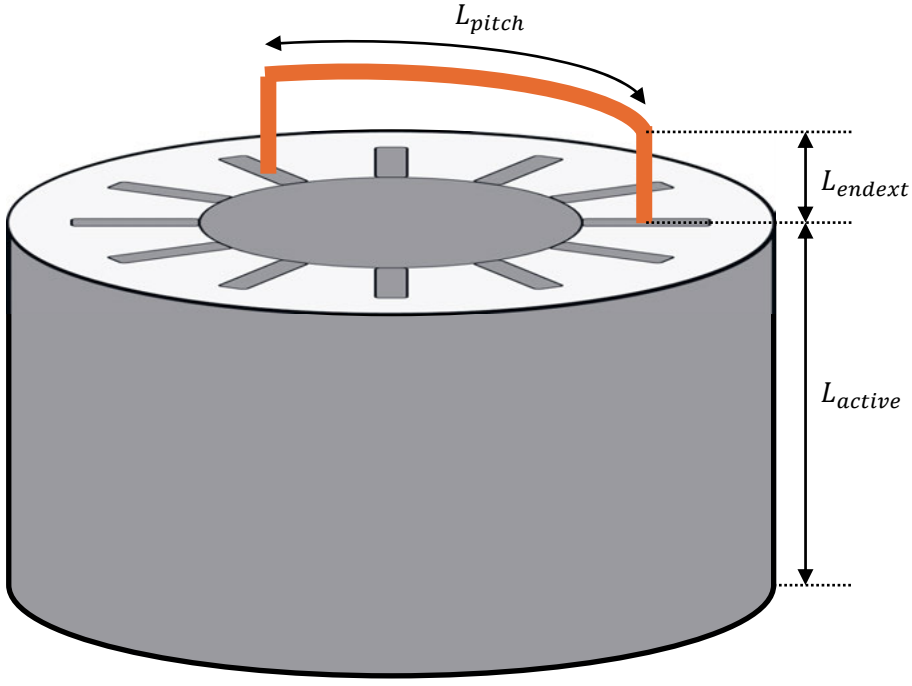


Figure E.1. Simplified geometry used to estimate the conductor length of one turn. From **Paper V** [94] under the CC BY 4.0 license.

Following the winding-geometry approximation in [100], [101], the circumferential coil-pitch length is estimated as

$$L_{pitch} = 1.2 \cdot \frac{2\pi r_{slot,center} N_{pitch}}{N_{slots}} \tag{E.1}$$

where $r_{slot,center}$ denotes the radius of the slot center, N_{pitch} denotes the coil pitch expressed in number of slots, and N_{slots} denotes the total number of stator slots. In the present study, the stator cross-section of the two motors is identical. Both motors therefore share the same values of $r_{slot,center} \approx 85.4\ mm$, $N_{pitch} = 9$, and $N_{slots} = 54$.

The end-winding length on one side of the motor is then expressed as

$$L_{ew} = 2 \cdot L_{end,extension} + L_{pitch} \tag{E.2}$$

where $L_{end,extension}$ represents the straight conductor length extending from the slot before bending into the end-winding region. A value of 25 mm is adopted following [101]. This parameter is treated as a representative design approximation rather than an exact reconstruction of the production winding geometry.

The total conductor length of one turn is therefore given by

$$L_{turn} = 2(L_{active} + L_{ew}) \tag{E.3}$$

where L_{active} denotes the axial active length of the stator.

The total conductor length per phase winding is then calculated as

$$\begin{aligned} L_{phase} &= L_{turn} \cdot N_{turns} \cdot \frac{N_{slots}}{N_{phase} \cdot N_{parallel} \cdot 2} \cdot N_{parallel} \\ &= L_{turn} \cdot N_{turns} \cdot \frac{N_{slots}}{6} \end{aligned} \tag{E.4}$$

where N_{turns} denotes the number of turns per phase, which is 5, N_{phase} denotes the number of phases, which is 3, and $N_{parallel}$ denotes the number of parallel paths, which is 3. The term $\frac{N_{slots}}{N_{phase} \cdot N_{parallel} \cdot 2}$ represents half of the number of slots passed by one parallel path.

Equation (E.4) gives the total conductor length of all parallel paths within one phase and is therefore appropriate for estimating copper mass. It cannot be used directly to calculate phase resistance because the electrical resistance of the parallel paths must be combined accordingly.

The total copper conductor length of the three-phase winding, including both active and end-winding regions, is then obtained as

$$L_{total} = 3 \cdot L_{phase} \tag{E.5}$$

The above expressions assume a single strand per conductor. When multiple strands are used, the total strand length is obtained by multiplying the resulting total length L_{total} by the number of strands.

Finally, the phase resistance accounting for parallel paths is calculated as

$$R_{phase} = \frac{\rho_{cu} \cdot N_{turn} \cdot L_{turn} \cdot \frac{N_{slots}}{N_{phase} \cdot N_{parallel} \cdot 2}}{N_{parallel} \cdot A_{turn}} = \frac{\rho_{cu} \cdot N_{turn} \cdot L_{turn} \cdot N_{slots}}{6 \cdot N_{parallel}^2 \cdot A_{turn}} \quad \text{(E.6)}$$

where ρ_{cu} denotes the electrical resistivity of copper at room temperature, which is $1.68 \cdot 10^{-8}\ \Omega \cdot \text{m}$ [115], and A_{turn} denotes the total cross-sectional area of all strands within one turn, which is $8.55 \cdot 10^{-6}\ \text{m}^2$.

Using the expressions above, the total copper conductor lengths are 78.65 m for the benchmark motor and 100.79 m for the selected ferrite-based motor. The corresponding equivalent phase resistances are $R_{\text{benchmark}} = 0.0057\ \Omega$ and $R_{\text{ferrite_based}} = 0.0073\ \Omega$, respectively. For the benchmark motor, the calculated value of 0.0057 Ω is close to the reported value of 0.0052 Ω in [60], providing a consistency check for the adopted analytical approximation. The recalculated resistance of the ferrite-based motor is subsequently used in the efficiency and use-phase calculations described in Section 4.4.3.

F. Detailed Modelling of Carbon-Fiber Sleeve Production

This appendix sets out the assumptions, process representations, parameter values, and data sources used to compile the life cycle inventory for the carbon-fiber composite rotor sleeve of the selected ferrite-based motor. It expands on the manufacturing-stage modelling in Section 4.4.2 and documents how the foreground inventory was established for this sleeve design.

F.1 Scope and system boundary

The carbon-fiber rotor sleeve is modelled using a one-step composite manufacturing route based on carbon-fiber prepreg. The system boundary includes upstream raw-material production, carbon-fiber production, resin production, prepreg formation, and sleeve manufacturing. The overall modelling structure is illustrated in Figure F.1, Figure F.2, and Figure F.3.

Carbon fiber and resin production are treated as background processes using externally sourced life cycle inventory datasets. Prepreg formation and sleeve manufacturing are modelled as foreground processes. Energy use during prepreg formation is excluded from the inventory, as it is assumed to be small compared with the energy required for carbon-fiber production and in-situ consolidation.

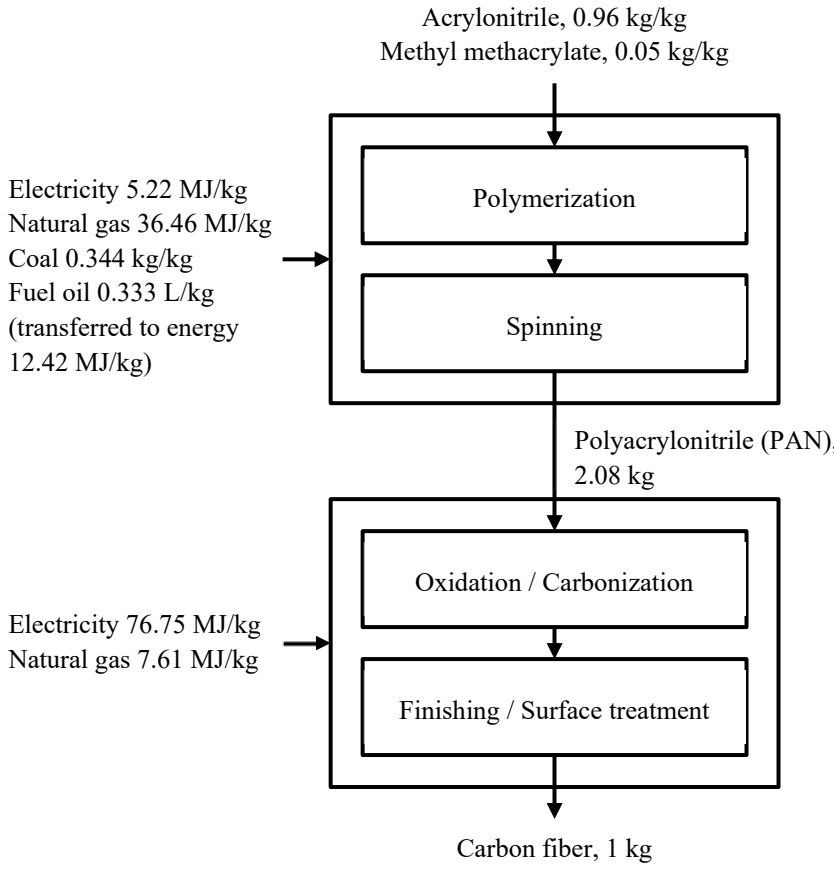


Figure F.1. Schematic representation of PAN-based carbon fiber production, based on [116], [117], [118], [119], [120]. From **Paper V** [94] under the CC BY 4.0 license.

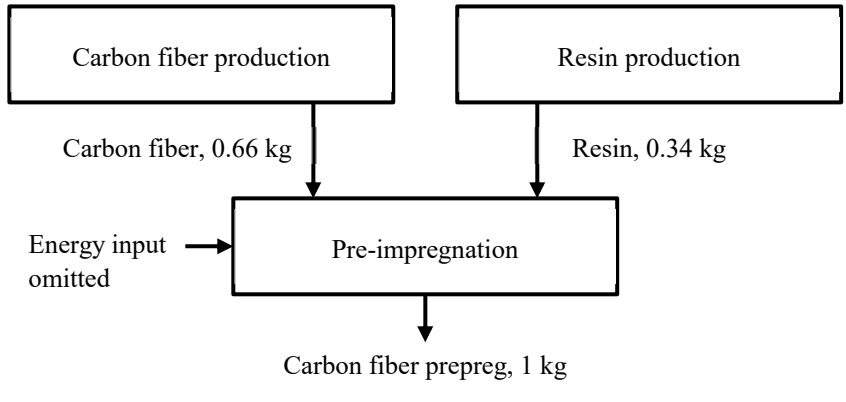


Figure F.2. Schematic representation of thermoplastic prepreg production, based on [121], [122]. From **Paper V** [94] under the CC BY 4.0 license.

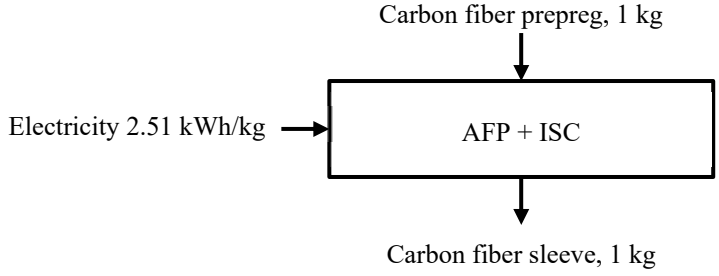


Figure F.3. Schematic representation of the assumed one-step carbon-fiber sleeve manufacturing process. From **Paper V** [94] under the CC BY 4.0 license.

F.2 Carbon fiber production

Carbon-fiber production is modelled using a polyacrylonitrile (PAN) precursor route, representing the industrial pathway assumed for the high-performance carbon fiber used in this study. As illustrated in Figure F.1, the production process includes polymerization and spinning of polyacrylonitrile precursor fibers, followed by stabilization, carbonization, and surface treatment.

The background life cycle inventory is obtained from the Federal LCA Commons using datasets provided by the National Renewable Energy Laboratory. The datasets represent PAN-based carbon-fiber production and include upstream material inputs such as acrylonitrile and auxiliary chemicals, together with electricity and fuel inputs associated with precursor production, stabilization, carbonization, and finishing.

The dataset is adopted without further modification, and carbon fiber is treated as a background material input to the prepreg production stage.

F.3 Resin production

The polymer matrix used in the carbon-fiber prepreg is represented by a thermoplastic resin. Background life cycle inventory data for resin production are obtained from Ecoinvent using market-average datasets consistent with the cut-off system model adopted in Section 4.2.3. The resin is treated as a background material input and is combined with carbon fiber according to the prepreg composition defined in Section E.4.

F.4 Carbon fiber prepreg production

The carbon-fiber prepreg is represented as a continuous carbon-fiber reinforced thermoplastic composite. A representative material system comprising TORAYCA™ T800GC-24K carbon fiber and Cetex® TC1200 resin is selected because its composition and material properties are documented in publicly available sources.

The resin mass fraction is set to 34%, with carbon fiber accounting for the remaining 66%. The prepreg density is assumed to be 1.59 g/cm³ based on manufacturer data.

Energy use during prepreg production is excluded from the model because the adopted inventory contains no process-specific energy data. This energy demand is assumed to be small compared with that of carbon-fiber production and sleeve consolidation.

F.5 Sleeve geometry and material composition

The rotor sleeve is assumed to have a uniform thickness of 1 mm. Based on the selected ferrite-based motor geometry defined in Section 4.3.2.4, the sleeve volume is calculated to be 99.04 cm^3. Using the adopted prepreg density of 1.59 g/cm³, the resulting sleeve mass is 157.47 g.

With a resin mass fraction of 34%, the sleeve consists of 53.54 g of resin and 103.93 g of carbon fiber. The resulting material composition is summarized in Table F.1.

Table F.1. Material composition of the carbon fiber rotor sleeve. From **Paper V** [94] under the CC BY 4.0 license.

Material	Weight [g]
Resin	53.54
Carbon fiber	103.93
Total	157.47

F.6 Manufacturing process selection

Carbon-fiber rotor sleeves can be produced using either separate sleeve manufacture followed by assembly, or direct winding and consolidation on the rotor. In the present study, a one-step route is assumed in which prepreg material is applied directly to the rotor and consolidated during placement.

This route is selected as a representative manufacturing process for the investigated high-speed rotor because it allows fiber tension and sleeve prestress to be controlled during winding. The two manufacturing routes are not compared environmentally in this study. The selected process should therefore be interpreted as a design-driven modelling assumption rather than an environmentally optimized production route.

F.7 Energy modelling for one-step sleeve manufacturing

The one-step sleeve manufacturing process is represented by automated fiber placement (AFP) combined with in-situ consolidation, as illustrated in Figure F.3. During this process, prepreg tapes are placed directly onto the rotor under controlled fiber tension, while an integrated heat source raises the material

temperature above the polymer melting point. Consolidation occurs during placement through the combined effects of heating and compaction pressure.

Because direct measurements of the energy consumption of AFP with in-situ consolidation applied specifically to motor rotor sleeves were not identified, the manufacturing energy demand is estimated using a simplified decomposition approach. The total energy consumption is divided into two principal contributions.

The first contribution represents the energy required for material transport, conveyor operation, and robotic motion. This contribution is approximated using literature data for filament-winding processes, with an energy intensity of 2.7 MJ per kg of processed composite material.

The second contribution is the energy required for in-situ consolidation. This demand is estimated at 20% of a reference autoclave energy consumption of 30 mmBTU per ton of processed composite material. The estimate is based on reported energy savings from induction heating compared with conventional autoclave curing.

Combining these two contributions gives a total manufacturing energy demand of approximately 2.51 kWh per kg of composite material. This value is implemented as a foreground process and scaled according to the calculated sleeve mass. The resulting energy demand should be interpreted as a process-based engineering estimate rather than a directly measured industrial value.

F.8 Limitations

The process representations adopted in this appendix are not intended to reproduce a specific industrial production line or proprietary manufacturing implementation. Variations in fiber architecture, tension profiles, heating technologies, tooling design, and production scale are not explicitly modelled. These simplifications affect only the ferrite-based motor and introduce uncertainty into the absolute manufacturing-stage results. However, they are applied consistently within the defined system boundary and do not change the basis for comparing the two motors in the LCA.

G. Detailed Calculations for End-of-Life Stage Modelling

This appendix provides the detailed calculations and assumptions used for the end-of-life modelling described in Section 4.4.4. The calculation covers material recovery, automobile shredder residue (ASR), ASR treatment, and the environmental factors used to report the potential benefits associated with recovered metals.

G.1 Recovered material mass calculation

Both traction motors are assumed to undergo shredding followed by material separation. The recovered mass of each metallic material is calculated as the product of its original mass in the motor, the shredding recovery rate, and the material-specific recovery efficiency. The adopted shredding and material-recovery efficiencies are summarized in Table G.1 and are based on literature values [107], [108], [109].

Table G.1. Shredding and material-recovery efficiencies adopted in the end-of-life model. From **Paper V** [94] under the CC BY 4.0 license.

Material	Shredding recovery [%]	Material recovery [%]
Steel	95	95.9
Aluminum	95	96.8
Copper	95	77.6

The resulting recovered material masses for the benchmark and ferrite-based motors are summarized in Table 4.7 in the main text.

G.2 Automobile shredder residue

The ASR mass is calculated as the total mass of magnets and non-metallic materials, including polymers, coatings, composites, and paints, plus the mass of other metals not recovered during shredding and separation. Recovery losses that do not enter the ASR stream are excluded from this calculation. The recovered materials and ASR therefore do not account for the full original motor mass.

G.3 ASR treatment

ASR is assumed to be incinerated. An average treatment energy demand of 13.18 MJ per kg of ASR is adopted from the literature [123]. In Ecoinvent, the process "market for residue from shredder fraction from manual dismantling" is used as a proxy for ASR treatment. Under the adopted dataset, this corresponds to approximately 2.47 kg CO_2-eq per kg of ASR, including upstream processes [95].

G.4 Environmental factors for recovered materials

Potential environmental benefits from metal recovery are estimated and reported separately, assuming that the recovered metals displace the production of equivalent quantities of market-average primary metals.
These avoided-production values are not added to the aggregated manufacturing- and use-phase results, in accordance with the cut-off system model adopted in the study.

Market-average Ecoinvent datasets are used as reference production processes for steel, copper, and aluminum. The greenhouse gas emission factors adopted in this study are summarized in Table G.2.

Table G.2. Climate change emission factors for the reference market-average material production datasets [95]. From **Paper V** [94] under the CC BY 4.0 license.

Material	CO_2-eq emissions to produce 1 kg material [kg CO_2-eq / kg]	Conservatively selected Ecoinvent processes as the reference
Steel	1.78	market for steel, unalloyed | steel, unalloyed | Cutoff, S - GLO
Copper	6.82	market for copper, cathode | copper, cathode | Cutoff, S - GLO
Aluminum	5.63	market for aluminium, cast alloy | aluminium, cast alloy | Cutoff, S - GLO

Using the recovered material masses, ASR treatment burdens, and reference production factors defined above, the estimated end-of-life climate change impacts are calculated and reported separately in Section 4.5.3.

H. Detailed Calculation of the Break-Even Electricity Carbon Intensity

For each motor concept, the total life cycle climate change impact per unit driving distance can be expressed as

$$GWP_{\text{total}} = GWP_{\text{manuf}} + CI_{\text{el}} \cdot E_{\text{use}} \tag{H.1}$$

where GWP_{manuf} is the manufacturing-stage climate change impact normalized by driving distance [g CO_2-eq/km], CI_{el} is the carbon intensity of electricity [g CO_2-eq/kWh], and E_{use} is the lifetime-normalized electricity consumption [kWh/km].

The break-even electricity carbon intensity is obtained by equating the total life cycle impacts of the benchmark and selected ferrite-based motors:

$$GWP_{manuf}^{Nd} + E_{use}^{Nd} \cdot CI_{el}^{*} = GWP_{manuf}^{Ferrite} + E_{use}^{Ferrite} \cdot CI_{el}^{*} \tag{H.2}$$

Solving for the break-even electricity carbon intensity CI_{el}^{*} yields:

$$CI_{el}^{*} = \frac{GWP_{manuf}^{Ferrite} - GWP_{manuf}^{Nd}}{E_{use}^{Nd} - E_{use}^{Ferrite}} \tag{H.3}$$

Using the results reported in Sections 4.5.1 and 4.5.2, the difference in manufacturing-stage climate change impact between the ferrite-based and benchmark motors is

$$GWP_{manuf}^{Ferrite} - GWP_{manuf}^{Nd} = 2.34 - 2.01 = 0.33 \text{ [g } CO_2\text{eq/km]} \tag{H.4}$$

The corresponding difference in lifetime-normalized electricity consumption is

$$\begin{aligned} E_{use}^{Nd} - E_{use}^{Ferrite} &= 114.33 - 108.78 = 5.55 \text{ [Wh/km]} \\ &= 0.00555 \text{ [kWh/km]} \end{aligned} \tag{H.5}$$

Substituting these values into (H.3) gives

$$CI_{el}^{*} = \frac{0.33}{0.00555} \approx 59.46 \text{ [g } CO_2\text{eq/kWh]} \tag{H.6}$$

The break-even electricity carbon intensity is therefore approximately 60 g CO_2-eq/kWh. Above this value, the selected ferrite-based motor has the lower combined manufacturing and use-phase climate change impact. Below it, the benchmark motor has the lower impact.

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